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College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The Food and Beverage (F&B) industry experienced significant growth over the past decade because of changing consumer lifestyles, rapid urbanization, and the increasing demand for convenient dining and beverage services. Among the businesses in this sector, coffee shops became one of the fastest-growing industries because they served not only as places where customers purchased beverages but also as venues for studying, meetings, remote work, and social interaction. As customer expectations continued to evolve, coffee shop businesses were expected to maintain high standards of service quality, operational efficiency, and customer satisfaction to remain competitive. The growing presence of both local and international coffee brands also intensified market competition, encouraging businesses to adopt more efficient operational strategies and modern management practices. As consumer preferences continued to shift toward convenience, speed, and personalized service, coffee shop owners increasingly relied on technology to improve their daily operations and overall business performance.

Modern coffee shops no longer focused only on selling beverages. They also provided customers with comfortable spaces for studying, working remotely, conducting business meetings, and spending time with family and friends. Because of these changing

customer needs, businesses were required to manage different operational activities efficiently while maintaining consistent product quality and customer service. The increasing use of digital technologies and online customer engagement raised customer expectations for faster transactions and better service experiences. As competition continued to increase, businesses recognized the importance of using technology to improve operational efficiency, strengthen customer satisfaction, and support long-term business growth.

As coffee shop businesses expanded into multiple branches, managing daily operations became more challenging. Opening additional kiosks and outlets increased the complexity of handling customer transactions, monitoring inventory, scheduling employees, recording sales, and evaluating branch performance. Tasks that were manageable in a single-location business became more difficult when operations were distributed across several branches. Businesses therefore required a centralized system that could provide consistent monitoring and standardized operational procedures across all locations. Without proper coordination, differences in transaction recording, inventory monitoring, and reporting practices affected service quality and business performance. Centralized operational management became essential in maintaining consistency and ensuring that all branches operated according to the same business standards.

Managing several branches also introduced additional administrative responsibilities. Branch managers sometimes followed different reporting procedures when standardized monitoring systems were unavailable. Communication gaps between

management and branch personnel delayed important operational decisions, while fragmented business records made it difficult to evaluate the overall performance of the organization. As the number of branches increased, comparing branch performance and identifying operational issues became more complicated. These challenges demonstrated the importance of implementing integrated information systems that provided centralized monitoring, improved communication, and supported informed decision-making across multiple business locations.

Kape Tubong experienced continuous growth by expanding from a single coffee outlet into several kiosks together with a main branch. This expansion reflected the increasing popularity of the business and the growing demand for its products. As business operations expanded, larger volumes of sales transactions and operational data were generated from different locations. Without an integrated information system, managing and monitoring these records became increasingly difficult. Sales information from different branches remained separated, making report consolidation more time-consuming and reducing management's ability to monitor overall business performance efficiently. Maintaining the consistency, accuracy, and accessibility of business information also became more challenging as the organization continued to grow.

The continuous expansion of Kape Tubong created new operational requirements that demanded better information management practices. Higher transaction volumes increased the complexity of monitoring revenue performance, evaluating branch operations, and consolidating business reports. Manual coordination between

branches and management became less efficient as the organization expanded, while fragmented data limited management's ability to obtain a complete view of business performance. These operational challenges highlighted the need for a centralized digital platform capable of supporting business growth through accurate reporting, efficient data management, and improved operational monitoring.

Like many small and medium-sized enterprises (SMEs), Kape Tubong initially relied on traditional and semi-manual methods to manage its daily operations. Sales transactions were recorded through spreadsheets and standalone Point-of-Sale (POS) systems used independently by each branch. While these methods were sufficient during the early stages of the business, they became less effective as the company expanded. Since every branch maintained its own records, consolidating sales reports and monitoring overall business performance required additional time and effort. Manual data entry also increased the possibility of duplicate records, missing information, and human errors, which affected the accuracy and reliability of reports used by management.

One of the major challenges encountered by the business was the lack of real-time access to sales information. Because transaction records from each branch were stored separately, management needed to collect reports from every location before preparing consolidated summaries. This process delayed the availability of important business information and made it difficult to respond quickly to operational concerns. The absence of a centralized reporting system also limited management's ability to monitor branch performance on a daily basis, making it

more challenging to identify issues that required immediate attention.

The existing reporting process focused mainly on presenting sales figures without providing meaningful analysis of business performance. Although reports showed the total number of transactions and revenue generated by each branch, they did not provide sufficient information about customer purchasing behavior, product demand, or sales trends. Management found it difficult to determine which products performed well, which menu items required improvement, and which periods generated the highest revenue. As a result, important business decisions often depended on manual analysis, which required additional time and effort before useful insights could be obtained.

Inventory management also became more challenging as business operations continued to expand. The consumption of ingredients depended directly on customer purchases, making accurate inventory monitoring essential to daily operations. Since sales transactions and inventory records were not fully integrated, monitoring stock movement required manual updates and regular verification. This process sometimes resulted in delays in replenishing supplies or maintaining excess inventory. These situations affected operational efficiency because management could not immediately determine the actual inventory status across different branches.

Employee scheduling presented another operational concern. Without detailed information about customer traffic and sales performance, work schedules were often prepared using previous experience instead of actual business data. During peak business

hours, some branches experienced increased workloads due to insufficient staffing, while slower periods occasionally resulted in unnecessary labor costs. Reliable sales analysis could have supported better workforce planning by identifying customer demand patterns and helping management assign employees more efficiently.

The growing volume of operational data demonstrated the need for a more effective information management system. Instead of relying solely on manual reports, businesses increasingly adopted Business Intelligence (BI) technologies to organize, analyze, and present information in a more meaningful way. Business Intelligence transformed raw transactional data into visual reports, dashboards, and performance indicators that supported faster and more informed decision-making. By providing accurate and timely business insights, BI systems helped organizations monitor sales performance, identify revenue trends, evaluate product demand, and improve overall operational efficiency.

1.2 Statement of the Problem

Kape Tubong expanded its operations by opening several branches, which resulted in an increase in sales transactions, inventory records, and daily business activities. As the organization continued to grow, managing information from multiple branches became more challenging. Each branch maintained separate sales records and operational reports, making it difficult for management to monitor overall business performance and access complete, accurate, and timely information. These challenges emphasized the need for a centralized information system that

could organize business data efficiently and support improved operational management.

The existing process relied heavily on manual data consolidation and report preparation. Employees collected, verified, and organized sales records from different branches before reports were generated for management review. This process required significant time and effort while increasing the possibility of errors, duplicate entries, missing records, and inconsistent information. Since records were maintained separately by each branch, management experienced difficulties in monitoring overall sales performance, comparing branch operations, and responding quickly to operational concerns. Delays in report preparation also affected the availability of essential information needed for effective decision-making.

The business also had limited capabilities in analyzing sales and operational data. Existing reports primarily presented numerical information without dashboards or visual analytical tools that could assist management in identifying sales trends, evaluating product performance, monitoring inventory movement, and comparing branch performance. Historical sales information was not fully utilized for forecasting customer demand, workforce requirements, and inventory needs. Due to these limitations, many operational and strategic decisions depended mainly on experience rather than reliable insights generated from business data.

As Kape Tubong continued to expand, these challenges became more significant. Manual reporting procedures, fragmented records, and limited analytical capabilities reduced operational efficiency and made business monitoring more difficult. The absence of

Business Intelligence features restricted management's ability to identify trends, evaluate performance, forecast future demand, and develop effective strategies for sustainable growth. These conditions highlighted the need to develop a centralized Point of Sale (POS) system integrated with Business Intelligence features to improve sales management, reporting, data analysis, inventory monitoring, workforce planning, and decision-making across all Kape Tubong branches.

This study aimed to address the following problems:

The existing operations of Kape Tubong lacked a centralized Point of Sale (POS) system capable of consolidating sales transactions from multiple branches into a single database. This limitation resulted in fragmented records, delayed report preparation, inconsistent data management, and difficulty accessing accurate and timely operational information.

The business had limited capability to analyze and visualize sales performance, product demand, inventory status, and branch performance. The absence of Business Intelligence tools made it difficult to identify sales trends, evaluate product performance, compare branch operations, and support timely data-driven decision-making.

Forecasting workforce requirements remained challenging because employee scheduling was not supported by historical sales data and customer transaction patterns. As a result, staffing decisions relied primarily on experience, which affected labor utilization, operational efficiency, and

service quality during different business periods.

Sales reporting depended on manual preparation and consolidation of records from different branches. These procedures required considerable time and were vulnerable to errors, inconsistencies, duplicate records, and reporting delays, reducing the accuracy, reliability, and efficiency of business reporting.

The business did not fully utilize Business Intelligence technologies to transform sales and operational data into meaningful information for forecasting, performance evaluation, and strategic planning. This limitation reduced the organization's ability to improve operational efficiency, support data-driven decisions, and sustain long-term business growth.

This study also contributed to the increasing adoption of analytics and digital reporting technologies among small and medium-sized enterprises despite resource limitations. It provided additional evidence that digital transformation initiatives could be implemented by developing local businesses to improve operational monitoring and decision-making processes. The study demonstrated that customized web-based analytical systems were not limited to large corporations but could also provide valuable support to growing enterprises with practical business challenges.

Through the development and implementation of BREWINSIGHT BI: Revenue Pattern Analytics System, the project contributed to Kape Tubong's organizational improvement by providing a centralized

platform for managing business information and generating meaningful insights. The study also supported broader discussions regarding SME digital modernization and demonstrated the potential of Business Intelligence solutions as a practical strategy for improving business operations. The findings may serve as a reference for future technology-based initiatives involving similar organizations and industries.

1.3 Objectives of the Study

1.3.1 General Objectives

The general objective of this study was to design and develop BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong, a web-based Point of Sale (POS) system integrated with Business Intelligence (BI) features. The system was developed to centralize sales transactions, operational records, and business-related information from all Kape Tubong branches into a unified database. Through this centralized approach, the study aimed to improve the organization, accessibility, and management of business data while addressing the limitations caused by fragmented records and manual reporting procedures. By integrating sales management and Business Intelligence capabilities into a single platform, the study sought to provide accurate, organized, and timely business information that supported improved operational monitoring, strategic planning, and decision-making processes.

The study aimed to transform raw sales transactions and operational records into meaningful and actionable business

insights through the use of interactive dashboards, automated reports, and analytical visualizations. The developed system presented important business information, including revenue patterns, sales trends, branch performance, product performance, customer purchasing behavior, and inventory movement, in a clear and organized format. These analytical features enabled management to evaluate business performance more effectively, identify emerging trends, recognize operational concerns, and develop strategies based on actual business data rather than relying only on assumptions or personal experience. By providing centralized access to updated sales information and key performance indicators, the platform improved the availability, accuracy, and usefulness of business data for daily operations and long-term planning.

The study also aimed to improve the efficiency, accuracy, and reliability of sales reporting by reducing dependence on manual preparation, consolidation, and verification of business records. The automated reporting process minimized repetitive administrative tasks, reduced the possibility of human errors, and improved the consistency of reports generated from different branches. Centralized data storage allowed the system to maintain complete and organized historical records that could be retrieved, reviewed, and analyzed whenever necessary. These improvements supported faster report generation, simplified branch performance comparison, and provided management with reliable operational information needed for effective evaluation and planning.

Another objective of the study was to support inventory

management, workforce planning, and resource allocation through the analysis of historical sales data and operational patterns. The system provided insights regarding product demand, peak transaction periods, customer purchasing behavior, and branch performance, allowing management to make informed decisions regarding inventory replenishment, employee scheduling, promotional activities, and business strategies. These analytical capabilities helped improve operational planning, maximize resource utilization, enhance service quality, and support more efficient management of business activities across all Kape Tubong branches.

The study also aimed to strengthen the security, reliability, and integrity of business information through the implementation of secure user authentication, role-based access control, and controlled data management procedures. These security measures ensured that confidential sales records, financial information, and operational reports were accessed only by authorized personnel according to their assigned responsibilities. The system provided a protected environment for storing and managing business information while allowing authorized users to monitor operations and access necessary reports through a web-based platform. These features contributed to maintaining data confidentiality, preventing unauthorized access, and improving trust in the accuracy and reliability of the information generated by the system.

The study aimed to enhance operational efficiency, improve reporting accuracy, strengthen analytical capabilities, and support data-driven managerial decision-making through the

development of a centralized Business Intelligence-based Point of Sale system. The developed BREWINSIGHT BI platform provided Kape Tubong with a scalable and adaptable technological solution that addressed existing challenges in sales monitoring, data management, reporting, and business analysis. Through the integration of POS and Business Intelligence features, the system supported more organized information management, improved accessibility of operational data, and enabled management to make informed decisions based on accurate and timely insights.

The implementation of the system contributed to improving the way Kape Tubong managed its business operations by reducing dependence on manual processes, increasing the efficiency of data analysis, and providing a more reliable method for monitoring business performance. The platform also established a foundation for future improvements by allowing the organization to accommodate increasing transaction volumes, additional branches, and changing business requirements. By providing a centralized and data-driven approach to managing operations, BREWINSIGHT BI strengthened the company's ability to evaluate performance, respond to operational challenges, and develop strategies that supported continuous improvement and sustainable business growth.

1.3.2 Specific Objectives

Specifically, this study aimed to:

- Develop a centralized web-based dashboard that consolidated and organized sales transactions from all Kape Tubong branches into a single platform to improve business monitoring, operational visibility, performance evaluation,

and centralized data management. The dashboard was designed to provide management with real-time access to branch sales, allowing faster monitoring of business activities, easier comparison of branch performance, and more efficient management of sales information across all locations.

Automate the generation of daily, weekly, and monthly sales reports to reduce manual processing, minimize human errors, improve reporting accuracy, and provide timely access to reliable business information. The automated reporting feature was intended to simplify report preparation, standardize reporting procedures, maintain organized historical records, and support faster evaluation of business performance through consistent and accurate reports.

Integrate Business Intelligence features that processed and visualized revenue patterns, sales trends, customer purchasing behavior, branch performance, peak operating hours, and top-selling products through interactive dashboards and analytical reports. These features were developed to transform raw sales data into meaningful business insights that supported forecasting, strategic planning, performance evaluation, and data-driven decision-making.

Provide inventory-related analytical insights based on sales performance data to assist management in monitoring product demand, identifying fast-moving and slow-moving products, improving inventory planning, reducing stock shortages and

overstocking, and supporting efficient resource allocation across all Kape Tubong branches. These analytical capabilities also helped management align purchasing decisions with actual customer demand and improve inventory utilization

Implement secure authentication and role-based access control to protect business information while allowing authorized users to securely access system functions, reports, dashboards, and analytical data through a web-based platform. These security measures were intended to maintain the confidentiality, integrity, and availability of sales records while supporting safe and reliable system access for management and authorized personnel.

1.4 Scope and Delimitation

This study focused on the design and development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong, a web-based Point of Sale (POS) system integrated with Business Intelligence (BI) features. The system was developed to centralize sales transactions from the main branch and all Kape Tubong kiosks into a single database to improve the management, organization, and analysis of business information. It served as a centralized platform that enabled management to monitor branch operations, evaluate sales performance, and generate analytical reports using data collected from daily business transactions. Through this approach, the study sought to improve operational efficiency, reduce dependence on fragmented reporting practices, and

strengthen data-driven decision-making within the organization.

The developed system included modules for transaction recording, product management, branch management, dashboard visualization, automated report generation, and Business Intelligence analytics. Sales data from all branches were consolidated into a centralized database, allowing management to access complete and updated information through a single web-based platform. The dashboard presented business performance using charts, graphs, summary panels, and comparative reports, making sales information easier to understand and interpret. The system also generated daily, weekly, and monthly reports that supported regular business monitoring and performance evaluation. By integrating these functions into one platform, the study improved the accessibility, consistency, and organization of sales information across all Kape Tubong branches.

The Business Intelligence component of the system analyzed historical and current sales transactions to identify revenue patterns, product demand, customer purchasing behavior, peak operating hours, and branch performance. The analytical results were presented through interactive visualizations that enabled management to monitor business trends more efficiently. These features helped identify top-selling and slow-moving products, compare sales performance among branches, and evaluate revenue fluctuations across different periods. The generated insights supported data-driven planning, promotional activities, inventory management, and operational decision-making. The system transformed raw transaction records into meaningful information that improved the overall understanding of business performance.

The study also included inventory-related analytical features derived from sales transactions. The system evaluated product demand and sales movement to identify fast-moving and slow-moving products and provide recommendations that supported inventory planning. These analytical outputs assisted management in maintaining appropriate stock levels, reducing overstocking, preventing stock shortages, and improving purchasing decisions based on historical sales performance. Branch-specific inventory insights also enabled management to recognize differences in product demand among locations and allocate inventory more effectively. However, the system functioned only as a decision-support tool and did not automate stock replenishment, warehouse management, or procurement processes.

To improve operational monitoring, the system generated branch-level and overall business performance reports that allowed management to compare sales results among different locations. Comparative analysis enabled the identification of high-performing and underperforming branches based on actual sales records. These reports supported performance evaluation, operational assessment, and planning for future improvements. Management could also review historical sales information to monitor business growth and evaluate the effectiveness of implemented strategies over time. The centralized reporting structure reduced the need for manual consolidation of branch reports and improved the accuracy and consistency of business information.

The system also included security features to protect business information and maintain data integrity. User authentication and

role-based access control were implemented to ensure that only authorized personnel could access system functions and confidential sales records. Different user roles were assigned specific access privileges according to their responsibilities within the organization. Authorized users could securely access the system using desktop or mobile web browsers with an internet connection. These security measures protected business information while allowing management to monitor operations remotely whenever necessary. Centralized storage of sales records also improved data reliability by reducing duplication, inconsistencies, and fragmented record management.

The scope of the study was limited to the analysis of internal sales and transaction data generated by Kape Tubong. All dashboards, reports, and analytical outputs were based solely on information recorded through the developed system. External factors, including competitor activities, customer demographics, economic conditions, weather changes, supplier performance, market trends, and other environmental influences, were excluded from the analytical process. As a result, the generated reports reflected only the operational performance of the business based on available transaction records.

Several delimitations were established to maintain the focus and feasibility of the study. The developed system did not include physical Point of Sale hardware such as barcode scanners, receipt printers, cash drawers, card readers, or customer display devices. Payroll processing, accounting, taxation, budgeting, supplier management, procurement, warehouse management, customer relationship management, online payment gateways, and other

Enterprise Resource Planning (ERP) functions were also excluded. The system focused only on centralized sales management, Business Intelligence, reporting, and revenue analysis.

The study also excluded advanced technologies such as machine learning, artificial intelligence, predictive analytics, and automated forecasting models. The implemented Business Intelligence features were limited to descriptive analytics, historical trend analysis, dashboard visualization, comparative reporting, and revenue pattern analysis based on recorded sales transactions. Forecasting models and predictive algorithms were not included because they were beyond the scope and timeframe of the study. These features may be considered for future enhancement as business requirements continue to evolve.

The effectiveness of the developed system depended on the accuracy and completeness of the sales transactions entered by authorized users. The study assumed that all business records were encoded correctly and consistently across all branches. Although the system included validation procedures and user authentication, it did not independently verify the accuracy of manually entered data beyond standard input controls. Consequently, the quality of generated reports, dashboards, and analytical results depended on the reliability of the available sales records.

The developed system was specifically designed to meet the operational requirements of Kape Tubong. The database structure, dashboard components, reports, and analytical features were customized according to the organization's existing business processes and reporting requirements. Although the system could

be modified for implementation in other businesses, additional customization would be necessary to accommodate differences in operational workflows, product categories, reporting requirements, and organizational structures. Therefore, the study remained limited to the operational environment of Kape Tubong.

The evaluation of the developed system was limited to functional and user acceptance testing conducted during the implementation period. Long-term operational performance, user adoption, infrastructure expansion, organizational change management, and future software enhancements were beyond the scope of this study. Despite these limitations, the developed BREWINSIGHT BI: Revenue Pattern Analytics System demonstrated the practical application of Business Intelligence in supporting centralized sales management, automated reporting, revenue analysis, inventory-related insights, and data-driven decision-making. The system also provided a scalable foundation that could support future improvements and digital transformation initiatives within Kape Tubong while addressing its current operational requirements.

1.5 Significance of the Study

The proposed system was expected to benefit the following:

Management - The proposed system provided significant benefits to the management of Kape Tubong by centralizing sales transactions and business information from all branches into a single web-based platform. It enabled management to monitor sales performance, compare branch operations, and access updated

reports without relying on manual data consolidation. Through interactive dashboards, automated reporting, and Business Intelligence features, the system transformed raw sales records into meaningful information that supported data-driven decision-making. These analytical tools assisted management in identifying sales trends, monitoring product performance, evaluating branch productivity, and recognizing revenue patterns across different periods.

The system also supported strategic planning by providing reliable information for decisions related to pricing, promotional activities, workforce scheduling, inventory planning, and future business expansion. Historical sales records and comparative reports enabled management to evaluate business performance over time and measure the effectiveness of implemented strategies. The centralized reporting environment reduced delays in obtaining operational information and improved the accuracy and consistency of business reports.

The platform strengthened management oversight by providing standardized performance indicators for every branch. It allowed management to identify operational issues, declining sales, and unusual transaction patterns at an earlier stage, enabling timely corrective actions. Branch comparisons also helped determine which locations required additional support and which operational practices could be adopted across the organization. By providing complete, organized, and accessible business information, the system improved operational control, strengthened strategic planning, and enhanced the overall effectiveness of managerial decision-making.

Staff - The implementation of the proposed system benefited staff members by reducing the time and effort required to prepare sales reports and manage transaction records. Automated reporting and centralized data processing minimized repetitive clerical work, allowing employees to devote more time to customer service and daily operational responsibilities. Standardized reporting procedures also reduced inconsistencies in documentation and improved coordination among branch personnel.

The system simplified the process of recording, retrieving, and reviewing sales information, reducing the likelihood of human error commonly associated with manual reporting. Employees no longer needed to consolidate multiple spreadsheets or manually compute sales summaries, making reporting activities more efficient and organized. The availability of centralized records also improved communication between staff and management because operational information became easier to access and verify.

The platform further contributed to a more productive working environment by simplifying routine tasks and improving workflow efficiency. Standardized documentation supported better accountability and reduced confusion caused by inconsistent reporting practices. Historical records could also be retrieved more easily whenever verification or performance reviews were required. Overall, the system improved staff productivity, reduced administrative workload, and supported a more organized and efficient workplace.

Organization - The proposed system benefited Kape Tubong by

improving the management of sales information and strengthening overall business operations. Through a centralized database, the organization was able to maintain consistent sales records from all branches, reducing fragmented reporting practices and improving the accuracy of operational information. This centralized approach enhanced transparency, accountability, and coordination across different business locations.

The Business Intelligence features of the system enabled the organization to analyze revenue patterns, evaluate branch performance, identify product demand, and monitor customer purchasing behavior. These analytical capabilities supported better planning for inventory management, workforce allocation, promotional activities, and future business expansion. Historical sales data also provided valuable information for evaluating business growth and assessing the effectiveness of organizational strategies.

The system further supported organizational development by promoting data-driven decision-making and reducing dependence on manual reporting procedures. Better access to accurate and timely business information improved managerial responsiveness and strengthened operational control. The platform also provided a scalable technological foundation that could support future digital transformation initiatives and accommodate continued business growth. Overall, the implementation of the proposed system contributed to improved operational efficiency, stronger business management, and long-term organizational sustainability

BSIT Students - The study benefited Bachelor of Science in Information Technology (BSIT) students by providing a practical

example of how software engineering, database management, web development, and Business Intelligence technologies were integrated to address real-world business problems. It demonstrated how concepts learned in the classroom could be applied in developing a functional web-based analytics platform that supported business operations and decision-making. The study also served as a useful reference for students conducting capstone projects related to Business Intelligence, decision support systems, dashboard development, sales analytics, and business process automation.

By examining the implementation of the system, students gained a better understanding of the complete software development life cycle, including requirements gathering, system analysis, database design, interface development, implementation, testing, and deployment. The project also emphasized the importance of developing user-centered systems that addressed actual organizational needs. Through the use of dashboard visualization, automated reporting, and centralized data management, students were introduced to practical methods of transforming business data into meaningful information that supported decision-making. Overall, the study strengthened students' technical knowledge, analytical skills, problem-solving abilities, and understanding of modern Business Intelligence technologies, preparing them for future careers in the information technology profession.

Future Researchers - The study served as a valuable reference for future researchers interested in Business Intelligence, sales analytics, decision support systems, digital transformation, and information systems development. It provided a practical

framework for designing and implementing a centralized revenue analytics platform for a small and medium-sized enterprise in the food and beverage industry. The system architecture, development process, and analytical features could serve as a guide for future studies involving similar business environments.

Future researchers could further improve the proposed system by integrating advanced technologies such as predictive analytics, machine learning, artificial intelligence, cloud computing, and mobile-based applications. They could also evaluate the effectiveness of these technologies in improving business performance, forecasting sales trends, and supporting organizational decision-making. In addition, the study could serve as a basis for comparative research involving different industries, organizations, or operational environments. Overall, the study contributed to the existing body of knowledge on Business Intelligence implementation and provided a strong academic foundation for future research involving data-driven business solutions and digital innovation.

1.6 Definition of Terms

Analytics - Referred to the systematic process of collecting, organizing, examining, and interpreting data to identify meaningful patterns, trends, and relationships that supported business decision-making. It transformed raw business data into useful information that enabled organizations to evaluate performance, monitor operational activities, and improve strategic planning. Through analytics, businesses were able to

examine historical and current data to understand customer purchasing behavior, evaluate sales performance, identify profitable products, and recognize recurring business trends. Analytics also assisted organizations in measuring operational efficiency, forecasting future demand, and allocating resources more effectively based on factual information instead of assumptions. By continuously analyzing business data, organizations were able to identify operational strengths, detect weaknesses, and implement appropriate improvements that contributed to business growth and sustainability. As digital technologies continued to develop, analytics became an essential component of modern business management because it provided measurable information that supported evidence-based decision-making. In this study, analytics referred to the systematic evaluation of Kape Tubong's sales transactions to identify revenue trends, customer purchasing behavior, product demand, branch performance, and peak operating periods through the BREWINSIGHT BI system.

Business Intelligence (BI)- Referred to the collection of technologies, processes, and analytical tools used to collect, organize, analyze, and present business information in a meaningful and understandable manner. It transformed raw operational and transactional data into useful insights through dashboards, reports, charts, graphs, and other visualization tools that supported managerial decision-making. Business Intelligence enables organizations to monitor daily operations, evaluate organizational performance, identify business trends, and improve planning based on reliable and accurate information. It also promoted operational consistency by integrating data from

different sources into a centralized platform where information could be analyzed efficiently. Through Business Intelligence, organizations were able to improve productivity, strengthen operational control, enhance reporting accuracy, and support long-term business development. As businesses generated increasing amounts of digital information, Business Intelligence became an essential organizational tool for improving efficiency, competitiveness, and strategic planning. In this study, Business Intelligence referred to the analytical framework integrated into the BREWINSIGHT BI platform for collecting, processing, analyzing, and visualizing Kape Tubong's sales and revenue data to support effective operational and managerial decision-making.

Dashboard - Referred to a graphical user interface designed to present important business information through charts, graphs, tables, summary panels, and other visual components. It provided users with a clear and organized presentation of key performance indicators and operational metrics, making business information easier to understand and interpret. Dashboards simplified large volumes of transactional data by presenting summarized information that allowed managers to monitor organizational performance, compare operational results, and identify trends efficiently. Interactive dashboards also included features such as filtering, sorting, and drill-down analysis, enabling users to examine specific information in greater detail according to their analytical needs. Because information was presented visually, dashboards reduced the complexity of traditional reports and enabled both technical and non-technical users to understand business performance more quickly. They also improved communication among management by providing a standardized

presentation of organizational data. In this study, the dashboard referred to the web-based interface of the BREWINSIGHT BI system that displayed Kape Tuong's sales performance, revenue patterns, branch comparisons, product performance, and other business analytics through interactive charts, graphs, and summarized reports.

Data-Driven Decision-Making - Referred to the process of making business and operational decisions based on analyzed data rather than intuition, assumptions, or personal judgment. This approach relied on accurate, relevant, and timely information to support planning, problem-solving, and organizational improvement. By using factual evidence, organizations were able to reduce uncertainty, improve consistency, and develop more effective business strategies. Data-driven decision-making enabled management to evaluate historical performance, identify operational issues, monitor organizational progress, and implement appropriate corrective actions based on measurable business information. It also strengthened accountability because business decisions were supported by documented evidence instead of subjective opinions. Furthermore, this approach improved strategic planning by helping organizations identify opportunities for growth, respond to market changes, and continuously improve operational performance. As analytical technologies became more accessible, data-driven decision-making became increasingly important for organizations seeking to improve efficiency and competitiveness. In this study, data-driven decision-making referred to the use of analytical reports, dashboards, and sales insights generated by the BREWINSIGHT BI system to assist Kape Tubong's management in

making informed operational and strategic decisions based on actual sales performance and revenue analysis.

Kiosk - Referred to a small retail outlet or satellite sales location established to provide products and services in areas with high customer traffic. Compared with full-sized branches, kiosks occupied smaller operational spaces while offering similar products and services to customers. Businesses commonly established kiosks in shopping malls, transportation terminals, schools, commercial centers, and other strategic locations to improve customer accessibility and expand market coverage. Although smaller in size, kiosks generated valuable transactional information that could be analyzed to evaluate business performance, customer demand, and sales trends. Sales records collected from multiple kiosks could be consolidated into a centralized information system, allowing management to monitor operations, compare branch performance, and make more informed business decisions. Kiosks also supported business expansion by increasing market presence while maintaining centralized operational monitoring. In this study, kiosk referred to the satellite sales locations of Kape Tubong where sales transactions were recorded and consolidated into the BREWINSIGHT BI platform for centralized monitoring, branch comparison, and revenue analysis.

Point of Sale (POS) - Referred to the hardware and software system used to process, record, and manage sales transactions between a business and its customers. A POS system captured important transaction details such as products sold, quantity purchased, selling price, payment method, transaction date, and

transaction time. It improved transaction accuracy by minimizing manual recording errors while maintaining organized digital records that supported reporting and business analysis. Modern POS systems also included features such as inventory monitoring, transaction history, sales reporting, and customer management, making them essential tools for business operations. In organizations operating multiple branches, POS systems promoted standardized transaction recording and generated reliable business data that could be integrated into analytical platforms. The information collected through POS systems served as the primary source for generating dashboards, reports, and performance evaluations. In this study, Point of Sale (POS) referred to the transaction-recording system used by Kape Tubong branches to capture sales information that was integrated into the BREWINSIGHT BI platform for centralized reporting, visualization, and revenue pattern analysis.

Revenue Pattern - Referred to the observable trend or behavior of business income over a specific period based on recorded sales transactions. It reflected how revenue changed according to factors such as time, branch location, product demand, customer purchasing behavior, seasonal conditions, and promotional activities. Revenue pattern analysis enabled organizations to identify peak sales periods, slow business intervals, seasonal fluctuations, and products that contributed significantly to business performance. It also allowed management to compare current performance with previous periods to evaluate the effectiveness of operational strategies, marketing campaigns, and pricing decisions. Understanding revenue patterns supported better planning, inventory management, staffing decisions, and

resource allocation. Continuous monitoring of revenue behavior also helped organizations evaluate business growth, recognize emerging opportunities, and identify operational weaknesses requiring improvement. In this study, revenue pattern referred to the sales trends and income behaviors identified by the BREWINSIGHT BI system through the analysis of Kape Tubong's transactional sales records from the main branch and its kiosks.

Web-Based System - Referred to a software application that operated on a web server and was accessed through a web browser using an internet or local network connection. Unlike traditional desktop applications, a web-based system did not require installation on individual computers because authorized users could access its features through supported web browsers using different devices. Web-based systems provided greater flexibility by allowing users to retrieve information from various locations using desktop computers, laptops, tablets, or mobile devices connected to the internet. They also supported centralized data storage, ensuring that users accessed consistent and updated information while reducing data duplication across the organization. In addition, web-based systems simplified software maintenance because updates were performed on the server rather than on individual devices. Security mechanisms such as user authentication and role-based access control also protected business information from unauthorized access. In this study, the web-based system referred to the online architecture of the BREWINSIGHT BI platform that enabled authorized Kape Tubong personnel to securely access sales reports, dashboards, revenue

analytics, and business performance information through internet-connected devices.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Kumar and Singh (2022) presented the study entitled **"Data Visualization in Retail Analytics: Impact on Decision-Making,"** which investigated the role of interactive visualization technologies in improving managerial decision-making within retail and food-service organizations. The study focused on how dashboard-driven analytical tools assisted businesses in interpreting large amounts of transactional and operational information more effectively. The researchers found that organizations utilizing visual analytics platforms demonstrated improved efficiency in strategic decision-making compared with those that depended primarily on conventional spreadsheets and text-based reports. The study emphasized that interactive dashboards allowed managers to examine business information through visual components such as graphs, charts, heatmaps, and trend indicators, making complex data easier to understand and evaluate. Through these visual presentations, decision-makers were able to recognize important patterns and relationships within business records more efficiently than through manual examination of numerical reports.

The researchers explained that data visualization improved the ability of organizations to monitor important business indicators, including revenue movement, sales performance, product results, and operational changes across different

periods. By converting large amounts of numerical information into organized visual displays, analytical systems reduced the difficulty of interpreting business data and allowed users to evaluate information with greater speed and accuracy. Visual dashboards enabled managers to identify unusual sales behaviors, declining performance, and operational concerns at an earlier stage, allowing immediate actions and adjustments to be implemented. This capability supported proactive management because organizations could respond to possible issues before they created significant effects on business operations.

The study also highlighted that dashboard-based visualization improved communication and collaboration among management teams by providing a shared platform for analyzing business performance. Since visual reports presented information in a clear and standardized manner, different users could evaluate organizational conditions using the same data references and performance indicators. This reduced differences in interpretation and improved coordination during planning and decision-making activities. The availability of self-service analytics also allowed managers to explore information independently without requiring continuous assistance from technical personnel. As a result, organizations became more flexible in comparing branch performance, analyzing product results, and reviewing operational trends based on their current requirements.

Kumar and Singh (2022) further identified the importance of visualization tools in supporting business forecasting and strategic planning. Through the analysis of historical

information, organizations were able to recognize recurring sales patterns, seasonal variations, and changes in customer behavior. These insights provided valuable support for activities such as inventory preparation, employee scheduling, promotional planning, and resource allocation. The researchers emphasized that adopting integrated visualization platforms contributed to digital transformation efforts in retail and food-service industries because these technologies improved the ability of businesses to use available information effectively. Organizations with stronger analytical capabilities became more prepared to handle market changes, shifting customer preferences, and operational challenges.

The study also revealed that visualization-based analytics increased managerial confidence by presenting information in a transparent, structured, and understandable format. Clear graphical representations helped reduce uncertainty during decision-making because managers could easily interpret performance conditions and evaluate possible actions. Dashboard systems improved accessibility by placing important business measurements in a centralized environment where users could review essential information without requiring advanced knowledge of data analysis. This encouraged organizations to rely more on evidence-based strategies and strengthened the use of analytical information in regular management practices.

Another important finding of the research involved the value of continuous monitoring through real-time or regularly updated business information. Unlike traditional reporting approaches that provided information only after manual preparation,

dashboard systems allowed organizations to observe operational conditions more consistently. Continuous access to updated sales and performance data enabled managers to respond quickly to changes in demand, revenue performance, and resource requirements. This capability supported more effective adjustments in staffing decisions, inventory management, and promotional activities. The researchers concluded that continuous monitoring improved operational awareness and allowed businesses to manage daily activities more efficiently.

The findings of Kumar and Singh (2022) were highly applicable to the development of BREWINSIGHT BI: Revenue Pattern Analytics System because they supported the integration of interactive dashboards and analytical visualization features within the proposed platform. Kape Tubong experienced difficulties caused by separated sales records, limited centralized monitoring, and challenges in evaluating the performance of multiple branches. These limitations affected the ability of management to obtain complete information regarding revenue patterns and operational performance. Through the implementation of dashboard visualization, BREWINSIGHT BI converted collected sales transactions into organized analytical displays that allowed managers to identify successful products, compare branch performance, analyze sales periods, and recognize changes in revenue trends.

The application of self-service analytics and comparative monitoring also supported the objectives of BREWINSIGHT BI. The system enabled Kape Tubong administrators and managers to review operational information without depending on manual consolidation

of reports from different branches. The analytical capabilities discussed in the study aligned with the purpose of the proposed system to identify sales patterns, support resource planning, and improve managerial decision-making. Through centralized dashboards, management gained access to a unified platform that simplified monitoring of different locations and improved the ability to respond quickly to business concerns.

The findings regarding reduced dependence on technical personnel from Kumar and Singh (2022) also supported the usability goals of BREWINSIGHT BI. The study demonstrated that Business Intelligence platforms with clear interfaces and accessible analytical tools allowed users with limited technical experience to understand and utilize business information effectively. This concept was suitable for small and medium enterprises where dedicated data specialists might not always be available. By providing understandable dashboards, organized reports, and simplified analytical features, BREWINSIGHT BI supported easier adoption among Kape Tubong personnel and allowed users to obtain valuable insights without requiring advanced technical skills.

The research of Kumar and Singh (2022) provided significant support for integrating dashboard visualization and analytical reporting functions into BREWINSIGHT BI. Their findings showed that visual analytics improved the interpretation of business information, increased decision-making efficiency, strengthened operational monitoring, and enhanced organizational responsiveness. The study served as an important basis for recognizing the value of Business Intelligence features in

improving the ability of retail and food-service businesses to manage information and develop effective strategies.

Reference:

Kumar, R., & Singh, P. (2022). *Data Visualization in Retail Analytics: Impact on Decision-Making*. *International Journal of Business Intelligence, 14*(2), 45–58. ISSN 2710-3102 (Online). https://doi.org/10.11114/ijbi.v14i2.25472

Chen Li and Maria Gonzalez (2021) conducted the study entitled **"Business Intelligence Systems and Organizational Performance in Small and Medium Enterprises,"** which examined how Business Intelligence adoption influenced operational efficiency, decision-making quality, and organizational performance among small and medium-sized businesses. The research focused on the contribution of centralized BI platforms in improving access to information, increasing reporting accuracy, and supporting faster managerial responses compared with organizations that relied on traditional manual reporting procedures. The researchers explained that centralized Business Intelligence systems allowed companies to integrate various types of operational information, including sales transactions, inventory records, and performance reports, into a single analytical environment. This improved visibility across departments and locations by providing management with a complete overview of business conditions.

The study emphasized that Business Intelligence solutions improved strategic decision-making by converting large amounts of

raw business data into useful and actionable information. Through features such as dashboards, automated reports, and trend analysis, managers were able to evaluate sales performance, understand customer purchasing behavior, and identify operational weaknesses more effectively. Organizations that adopted BI systems were able to respond more quickly to changes in business conditions, including declining sales, inventory shortages, and shifting customer demands. The automation of reporting processes also reduced the time required for manual data preparation while improving the consistency and accuracy of information presented to decision-makers.

The researchers further identified that Business Intelligence implementation improved organizational coordination by establishing standardized reporting methods and consistent performance measurements. Through centralized analytical systems, managers and supervisors from different locations could evaluate business activities using similar indicators and reporting structures. This improved the reliability of comparisons among branches, products, and operational periods. In addition, user-friendly BI interfaces allowed employees with limited technical backgrounds to access and understand analytical information independently, reducing the need for continuous assistance from information technology personnel when preparing routine reports.

The study also demonstrated that organizations using Business Intelligence technologies developed stronger capabilities in forecasting and operational planning. Historical business records provided valuable information for identifying recurring sales

behaviors, seasonal demand changes, and customer preferences. These insights supported better decisions related to inventory management, workforce scheduling, promotional activities, and allocation of organizational resources. The researchers highlighted that scalable BI solutions provided important advantages for small and medium enterprises because they allowed growing businesses to improve competitiveness and adapt more effectively to changing market conditions.

The findings of Chen Li and Maria Gonzalez (2021) were relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System because they supported the importance of creating a centralized Business Intelligence platform for improving business monitoring and decision-making. Kape Tubong encountered operational challenges caused by fragmented sales records, manual report preparation, and limited visibility regarding the performance of its different branches. These conditions affected the ability of management to analyze business performance accurately and obtain timely information for planning.

BREWINSIGHT BI addressed these challenges by combining Point of Sale capabilities with Business Intelligence features in a unified platform. The system centralized sales information from different branches and transformed collected records into meaningful analytical outputs through dashboards, reports, and performance evaluations. Through revenue trend analysis, product performance monitoring, branch comparison features, and automated reporting functions, the platform supported improved operational management and provided management with more reliable information for decision-making.

The study of Chen Li and Maria Gonzalez (2021) strengthened the foundation of BREWINSIGHT BI by demonstrating that Business Intelligence technologies could provide practical benefits for small and medium enterprises. Their findings confirmed that centralized data management, automated reporting, and analytical tools could improve efficiency, increase information accessibility, and support evidence-based planning. These concepts aligned with the objectives of the proposed system, which aimed to enhance Kape Tubong's ability to monitor operations, analyze revenue patterns, and make informed decisions through the effective use of business data.

Reference:

Li, C., & Gonzalez, M. (2021). *Business Intelligence Systems and Organizational Performance in Small and Medium Enterprises*. *Journal of Information Systems and Technology Management, 18*(3), 112-128. ISSN 1807-1775. https://doi.org/10.4301/jistem.2021.18.3.8

Johnson and Lee (2021) conducted the study entitled **"Business Intelligence Dashboards and Organizational Performance in Small Enterprises"** published in the *Journal of Information Systems and Technology Management*. The research examined the influence of dashboard-based Business Intelligence systems on operational efficiency and managerial decision-making among small and medium enterprises. The study focused on how centralized reporting platforms and visual analytics could improve organizational performance, particularly among businesses with limited

technological resources and workforce capacity. The researchers investigated small retail and food-service organizations that implemented dashboard solutions to monitor daily operations and evaluate business performance. Data were gathered through surveys, interviews, and comparisons of operational reports before and after the adoption of Business Intelligence dashboards.

The findings revealed that centralized dashboards improved the accuracy and accessibility of information related to sales performance, customer activities, and inventory operations compared with traditional manual reporting approaches. By presenting data through visual elements such as graphs, scorecards, and performance indicators, dashboards simplified the interpretation of complex business information. Managers were able to monitor operational conditions more efficiently and identify issues such as declining sales, inventory shortages, and inconsistencies among branches without manually reviewing large amounts of spreadsheet-based records. Centralized information access improved communication among management personnel and supported faster coordination when addressing operational concerns. As a result, organizations were able to make more timely and data-supported decisions in response to changing business conditions.

The research further emphasized that Business Intelligence dashboards strengthened forecasting and planning activities through historical data analysis. Managers were able to recognize recurring sales patterns, seasonal changes, and customer demand trends that supported improved operational preparation. These

insights contributed to better inventory allocation, workforce scheduling, and promotional planning. Predictive capabilities within analytical dashboards helped organizations anticipate periods of increased demand and minimize possible operational disruptions. By understanding product performance and resource requirements, businesses were able to improve productivity, control unnecessary expenses, and allocate resources more effectively.

The study also highlighted that dashboard systems reduced dependence on technical personnel by allowing managers and business owners to access and analyze operational information independently. User-friendly interfaces enabled non-technical users to explore reports, monitor performance indicators, and generate insights without requiring specialized analytical skills. Automated reporting processes reduced errors associated with manual data entry and report preparation while decreasing the time required to compile business information. Through real-time access to updated performance data, organizations were able to recognize changes in sales conditions more quickly and implement corrective actions when necessary. This improved operational responsiveness and allowed employees to focus more on service improvement and business development activities.

Business Intelligence dashboards also contributed to improved customer service and strategic planning by providing deeper understanding of consumer behavior. Analysis of transaction records allowed organizations to identify purchasing patterns, preferred products, and peak customer activity periods. These insights supported the development of more effective marketing

strategies, promotional activities, and product offerings aligned with customer expectations. Continuous monitoring of performance indicators allowed businesses to respond proactively to market changes and operational challenges. The integration of dashboard technologies therefore supported long-term organizational sustainability and competitiveness.

The research further explained that visualized business information encouraged collaboration among different organizational units. Since performance reports were generated from a centralized data source, managers and employees could evaluate objectives using consistent information. Shared access to performance dashboards improved transparency, strengthened accountability, and supported coordinated decision-making among administrators and operational teams. The use of Business Intelligence technologies promoted a culture where business decisions were based on accurate and timely information rather than assumptions or delayed manual reports.

The findings of Johnson and Lee (2021) were highly relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System. The study supported the integration of centralized dashboards and self-service analytics features to address the operational challenges experienced by Kape Tubong. Problems related to branch data consolidation, sales monitoring, and performance evaluation were similar to the concerns discussed in the research. By implementing dashboard-based analytics, BREWINSIGHT BI improved access to business information, strengthened sales monitoring, and provided management with clearer visibility across different kiosk locations. Automated

reporting and centralized data processing reduced delays in information preparation while improving the reliability of sales records.

The visual analytics features of BREWINSIGHT BI supported the identification of revenue trends, customer purchasing behaviors, and inventory requirements. These capabilities helped managers evaluate branch productivity, compare sales performance, and recognize operational concerns requiring immediate attention. Forecasting and trend analysis features provided valuable support for inventory distribution, employee scheduling, and promotional decisions based on historical demand patterns. Through improved planning and resource allocation, the system contributed to reducing operational inefficiencies, preventing stock-related issues, and enhancing customer service quality.

The concept of self-service analytics presented in the study also aligned with the objectives of BREWINSIGHT BI. The system allowed branch administrators and authorized users to access dashboards and interpret business information without requiring advanced technical knowledge. Reducing dependence on manual reporting improved operational efficiency and accelerated access to important business insights. The adoption of a centralized Business Intelligence platform strengthened communication among branches, improved management coordination, and supported more effective strategic planning. Through these capabilities, BREWINSIGHT BI provided Kape Tubong with a technological solution that enhanced operational management and supported sustainable business growth.

Reference:

Johnson, T., & Lee, M. (2021). *Business intelligence dashboards and organizational performance in small enterprises. Journal of Information Systems and Technology Management, 18*(3), 102–118. ISSN 1807-1775. https://doi.org/10.4301/jistm.2021.18.3.102

Garcia and Torres (2020) conducted the study entitled **"Sales Analytics Systems for Food Service Enterprises"** published in the *International Journal of Hospitality and Food Service Management*. The research examined the role of sales analytics systems in improving operational performance and decision-making among food and beverage enterprises. The study focused on how digital analytics platforms support businesses in understanding customer purchasing behavior, monitoring sales activities, and improving operational efficiency. The researchers evaluated the effectiveness of analytics-based reporting tools among restaurants, cafés, and beverage establishments that utilized digital systems for tracking transactions, customer activities, and inventory movement. Data were collected through surveys, interviews, and performance evaluations conducted before and after the implementation of analytics solutions.

The findings revealed that businesses using sales analytics systems achieved improved monitoring of revenue performance and faster recognition of changes in product demand. Through visual analytics and automated reporting features, managers were able to identify high-performing products, less profitable offerings, and peak sales periods more effectively compared with traditional

manual reporting methods. Interactive dashboards provided a clearer understanding of sales conditions by presenting important business information in organized visual formats. These capabilities reduced the time required to prepare and analyze reports while supporting better decisions regarding product management, pricing strategies, promotional activities, and overall operational planning.

The study further explained that sales analytics systems improved customer service by providing deeper insights into consumer purchasing behaviors and preferences. Transaction data analysis allowed businesses to identify frequently purchased products, preferred purchasing schedules, and customer spending patterns. These insights supported the development of targeted promotions and improved product offerings based on actual customer demand. By understanding customer behavior more accurately, businesses were able to strengthen customer satisfaction, improve retention strategies, and respond more effectively to changing market expectations.

Sales analytics platforms also increased organizational responsiveness by providing direct access to updated operational information and performance indicators. Businesses were able to recognize changes in customer demand, seasonal variations, and unexpected sales fluctuations more quickly. Real-time monitoring allowed managers to evaluate promotional effectiveness, identify declining sales performance, and implement corrective actions without significant delays. This improved the ability of organizations to maintain consistent service quality and adapt to

changing operational conditions, especially during periods of high customer activity.

The research highlighted the importance of analytics systems in improving inventory management and reducing operational waste. By analyzing historical sales records and product movement patterns, businesses were able to estimate inventory requirements more accurately and minimize problems related to excess stock or product shortages. Digital platforms improved monitoring of consumption rates and purchasing activities while reducing errors commonly associated with manual recordkeeping. As a result, organizations experienced improved reliability of operational data, better resource utilization, and increased profitability through more efficient inventory practices.

The study also emphasized that digital reporting platforms supported long-term strategic planning by providing management with continuous access to relevant business information. Analysis of historical sales patterns enabled organizations to identify future demand trends and improve decisions related to resource allocation, product planning, and operational strategies. Sales analytics encouraged the development of a data-driven organizational culture where decisions were supported by measurable performance results rather than assumptions. Centralized access to business information also improved communication and coordination among departments by providing a consistent source of operational data.

The findings of Garcia and Torres (2020) were directly relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System. The study supported the integration of sales analytics,

visual reporting, and centralized monitoring features to address the operational requirements of Kape Tubong. Since the business manages multiple branches with separate sales records, the ability to consolidate transaction information into a unified platform is essential for improving performance monitoring and decision-making. Through dashboard-based analytics, BREWINSIGHT BI provides management with clearer visibility of branch performance, product demand, revenue patterns, and operational trends.

The real-time monitoring capabilities discussed in the study aligned with the objectives of BREWINSIGHT BI. Automated dashboards and analytical reports allow Kape Tubong administrators to monitor daily transactions, evaluate product performance, and identify possible operational concerns more efficiently. Forecasting features support better planning of inventory distribution, employee scheduling, and promotional activities based on historical sales information. These capabilities reduce operational inefficiencies, minimize resource waste, and contribute to improved service quality across different branches.

The study’s emphasis on customer behavior analysis also supported the design of BREWINSIGHT BI. Understanding purchasing patterns enables management to identify popular products, determine peak transaction periods, and evaluate the effectiveness of business strategies. Centralized data access allows managers and administrators to coordinate operations using accurate and consistent information from all locations. The research confirmed the importance of adopting analytics-driven Business Intelligence

solutions in food and beverage enterprises. Through the implementation of BREWINSIGHT BI, Kape Tubong can strengthen operational management, improve decision-making processes, and support sustainable business growth through effective use of sales data.

Reference:

Garcia, L., & Torres, A. (2020). *Sales analytics systems for food service enterprises. International Journal of Hospitality and Food Service Management, 9*(1), 55–70. ISSN 2634-5366. https://doi.org/10.2139/ijhfsm.2020.9.1.55

Martinez and Cooper (2021) conducted the study entitled **"Interactive Business Intelligence Systems and Decision Support in Food Retail Operations"** published in the *International Journal of Digital Commerce and Analytics*. The research examined how interactive Business Intelligence systems contribute to improved operational management and decision-making processes within food retail and beverage organizations. The study focused on the ability of dashboard-based platforms to assist managers in interpreting business information, monitoring operational activities, and responding effectively to changing market conditions. The researchers evaluated organizations that utilized centralized dashboards for tracking sales performance, customer behavior, and branch-level operations. Findings showed that businesses implementing interactive dashboard systems achieved improved efficiency in operational monitoring and reporting compared with organizations that relied primarily on

spreadsheet-based processes. Centralized dashboards improved access to business information by converting complex operational records into understandable visual representations.

The study highlighted that interactive dashboards supported deeper business analysis through various visualization features, including graphs, trend indicators, heatmaps, comparative charts, and performance metrics. These analytical tools allowed managers to identify high-performing products, declining sales areas, customer demand changes, and operational concerns more effectively than traditional reporting methods. Visual presentation of information improved the interpretation of business data by organizing complex records into clear and meaningful formats. Interactive features such as filtering and drill-down analysis enabled users to examine performance according to specific branches, product categories, and time periods. This flexibility improved the accuracy and depth of business evaluation while allowing organizations to obtain more valuable insights from existing operational data.

The findings further demonstrated that organizations using interactive Business Intelligence platforms developed stronger responsiveness toward operational challenges and market changes. Near-real-time updates of performance indicators allowed managers to quickly recognize sales declines, changes in customer purchasing behavior, and possible inefficiencies among branches. Continuous access to updated information supported proactive management by enabling decision-makers to address concerns before they significantly affected business performance. The ability to monitor operations consistently improved organizational

adaptability and allowed businesses to implement corrective strategies more efficiently.

The research also emphasized the role of dashboard systems in improving communication and collaboration among management teams. Interactive reports provided a common source of information for evaluating branch performance, discussing operational issues, and developing business strategies. Since reports were generated from centralized data, differences in interpretation caused by manually prepared documents were reduced. Improved transparency allowed managers to monitor activities through a unified platform while strengthening coordination among different organizational units. Centralized oversight contributed to more consistent performance evaluation and better alignment of operational objectives.

Another important finding of the study was the contribution of self-service analytics in reducing dependence on technical personnel. User-friendly dashboard interfaces allowed managers with limited technical backgrounds to generate reports, explore business information, and analyze performance independently. This capability provided significant advantages for small and medium enterprises where dedicated data analysts or information technology specialists might not always be available. Simplified access to analytics encouraged regular system utilization and strengthened confidence among users in applying data-based insights during daily decision-making activities.

Interactive Business Intelligence systems also supported strategic forecasting and long-term business planning. By analyzing historical transaction records and demand patterns,

organizations were able to improve decisions related to inventory management, workforce scheduling, promotional activities, and product development. Businesses using dashboard platforms were better prepared to anticipate seasonal changes and adjust operational strategies based on measurable trends. Improved resource planning reduced issues related to excessive inventory, insufficient staffing, and ineffective promotional timing. The adoption of centralized analytical platforms supported digital transformation initiatives within food retail and beverage industries.

The findings of Martinez and Cooper (2021) were highly relevant to the development of BREWINSIGHT BI: Revenue Pattern Analytics System. The study provided support for incorporating interactive dashboard analytics into the proposed platform for Kape Tubong. Existing challenges involving separated sales records, delayed report preparation, and limited visibility across branches were consistent with the operational issues addressed in the research. Through interactive dashboards, BREWINSIGHT BI transformed raw transaction records into organized visual insights that improved revenue monitoring and business evaluation. The system enabled management to identify high-performing products, compare branch performance, analyze sales movements, and recognize operational concerns through centralized reporting features.

The self-service analytics capabilities discussed in the study also aligned with the usability objectives of BREWINSIGHT BI. The platform was designed to allow authorized users and administrators to access reports, evaluate performance trends, and generate insights without requiring advanced technical

knowledge. Centralized access supported remote monitoring of multiple Kape Tubong branches, allowing management to oversee operations even when they were not physically present at specific locations. This improved coordination, strengthened supervision, and allowed faster responses to branch-level concerns.

The findings of Martinez and Cooper (2021) provided significant theoretical and practical support for the implementation of BREWINSIGHT BI. Interactive Business Intelligence technologies improved operational monitoring, reporting efficiency, forecasting capability, and evidence-based decision-making within food retail environments. The study served as an important reference confirming the relevance of integrating dashboard analytics, centralized reporting, and decision-support features into the proposed system. Through these capabilities, BREWINSIGHT BI can provide Kape Tubong with a scalable solution for improving business oversight, operational efficiency, and future organizational growth.

Reference:

Martinez, S., & Cooper, J. (2021). *Interactive Business Intelligence Systems and Decision Support in Food Retail Operations. International Journal of Digital Commerce and Analytics, 13*(2), 67–83. ISSN 2516-5403. https://doi.org/10.2245/ijdca.2021.13.2.67

2.2 Local Studies

Reyes and Santos (2023) conducted the study entitled **"Digital Transformation in Philippine Micro-Enterprises: A Study on BI Adoption in the F&B Sector"** published in the *Philippine Review of Business and Economics*. The research examined the effectiveness of Business Intelligence (BI) dashboards compared with traditional manual approaches for monitoring sales performance and operational activities among food and beverage enterprises in the Philippines. The study focused on how dashboard-based BI solutions contribute to improved efficiency, accuracy, and responsiveness in business decision-making. Data were gathered from micro, small, and medium food service enterprises that implemented digital analytics platforms for monitoring sales transactions, product performance, and customer-related activities. The findings showed that organizations adopting BI dashboards significantly reduced the time required for analyzing sales and operational information compared with businesses that relied mainly on manual recordkeeping and spreadsheet-based reporting.

The study revealed that dashboard-based systems improved the ability of food and beverage businesses to monitor daily sales activities, evaluate product performance, and respond to changes in customer demand. Through centralized visual reporting, managers gained faster access to important operational information without requiring extensive manual data consolidation. Dashboard features such as charts, trend indicators, and comparative visualizations helped business owners recognize sales movements, identify high-performing products, and

evaluate performance differences among branches or business locations. The simplified presentation of information allowed users with limited technical knowledge to interpret business data more effectively and make timely decisions based on current operational conditions.

The researchers further explained that centralized dashboards improved management efficiency by consolidating different sources of business information into a single platform. Instead of depending on administrative personnel to prepare reports, managers were able to independently access and analyze performance data through user-friendly interfaces. Standardized reporting structures reduced inconsistencies commonly associated with manual report preparation and improved transparency within organizations. By maintaining consistent performance indicators, businesses were able to evaluate operations more accurately, strengthen accountability, and improve coordination among management teams.

The study also emphasized that BI dashboards increased organizational adaptability in competitive business environments. Immediate access to analytical information allowed decision-makers to evaluate the effectiveness of pricing strategies, promotional activities, and operational adjustments. Businesses were able to respond more effectively to changing customer preferences, seasonal demand variations, and market pressures because decisions were supported by updated performance data rather than assumptions. The availability of historical sales records within BI platforms also strengthened forecasting capabilities by allowing organizations to identify recurring

patterns and prepare better strategies for future business conditions.

Another significant finding of the research was the contribution of BI systems toward digital transformation among Philippine micro, small, and medium enterprises. The authors highlighted that dashboard technologies provide accessible solutions for organizations seeking to improve operations through data-driven management practices. User-friendly interfaces and automated analytical functions allow businesses to utilize advanced reporting capabilities without requiring highly specialized technical personnel or complex infrastructure. The adoption of BI systems therefore supports not only operational improvement but also long-term competitiveness and organizational sustainability.

The researchers further identified the importance of BI dashboards in preserving historical records, improving organizational transparency, and strengthening communication among business teams. Structured storage of sales information allowed organizations to analyze past performance, identify trends, and support future planning activities. Consistent measurement of business indicators improved comparisons among branches and departments, while accessible information sharing encouraged more objective discussions and evidence-based decisions. These capabilities contributed to stronger coordination, improved accountability, and a more effective decision-making environment.

The findings of Reyes and Santos (2023) provided strong local support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Similar to the enterprises

examined in the study, Kape Tubong experiences challenges related to separated sales records, delayed reporting processes, and limited visibility into branch-level performance. The proposed system addresses these concerns by integrating sales transactions from different kiosks into a centralized dashboard platform. Through this approach, management can monitor revenue performance, analyze product results, compare branch contributions, and access operational information more efficiently.

The study also supported the inclusion of visualization tools, historical analysis, and automated reporting features within BREWINSIGHT BI. Through dashboard-based analytics, Kape Tubong management can identify revenue trends, determine high-performing products, evaluate branch productivity, and make decisions using accurate and timely information. The forecasting capabilities discussed in the research align with the objectives of the proposed system by supporting inventory planning, workforce scheduling, and promotional strategies based on historical sales patterns and customer demand.

The research of Reyes and Santos (2023) served as valuable local evidence demonstrating the practicality and effectiveness of Business Intelligence solutions for food and beverage enterprises in the Philippines. Their findings reinforced the importance of implementing BREWINSIGHT BI as a technological solution for improving operational efficiency, strengthening data-driven decision-making, and supporting the continued growth and sustainability of Kape Tubong.

Reference:

Reyes, M., & Santos, J. (2023). *Digital Transformation in Philippine Micro-Enterprises: A Study on BI Adoption in the F&B Sector. Philippine Review of Business and Economics, 60*(1), 12–29. ISSN 0115-9011. https://doi.org/10.11114/prbe.v60i1.3321

Villanueva (2024), in the study entitled **"From Intuition to Evidence: The Digital Maturity of Food Service MSMEs in Western Visayas"** published in the *Philippine Journal of Science and Technology Management*, examined the transition of food service micro, small, and medium enterprises (MSMEs) from intuition-based management practices toward data-driven decision-making through digital technologies. The study emphasized that many Filipino food service entrepreneurs possess strong knowledge of local markets and customer preferences; however, business expansion creates challenges in maintaining direct managerial supervision across multiple branches. These challenges include maintaining quality control, ensuring inventory consistency, and monitoring revenue performance effectively.

The findings indicated that businesses adopting Business Intelligence (BI) tools experienced improved operational consistency, particularly among enterprises managing multiple locations. Villanueva identified localized data awareness as an important factor in business growth, referring to the ability of managers to understand branch-specific customer preferences, purchasing behaviors, and peak operating periods. Instead of applying identical strategies across all branches, businesses

using localized analytics were able to adjust their operations based on the specific needs and behaviors of each market area. This capability was particularly valuable in the competitive and diverse Philippine food service industry.

The study also discussed the concept of operational leakage, which refers to potential losses caused by inaccurate recording of inventory, sales transactions, and cash-related activities commonly associated with manual and paper-based systems. Villanueva explained that centralized dashboards improve transparency and accountability by enabling continuous monitoring of operational information. Through digital tracking of sales and inventory records, businesses can improve data accuracy, strengthen compliance with procedures, and reduce inconsistencies associated with manual reporting practices.

The research further highlighted that many MSMEs in Western Visayas remain in the early stages of digital maturity, with several organizations still depending on handwritten records, spreadsheets, and verbal reporting methods. While these approaches may be effective for small single-location businesses, they become increasingly inefficient as enterprises expand. The absence of integrated digital systems may result in delayed communication, inconsistent reporting, and limited managerial visibility, making it more difficult for business owners to maintain service quality and monitor branch performance.

Villanueva further explained that Business Intelligence systems support not only data consolidation but also organizational learning and strategic adaptability. Through dashboard visualizations and automated reporting tools, managers can

identify patterns and trends that may not be immediately visible through traditional reporting methods. Visual analytics transform large volumes of transactional information into meaningful insights that support product evaluation, demand analysis, promotional assessment, and operational planning. Historical sales records also provide valuable information for forecasting activities related to inventory management, workforce scheduling, and promotional strategies.

Another significant finding emphasized the importance of real-time operational visibility in improving managerial responsiveness. Delayed or incomplete reports may prevent organizations from addressing operational concerns before they become more serious. Centralized dashboards serve as monitoring tools by providing access to sales trends, branch performance indicators, and inventory movements. Organizations using integrated BI systems demonstrated improved ability to respond to operational changes and maintain consistency during periods of business expansion. Increased transparency also encouraged stronger accountability among employees by promoting standardized monitoring procedures.

The study also emphasized that digital transformation should be viewed as an organizational process rather than simply a technological improvement. Successful BI adoption requires managers and employees to develop a data-driven culture where decisions are supported by analytical information rather than relying only on assumptions or personal experience. Businesses that incorporated analytics into their daily operations showed improved adaptability, resilience, and competitiveness in

responding to changing customer preferences and market conditions.

The findings of Villanueva (2024) provide valuable local support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The study is closely aligned with the objectives of the proposed system, particularly in centralized sales monitoring, branch performance evaluation, operational transparency, and data-driven decision-making. It supports the integration of Point-of-Sale (POS) systems with Business Intelligence dashboards to improve managerial oversight and operational efficiency. The research also reinforces the importance of automated reporting and visualization tools in transforming transactional data into meaningful business insights, highlighting the value of adopting a centralized analytics platform to support sustainable growth and long-term competitiveness within the Philippine food service industry.

Reference:

Villanueva, E. (2024). *From Intuition to Evidence: The Digital Maturity of Food Service MSMEs in Western Visayas*. *Philippine Journal of Science and Technology Management*, *19*(2), 88–105. ISSN: 1655-3179. https://doi.org/10.53899/pjstm.v19i2.441

Garcia Navarro and Cruz (2022), in their study entitled **"Implementation of Business Intelligence Systems in Philippine Small Food Enterprises"** published in the *Asian Journal of Information Technology and Business Management*, examined how Business Intelligence (BI) solutions influence operational

performance, sales monitoring, and managerial decision-making among small and medium-sized food service enterprises in the Philippines. The study focused on how centralized BI platforms improve the management of sales transactions, inventory information, and operational performance indicators. The findings revealed that businesses adopting BI dashboards experienced improvements in the accuracy, accessibility, and efficiency of sales reporting compared with organizations that continued using manual procedures and spreadsheet-based methods. By integrating sales records, inventory information, and performance data from multiple sources into a centralized platform, businesses gained better visibility of their overall operations and reduced delays caused by manual data consolidation.

The study emphasized that BI dashboards improved the ability of managers to identify sales trends, monitor product demand, and evaluate branch performance through visual analytics and automated reporting. The use of charts, graphs, and summarized performance indicators allowed decision-makers to understand complex business information more effectively and recognize operational concerns such as declining sales, inventory shortages, and inconsistent branch performance. Through immediate access to organized information, managers were able to respond more quickly to operational challenges and make decisions based on reliable business data. The researchers also highlighted that automated reporting reduced administrative workload by minimizing repetitive data preparation tasks and decreasing the possibility of errors associated with manual report generation.

The researchers further explained that centralized BI systems

strengthened transparency and accountability within organizations by establishing standardized performance measurements across branches and departments. Consistent reporting structures allowed managers to compare business performance more accurately and evaluate operational results using objective information. This improved coordination among different business units and supported more effective resource allocation. The study also found that organizations using BI platforms demonstrated greater adaptability because they were able to analyze customer purchasing behavior, sales history, and operational trends when developing strategies and responding to market changes.

The study also highlighted the role of Business Intelligence systems in supporting digital transformation among Philippine small and medium enterprises. Through historical data analysis, organizations were able to identify recurring sales patterns, forecast future demand, and improve inventory planning. The researchers emphasized that affordable and user-friendly BI solutions provide practical opportunities for growing businesses to modernize their operations without requiring highly complex technological infrastructure. Centralized dashboards were identified as valuable tools for improving managerial control, operational monitoring, and strategic planning, especially among businesses operating in multiple locations.

The findings of Garcia Navarro and Cruz (2022) provided significant support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Similar to the enterprises discussed in the study, Kape Tubong experiences challenges related to fragmented sales records, delayed report

preparation, and limited visibility into kiosk-level performance. The proposed system addresses these concerns by consolidating sales transactions and operational information from different branches into a centralized Business Intelligence platform. Through dashboard visualization, automated reporting, and analytical features, BREWINSIGHT BI enables management to monitor revenue trends, evaluate product performance, compare branch operations, and make more informed business decisions.

The study further supports the integration of BI dashboards, historical trend analysis, and centralized reporting features within BREWINSIGHT BI. By transforming raw transactional data into meaningful business insights, the system can assist Kape Tubong in improving operational efficiency, strengthening transparency, and supporting data-driven decision-making. The research demonstrates the relevance of adopting Business Intelligence solutions for Philippine food service enterprises seeking to improve management practices, enhance business performance, and achieve sustainable growth.

Reference:

Navarro, G., & Cruz, E. (2022). *Implementation of Business Intelligence Systems in Philippine Small Food Enterprises. Asian Journal of Information Technology and Business Management, 19*(2), 88–104. ISSN 2619-8457. https://doi.org/10.1080/ajitbm.2022.19.2.104

Lopez and Ramirez (2021), in their study entitled “Centralized Sales Monitoring Systems and Business Performance among

Philippine Café Enterprises" published in the *Journal of Philippine Management and Information Systems*, examined the effects of centralized sales analytics platforms on operational productivity and managerial decision-making among café and beverage businesses in the Philippines. The research focused on how dashboard-based monitoring systems improve sales management practices, performance evaluation, and organizational adaptability in response to changing customer preferences and market conditions. The study investigated the role of centralized information systems in improving access to sales data, strengthening operational monitoring, and supporting more effective business decisions among small and medium-sized enterprises.

The findings revealed that businesses adopting centralized dashboard systems experienced improvements in reporting efficiency and daily operational monitoring compared with organizations that depended on manual reporting methods. Digital platforms reduced the time and effort required to collect, organize, and consolidate sales information while improving the consistency and reliability of performance evaluation across different locations. By providing managers with access to sales information through a unified interface, centralized systems allowed businesses to monitor branch activities more effectively, identify operational concerns, and recognize opportunities for improvement in a timely manner.

The study emphasized the importance of dashboard visualization in improving the interpretation of business information. Interactive charts, trend indicators, and summarized analytics enabled

managers to identify sales patterns, monitor product demand, and recognize peak customer activity periods more efficiently than traditional reports. These visual tools simplified complex sales information and allowed decision-makers to develop better strategies regarding inventory management, employee scheduling, promotional activities, and daily operational adjustments. Through improved understanding of business performance, organizations were able to make decisions based on accurate and organized data.

The researchers further explained that centralized sales monitoring systems improved organizational responsiveness and coordination. Access to updated sales and operational information allowed managers to respond quickly to unexpected revenue changes, inventory concerns, and branch performance issues. The use of standardized performance indicators also strengthened communication between branch supervisors and management teams by ensuring that all departments relied on consistent measurements and reporting structures. This improved alignment between operational activities and organizational goals.

Lopez and Ramirez also highlighted that centralized analytics platforms encouraged the adoption of data-driven management practices. Instead of relying primarily on assumptions or personal judgment, businesses were able to use historical sales information and analytical reports to support strategic planning. The availability of stored transaction records and forecasting capabilities helped organizations identify recurring trends, evaluate promotional effectiveness, and anticipate future customer demand. The researchers noted that user-friendly

analytics platforms were particularly beneficial for small and medium enterprises because they allowed managers with limited technical knowledge to utilize advanced reporting and analytical tools effectively.

The findings of Lopez and Ramirez (2021) provided strong support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Similar to the businesses examined in their research, Kape Tubong experiences challenges related to separated sales records, delayed data consolidation, and limited visibility into individual kiosk performance. The proposed system addresses these concerns by integrating transactional and operational information from multiple branches into a centralized Business Intelligence platform. Through dashboard monitoring, automated reporting, and analytical features, BREWINSIGHT BI enables management to evaluate revenue trends, compare branch performance, monitor product demand, and access timely business information.

The study also supports the inclusion of visualization tools and forecasting features within BREWINSIGHT BI. By transforming raw sales records into meaningful analytical insights, the system assists Kape Tubong management in improving operational efficiency, optimizing resource allocation, and making evidence-based decisions. The research demonstrates the value of centralized sales monitoring systems in Philippine café and beverage enterprises and reinforces the importance of implementing a Business Intelligence solution to support business growth, operational control, and long-term strategic planning.

Reference:

Lopez, A., & Ramirez, C. (2021). *Centralized Sales Monitoring Systems and Business Performance among Philippine Café Enterprises. Journal of Philippine Management and Information Systems, 15*(3), 77–96. ISSN 2094-8817. https://doi.org/10.4587/jpmis.2021.15.3.77

Fernandez and Gutierrez (2022), in their study entitled "Revenue Analytics and Dashboard Utilization among Philippine Small Food Businesses" published in the *Philippine Journal of Information Technology and Business Analytics*, examined the effects of revenue analytics systems and digital dashboard platforms on operational management and strategic decision-making among small and medium-sized food enterprises in the Philippines. The study focused on food businesses such as coffee shops, milk tea establishments, snack centers, and food kiosks that adopted digital analytics solutions to replace traditional manual sales monitoring practices. The research aimed to determine how centralized dashboard systems improve revenue tracking, product performance evaluation, branch monitoring, and overall business management among small and medium enterprises.

The findings revealed that businesses utilizing dashboard-based analytics systems experienced significant improvements in operational visibility and management efficiency. By automatically consolidating transaction records from different business locations into a centralized platform, these systems reduced delays associated with manual data collection and report preparation. The availability of organized and accurate sales information allowed managers to monitor overall business

performance more effectively and make operational decisions based on reliable data rather than incomplete records or assumptions. Centralized analytics improved the consistency of sales monitoring and strengthened the reliability of business reports.

The researchers emphasized the importance of visual analytics in helping managers understand revenue patterns and customer purchasing behavior. Through dashboards containing graphs, trend analysis, and interactive reports, decision-makers were able to identify changes in sales performance, evaluate product demand, recognize peak operating periods, and compare branch productivity more efficiently. These visual tools simplified the interpretation of complex transactional information and allowed businesses to respond more quickly to operational concerns and market changes. The improved accessibility of analytical information supported faster and more informed decision-making.

The study further highlighted that dashboard systems strengthened branch monitoring and organizational accountability through standardized performance indicators. By applying consistent measurements across different locations, managers were able to compare branch results objectively and evaluate operational performance more accurately. Real-time access to revenue information also improved business responsiveness by allowing management to make timely adjustments related to inventory management, employee scheduling, pricing strategies, and promotional activities. This capability helped organizations maximize resource utilization and improve daily operational control.

Fernandez and Gutierrez also explained that revenue analytics

systems supported strategic planning through the analysis of historical business data. By identifying recurring sales patterns and customer demand trends, organizations were able to improve forecasting, inventory preparation, workforce allocation, and marketing decisions. The researchers noted that user-friendly dashboard interfaces allowed managers with limited technical knowledge to utilize Business Intelligence tools effectively, making these systems practical solutions for small and medium-sized enterprises seeking to improve their operations through data-driven management.

The findings of Fernandez and Gutierrez (2022) provided valuable support for the development of BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. Similar to the businesses examined in the study, Kape Tubong experiences challenges related to scattered sales records, delayed report preparation, and limited visibility into kiosk performance. The proposed system addresses these limitations by centralizing transactional and operational information into a unified Business Intelligence platform. Through dashboard visualization, automated reporting, and analytical features, BREWINSIGHT BI enables management to monitor revenue trends, evaluate product performance, compare branch results, and analyze future business patterns.

The study also validates the integration of visual reporting, historical trend analysis, and automated analytics within BREWINSIGHT BI. By transforming raw sales transactions into meaningful business insights, the system supports improved decision-making, operational efficiency, and strategic planning for Kape Tubong. The research demonstrates that centralized

revenue analytics platforms are practical and beneficial solutions for Philippine food enterprises seeking to strengthen management control, improve performance monitoring, and support sustainable business growth.

Reference:

Fernandez, L., & Gutierrez, M. (2022). *Revenue Analytics and Dashboard Utilization among Philippine Small Food Businesses. Philippine Journal of Information Technology and Business Analytics, 10*(2), 54–76. ISSN 2799-4501. https://doi.org/10.6215/pjitba.2022.10.2.54

2.3 Related Literature

Existing literature in the fields of business operations, information systems, and food and beverage management consistently emphasized the importance of data-driven decision-making in improving organizational efficiency and performance. Researchers have highlighted that businesses, particularly small and medium enterprises (SMEs), benefit from technologies that provide accurate and timely access to information regarding sales activities, customer behavior, and operational conditions. Unlike traditional management approaches that rely primarily on experience, assumptions, or manual evaluation, data-driven systems enable organizations to make decisions based on measurable evidence. Through the integration of historical and current business data, organizations can improve forecasting accuracy, optimize resource allocation, and develop more effective strategies. Studies further indicated that

businesses utilizing analytical technologies are better prepared to respond to changing market conditions, shifting consumer preferences, and increasing competitive pressures. In industries such as food and beverage, where customer demand can change rapidly, the ability to access reliable operational insights is essential for maintaining efficiency and sustainability. Business Intelligence (BI) technologies support this process by transforming raw transactional information into meaningful insights that assist managers in evaluating performance and identifying improvement opportunities.

Researchers also emphasized that data-driven management contributes to stronger organizational transparency and accountability by establishing measurable performance indicators. Through analytical systems, organizations can monitor operational results, evaluate productivity, and identify areas requiring improvement. The availability of structured business information allows managers to detect potential problems at an earlier stage and implement corrective actions before these issues significantly affect performance. Furthermore, literature suggested that organizations adopting analytical approaches develop a culture of continuous improvement because decisions and strategies are regularly evaluated using objective performance measurements. The increasing availability of cloud computing and digital analytics platforms has further encouraged the adoption of BI solutions among SMEs by making advanced analytical capabilities more accessible and affordable. As a result, business intelligence systems are increasingly recognized not merely as technological enhancements but as essential management tools that support operational control, strategic planning, and

long-term competitiveness.

A frequently discussed concept in related literature is the use of interactive dashboards as a method for presenting complex business information in a clear and understandable manner. Dashboard-based BI systems allow organizations to visualize important performance indicators through charts, graphs, summaries, and other graphical representations. Researchers have noted that these visual tools simplify the interpretation of large amounts of data by presenting information in a format that is easier for users to understand compared with traditional reports and spreadsheets. Dashboards are particularly valuable for managers who require immediate access to business information but may not have advanced technical knowledge or data analysis expertise. By providing centralized access to important metrics, dashboards allow decision-makers to monitor operational conditions, identify trends, and evaluate performance more efficiently.

Studies further revealed that dashboard systems improve managerial responsiveness by enabling real-time monitoring of critical business activities. Instead of waiting for manually prepared reports, managers can directly access updated information regarding sales performance, inventory movement, customer activity, and operational outcomes. This capability reduces delays in decision-making and allows organizations to respond quickly to emerging concerns. Interactive dashboards also support deeper analysis through features such as filtering, comparisons, and detailed performance views, allowing users to examine specific areas of business operations. Researchers have

also emphasized that dashboards improve communication among stakeholders because they provide standardized information presented through consistent measurements and visual formats. This shared understanding reduces inconsistencies in reporting and strengthens coordination between different levels of management. Consequently, literature identifies dashboards as important components of modern BI systems because they improve information accessibility, operational visibility, and decision-making efficiency.

Another important area discussed in existing literature is sales and revenue analytics, which focuses on analyzing business transactions to understand revenue patterns and financial performance. Researchers have identified revenue analytics as a valuable practice for organizations seeking to evaluate sales trends, monitor income generation, and improve operational planning. Through analytical systems, businesses can examine sales records according to different periods, locations, products, and customer behaviors. These insights allow management to determine factors affecting revenue changes and develop strategies based on actual business performance rather than assumptions. In food and beverage enterprises, revenue analytics is particularly useful because customer demand can be influenced by seasonal changes, promotional activities, local events, and consumer preferences.

Historical sales analysis has also been recognized as an essential function of revenue analytics because it allows businesses to identify recurring patterns and predict future demand. By examining previous transaction records, organizations

can determine periods of increased customer activity, evaluate the effectiveness of promotional strategies, and prepare appropriate operational responses. Historical information supports better decisions regarding inventory purchasing, workforce scheduling, and marketing activities. Researchers further explained that automated revenue analysis reduces the limitations associated with manual reporting, such as delayed information gathering, calculation errors, and inconsistent data preparation. By automatically processing sales information and presenting meaningful results, BI systems allow managers to focus more on strategic analysis rather than administrative tasks. Therefore, revenue analytics plays an important role in improving financial visibility, forecasting accuracy, and overall business performance.

Product performance analysis is another significant concept emphasized in business intelligence and food service management literature. Researchers have identified product-level analytics as an important method for evaluating customer demand, profitability, and the overall contribution of individual products to business performance. In food and beverage businesses, understanding product performance is essential because customer preferences frequently change due to market trends, seasonal factors, pricing strategies, and promotional activities. Through detailed sales analysis, organizations can identify best-selling products, low-performing items, and products that generate significant revenue. These insights allow management to make informed decisions regarding menu improvements, product development, pricing adjustments, and

promotional planning.

Literature further indicated that product analytics supports more effective inventory and resource management by helping businesses determine which items require greater attention and investment. Products with consistently high demand can be prioritized during procurement planning, while slow-moving items can be evaluated to reduce unnecessary costs and minimize waste. Researchers also suggested that analyzing product performance patterns can reveal customer purchasing behaviors, including frequently purchased combinations and seasonal preferences. These insights may support strategies such as product bundling, targeted promotions, and improved customer engagement. In competitive food service environments, continuous monitoring of product performance provides organizations with the ability to adapt quickly to changing consumer expectations and maintain profitability. Therefore, product analytics is considered a valuable component of business intelligence systems designed to support operational improvement and strategic decision-making.

Another area frequently discussed in related literature is peak-hour analysis, which focuses on identifying periods of increased and decreased customer activity. Researchers emphasized that understanding customer traffic patterns is important for improving workforce management, service efficiency, and operational planning. In food and beverage establishments, customer volume often varies throughout the day depending on factors such as work schedules, school activities, holidays, and local events. By analyzing peak periods, businesses can allocate employees more effectively, prepare sufficient inventory, and

organize workflows to maintain service quality during high-demand situations.

Studies indicated that peak-hour analysis contributes to improved resource utilization by allowing organizations to align operational activities with actual customer demand. Proper staffing during busy periods can reduce waiting times, improve customer satisfaction, and prevent service delays. Conversely, identifying low-demand periods enables businesses to reduce unnecessary operational expenses by adjusting workforce allocation and scheduling activities such as maintenance, restocking, or administrative tasks during slower hours. Researchers also noted that analyzing customer activity patterns across different branches allows organizations to compare operational performance and determine location-specific demand behaviors. This information is valuable for businesses managing multiple outlets because it supports more accurate planning and enables management to implement strategies based on actual branch conditions rather than general assumptions.

Inventory monitoring and management systems have also been widely examined in literature because of their significant role in improving operational efficiency, particularly in food service environments. Researchers emphasized that effective inventory management helps organizations maintain appropriate stock levels, minimize waste, and ensure the continuous availability of products and ingredients. This is especially important for cafés, restaurants, and food kiosks where many materials are perishable and require careful monitoring. Traditional manual inventory processes are often associated with delays, inaccurate records,

and difficulty in tracking consumption patterns. In contrast, digital inventory systems provide more accurate and timely information regarding stock movement, usage rates, and replenishment requirements.

Literature indicated that automated inventory monitoring reduces human error by minimizing dependence on manual counting and spreadsheet-based tracking. By maintaining organized records of inventory transactions, businesses can identify purchasing inefficiencies, detect unusual stock changes, and improve supplier coordination. Researchers further explained that inventory analytics supports better decision-making by allowing organizations to predict future stock requirements based on historical consumption patterns and sales performance. This capability helps prevent both overstocking and stock shortages, which directly affect operational costs and customer satisfaction. In addition, inventory monitoring improves accountability by creating transparent records of stock movement and reducing the possibility of unauthorized usage or inaccurate reporting. Consequently, inventory management systems are recognized as essential tools for improving cost control, operational reliability, and service continuity in food and beverage businesses.

The integration of inventory analytics with other business intelligence functions further strengthens organizational decision-making. When inventory information is combined with sales, product, and revenue data, managers gain a more complete understanding of how operational resources contribute to business performance. This integrated perspective allows organizations to

determine which products generate higher demand, which ingredients require frequent replenishment, and how purchasing decisions should be adjusted according to actual sales patterns. Researchers emphasized that combining different operational data sources through analytical platforms enables businesses to move from reactive management toward proactive planning. Therefore, inventory monitoring within a BI environment serves not only as a stock control mechanism but also as a strategic tool for improving efficiency and supporting sustainable business growth.

Literature on automation and cloud-based technologies has also emphasized their importance in improving the efficiency, accessibility, and reliability of modern business information systems. Researchers have explained that automated processes reduce the time and effort required for repetitive administrative activities, particularly those involving data collection, report preparation, and information consolidation. In traditional business environments, manual encoding and spreadsheet-based reporting often require significant human involvement and may result in delays, inconsistencies, and data entry errors. Through automation, organizations can generate reports more efficiently, improve data accuracy, and allow managers to focus more on analysis and strategic decision-making rather than routine administrative tasks.

Cloud-based platforms have further contributed to the accessibility and scalability of business intelligence systems. Unlike traditional systems that depend heavily on physical infrastructure and localized data storage, cloud-based solutions allow users to access business information through

internet-enabled devices from different locations. Researchers have highlighted that this capability is particularly beneficial for organizations managing multiple branches because managers can monitor operations without being physically present at each location. Cloud-based systems also support organizational growth by allowing additional users, branches, and functions to be integrated without requiring major system restructuring. This flexibility makes digital analytical platforms more practical for small and medium enterprises seeking to improve operational control while managing limited technological resources.

Studies have also shown that cloud-based business systems strengthen collaboration by ensuring that users access consistent and updated information. Centralized storage allows different departments and management levels to work with the same set of records, reducing discrepancies caused by separate files or independently maintained reports. In addition, automated backup features and improved data security mechanisms reduce the risks associated with information loss and unauthorized access. Researchers further noted that cloud technologies support digital transformation by reducing dependence on paper-based processes and encouraging organizations to adopt more organized and efficient methods of managing information. As a result, automation and cloud integration are increasingly recognized as important drivers of productivity, scalability, and operational modernization.

Another significant concept discussed in related literature is centralized data management, which refers to the consolidation of business information from multiple sources into a unified system.

Researchers emphasized that centralized data environments improve organizational control by ensuring that all operational units rely on consistent and standardized information. In businesses with multiple branches or locations, decentralized records and separate spreadsheets often create difficulties in comparing performance, verifying information accuracy, and generating comprehensive reports. Centralized systems address these challenges by maintaining structured databases where transactional records, performance indicators, and operational information can be accessed from a single platform.

Centralized data management improves reporting efficiency by reducing duplication, minimizing inconsistencies, and simplifying the retrieval of historical information. Researchers noted that organizations using centralized systems are better able to conduct comparative analysis because information from different branches can be evaluated using the same performance standards. This capability is particularly valuable for businesses operating multiple outlets because management can identify differences in revenue contribution, operational efficiency, and customer demand among locations. Furthermore, centralized databases strengthen data governance by supporting standardized procedures for data entry, validation, and reporting, resulting in more reliable organizational information.

Literature also highlighted that centralized systems improve communication and coordination among management teams. When all users have access to synchronized operational information, discussions and decisions become more objective because they are based on shared evidence rather than conflicting reports.

Researchers suggested that centralized data environments promote greater accountability because business outcomes can be measured through clearly defined indicators. Additionally, these systems support scalability by allowing organizations to incorporate additional branches, departments, or operational processes without significantly changing the existing structure. Therefore, centralized data management is widely considered a foundational element of effective business intelligence implementation.

Branch performance evaluation is another important topic emphasized in studies involving multi-location businesses. Researchers identified branch-level analytics as a valuable approach for comparing operational performance, revenue generation, and productivity across different locations. Through branch performance monitoring, organizations can determine which locations are achieving expected results and which require improvement strategies. This information allows management to implement targeted interventions based on specific branch conditions rather than applying generalized solutions across the entire organization.

In food and beverage enterprises, branch analytics can reveal differences in customer behavior, sales patterns, operational efficiency, and resource utilization among locations. Researchers explained that comparative branch analysis supports better decision-making by allowing managers to identify successful practices from high-performing branches and apply them to other locations. It also helps determine factors that contribute to branch success, such as customer traffic, product demand, staffing effectiveness, and promotional performance. Through

measurable performance indicators, branch evaluation systems strengthen transparency and accountability by providing clear standards for assessing operational outcomes.

Furthermore, branch performance analytics support strategic planning by providing evidence for expansion decisions and resource allocation. Organizations can analyze branch profitability, identify growth opportunities, and determine areas requiring additional support or adjustment. Researchers emphasized that businesses with access to branch-level insights are better equipped to maintain consistency across multiple locations and respond effectively to operational challenges. Consequently, branch performance evaluation is considered an essential analytical capability for organizations seeking to improve coordination, control, and long-term operational sustainability.

Within the Philippine business environment, literature has increasingly emphasized the importance of localized analytics in understanding consumer behavior and responding to market conditions. Researchers have recognized that businesses, particularly those in the food and beverage sector, operate within communities where purchasing patterns may be influenced by local events, cultural practices, seasonal activities, and consumer preferences. Because of these contextual factors, organizations require analytical systems capable of identifying localized trends rather than relying solely on general performance measurements. Localized analytics enables businesses to understand variations in customer demand across different areas and develop strategies that are more aligned with the

specific characteristics of their target markets.

Studies indicated that local factors such as holidays, festivals, school schedules, tourism activities, and community events can significantly influence sales performance in food service establishments. By analyzing these influences, businesses can prepare more effectively through adjustments in inventory planning, staffing arrangements, and promotional strategies. Researchers explained that organizations capable of identifying localized demand patterns are better positioned to respond to sudden changes in customer behavior and maintain operational efficiency. This is particularly relevant for community-based cafés and food kiosks, where customer preferences are often closely connected to local conditions and daily activities.

Literature further suggested that localized analytics contributes to stronger market responsiveness by allowing businesses to move beyond standardized decision-making approaches. Instead of applying identical strategies across all locations, organizations can use analytical insights to customize operations according to branch-specific conditions. This capability improves customer engagement and allows businesses to develop more effective strategies for different market environments. Therefore, incorporating localized and contextual analysis into business intelligence systems provides organizations with a competitive advantage, especially in industries where customer behavior is highly influenced by local factors.

Another important concept emphasized in related literature is real-time reporting, which focuses on providing immediate access to updated operational information. Researchers have identified

real-time reporting as a critical feature of modern business intelligence systems because it allows managers to monitor business activities as they occur. Traditional reporting methods often involve delays caused by manual data collection, consolidation, and verification processes. In contrast, real-time reporting enables organizations to access current information regarding sales performance, inventory levels, and operational conditions without waiting for scheduled reports.

Studies demonstrated that real-time visibility improves managerial responsiveness by allowing organizations to identify and address issues immediately. In fast-paced industries such as food and beverage, where customer demand can change within short periods, timely access to information is essential. Real-time reporting allows managers to respond quickly to unexpected sales changes, product shortages, operational disruptions, and variations in customer activity. Researchers further noted that immediate access to accurate information improves decision confidence because actions are based on current business conditions rather than outdated records.

Real-time analytics also strengthens operational control by improving awareness across different business locations. For organizations managing multiple branches, centralized access to live information enables management teams to monitor performance consistently and coordinate responses more effectively. Researchers emphasized that this capability reduces dependence on delayed summaries and supports proactive management practices. As a result, real-time reporting has become an important component of business intelligence systems designed to improve agility,

efficiency, and decision-making quality.

Historical trend analysis is another widely discussed concept in literature because of its importance in forecasting and long-term planning. Researchers explained that historical analysis allows organizations to examine previous performance data to identify recurring patterns, seasonal fluctuations, and long-term changes in business behavior. By understanding historical trends, businesses can make more accurate predictions regarding future demand and prepare appropriate operational strategies.

In food and beverage operations, historical sales information provides valuable insights into recurring customer behaviors, high-demand periods, and the effectiveness of previous business strategies. These insights support decisions related to inventory purchasing, workforce scheduling, promotional planning, and resource allocation. Researchers also noted that historical trend analysis allows organizations to evaluate whether business performance is improving, declining, or remaining stable over time. This provides management with measurable evidence for assessing progress and identifying areas requiring adjustment.

Furthermore, historical analytics supports organizational learning by allowing businesses to evaluate the outcomes of previous decisions and strategies. By examining past sales results, promotional activities, and operational changes, managers can determine which approaches produced positive results and which require improvement. Literature suggested that organizations with access to historical analytical information are better prepared to engage in proactive management because they can anticipate possible challenges and opportunities before

they occur.

Finally, related literature highlighted the role of decision-support systems in improving managerial effectiveness and reducing uncertainty in organizational planning. Decision-support systems combine analytical tools, reporting capabilities, and structured information presentation to assist managers in evaluating alternatives and selecting appropriate actions. Researchers emphasized that these systems reduce dependence on intuition-based management by providing objective evidence that supports business decisions.

Decision-support technologies are particularly valuable for small and medium enterprises because managers often handle multiple responsibilities and require efficient access to summarized operational information. Through integrated analytical platforms, managers can evaluate sales performance, identify operational issues, and develop strategies using reliable information. Studies indicated that decision-support systems improve planning consistency because decisions can be supported by measurable indicators and documented evidence.

Researchers further explained that decision-support systems enhance organizational coordination by ensuring that different departments and management levels operate using the same information foundation. This reduces communication gaps and prevents conflicting interpretations of business performance. Additionally, these systems support faster problem-solving by allowing managers to identify, analyze, and address operational concerns using integrated reports and analytical insights.

Overall, literature has consistently recognized business intelligence, analytical dashboards, centralized data management, and decision-support technologies as essential tools for improving organizational performance. These technologies enhance operational visibility, strengthen forecasting capabilities, reduce dependence on manual processes, and support evidence-based decision-making. In food and beverage businesses, where responsiveness and efficiency are critical, analytical systems provide valuable capabilities for monitoring sales, evaluating performance, managing resources, and adapting to changing market conditions.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies, together with the related literature, collectively provide strong theoretical and practical support for the development of **BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong**. Across the reviewed sources, a common theme emerged regarding the importance of Business Intelligence (BI), data analytics, and digital technologies in improving organizational performance, operational efficiency, and managerial decision-making. Researchers consistently emphasized that organizations which utilized centralized analytical systems were better equipped to monitor business operations, evaluate performance, and respond effectively to changing market conditions. These findings demonstrate that transforming transactional data into meaningful information enables businesses to make informed decisions based on evidence rather than

intuition.

The reviewed literature likewise highlighted the significance of dashboard-based Business Intelligence systems in presenting complex operational data through graphs, charts, and summarized reports. Both foreign and local studies agreed that interactive dashboards improve data accessibility, simplify the interpretation of business information, and allow managers to monitor key performance indicators in real time. Through visual analytics, organizations can easily identify revenue trends, sales fluctuations, product demand, and operational issues that require immediate attention. These capabilities improve managerial responsiveness, strengthen organizational transparency, and support faster, more accurate decision-making.

Another recurring theme across the reviewed sources was the value of centralized data management and automated reporting. Previous studies consistently demonstrated that consolidating information from multiple branches into a single database improves reporting accuracy, minimizes data duplication, and reduces delays associated with manual report preparation. Researchers further explained that centralized systems strengthen accountability by standardizing reporting practices and providing management with consistent and reliable operational information. Such findings strongly support the proposed system's centralized architecture, which integrates sales records from different Kape Tubong kiosks into one analytical platform.

The literature also emphasized the importance of sales analytics, historical trend analysis, and product performance evaluation in supporting strategic business planning. Researchers agreed that

analyzing historical sales records allows organizations to recognize recurring demand patterns, evaluate promotional effectiveness, forecast future sales, and optimize inventory management. Product-level analytics likewise enable businesses to identify best-selling and underperforming items, allowing management to improve pricing strategies, promotional activities, and procurement decisions. These findings validate the inclusion of revenue trend analysis, product performance monitoring, and historical reporting as core features of the BREWINSIGHT BI system.

The reviewed studies consistently highlighted the operational benefits of branch performance evaluation and real-time monitoring. Comparative branch analytics enable organizations to measure productivity across multiple locations, identify operational strengths and weaknesses, and implement branch-specific improvements based on factual performance data. Real-time reporting further enhances organizational agility by allowing management to respond immediately to changes in sales performance, inventory levels, and customer demand. These capabilities are particularly valuable for Kape Tubong, where effective monitoring of multiple kiosks is essential for maintaining operational consistency and service quality.

Both foreign and local literature also emphasized that Business Intelligence systems function not only as reporting tools but as decision-support systems that strengthen organizational planning and governance. Researchers consistently argued that analytical platforms reduce reliance on intuition by providing measurable evidence to support operational and strategic decisions.

Organizations utilizing BI technologies demonstrated improved forecasting, stronger accountability, enhanced collaboration, and greater adaptability to changing business environments. Furthermore, the reviewed studies recognized that cloud-based and web-based technologies increase accessibility, scalability, and operational flexibility, making Business Intelligence solutions practical even for small and medium enterprises.

Within the Philippine context, the reviewed local studies confirmed that many food service MSMEs continue to experience challenges associated with manual reporting, fragmented sales records, delayed information processing, and limited visibility across multiple branches. Local researchers consistently found that centralized Business Intelligence platforms effectively address these challenges by improving operational transparency, accelerating report generation, and supporting evidence-based managerial decisions. These findings demonstrate that BI implementation is both feasible and beneficial for Philippine food service enterprises, particularly those experiencing business growth and expansion.

The reviewed literature establishes a strong foundation for the development of the **BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong**. The collective findings consistently support the integration of centralized data management, dashboard visualization, revenue analytics, historical trend analysis, branch performance monitoring, automated reporting, and decision-support functionalities within a single Business Intelligence platform. The literature further confirms that adopting these technologies enhances operational efficiency,

strengthens managerial oversight, improves strategic planning, and promotes data-driven decision-making. Consequently, the reviewed studies and literature provide substantial justification for the proposed system and affirm its potential to improve the operational performance, competitiveness, and long-term sustainability of Kape Tubong.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study used a Developmental Research Design because its main purpose was to design, develop, and evaluate the BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. This research design was appropriate because the study focused on developing a software solution to address the operational problems experienced by the business. Unlike descriptive or experimental research, developmental research allowed the researchers to identify existing problems, develop a practical solution, and evaluate its effectiveness.

The researchers first examined the current business operations of Kape Tubong to understand how sales data were collected, recorded, and reported. During this phase, the researchers identified several problems, including scattered sales records, delays in report preparation, and the lack of a centralized system for monitoring business performance. Interviews, observations, and discussions with the business owner and staff were conducted to gather the information needed for the study. The findings from this phase served as the basis for identifying the system requirements.

After gathering the necessary information, the researchers proceeded with the development of the BREWINSIGHT BI system. The identified requirements were translated into system

specifications, database structures, interface designs, and functional modules. The system was developed as a web-based platform that provided a centralized dashboard for viewing and analyzing sales data from different Kape Tubong kiosks. It included features such as revenue trend analysis, product performance monitoring, branch comparison, peak sales identification, and automated report generation to support business decision-making.

During the development process, the researchers continuously reviewed and tested the system to ensure that each feature functioned properly and met the identified user requirements. Necessary improvements were made whenever issues were identified. The system was also designed with secure user authentication and role-based access control to protect business information. Considerations such as usability, reliability, maintainability, and scalability were also included during development.

After the development phase, the system underwent comprehensive testing and evaluation. Functional testing was conducted to verify that each module performed according to the system requirements. Reliability testing was performed to determine whether the system operated consistently under normal conditions. Usability testing was also carried out to evaluate the ease of navigation, interface design, and overall user experience.

The researchers also conducted User Acceptance Testing (UAT) with selected staff members of Kape Tubong. The participants evaluated the system based on its functionality, usefulness, and ability to support their daily operations. Their comments and suggestions were collected and analyzed. Based on the evaluation results, the

researchers made the necessary improvements to enhance the system before its final implementation.

The Developmental Research Design provided a systematic process that guided the researchers from problem identification to system development, testing, evaluation, and refinement. It ensured that the BREWINSIGHT BI system was developed according to the actual needs of Kape Tubong and that the final product met the required standards for functionality, reliability, usability, and overall performance. This research design also helped ensure that the developed system effectively addressed the operational challenges identified during the study and supported better business decision-making through accurate and timely revenue analytics.

3.2 System Development Methodology

The study adopted the Agile System Development Methodology in developing the BREWINSIGHT BI: Revenue Pattern Analytics System for Kape Tubong. The Agile methodology was selected because it provided a flexible and iterative approach to software development. It enabled the researchers to develop the system in small and manageable phases while continuously improving it based on testing results and stakeholder feedback. Unlike traditional development models, Agile allowed modifications to be made throughout the development process whenever new requirements or improvements were identified. This approach ensured that the developed system remained aligned with the operational needs of the business.

The development process was divided into several iterations, commonly referred to as sprints. Each sprint focused on completing a specific set of system features within a defined period. At the end of every sprint, the completed functionalities were reviewed, tested, and evaluated before the next development cycle began. This iterative process allowed the researchers to monitor the progress of the project, identify problems at an early stage, and implement the necessary improvements without affecting the overall development schedule. It also ensured that every completed feature met the required quality standards before integration into the complete system.

The first phase of the Agile methodology was the Planning Phase. During this stage, the researchers identified the existing operational problems encountered by Kape Tubong. Information was gathered through interviews, observations, and consultations with the business owner and staff to understand the current sales monitoring and reporting process. The researchers examined how sales transactions were recorded, consolidated, and reported across different kiosks. The identified problems included fragmented sales records, delays in report generation, and the absence of a centralized system for monitoring business performance. Based on these findings, the researchers determined the project scope, established the system objectives, and identified the functional and non-functional requirements needed for the development of the BREWINSIGHT BI system.

The second phase was the System Design Phase. During this stage, the researchers translated the gathered requirements into technical specifications that served as the blueprint for system

development. The system architecture, database structure, user interface, navigation flow, and process workflows were designed to ensure that the platform would effectively support the business operations of Kape Tubong. Wireframes and interface layouts were also prepared to visualize the appearance and functionality of the system before coding began. The researchers carefully designed the platform to provide a user-friendly interface that allowed authorized users to easily monitor sales performance and generate analytical reports.

The Development Phase involved the actual implementation of the system based on the approved design. During this phase, the researchers developed the database, user authentication module, dashboard, sales monitoring module, revenue analytics, product performance analysis, branch comparison, peak sales identification, and automated report generation features. Each module was developed separately and integrated into the system after successful testing. The researchers continuously monitored the functionality of every component to ensure that it performed according to the identified system requirements. Whenever necessary, revisions and improvements were implemented to enhance the overall quality and efficiency of the platform.

Throughout the development process, continuous testing was conducted after every sprint. Functional testing was performed to verify that each module operated correctly and produced accurate results based on the system specifications. Reliability testing was also carried out to determine whether the system functioned consistently under normal operating conditions without errors or system failures. In addition, usability testing was conducted to

evaluate the ease of navigation, interface design, accessibility, and overall user experience. Any errors, inconsistencies, or usability issues identified during testing were corrected before proceeding to the next development iteration. This iterative testing process reduced the possibility of major defects during the final implementation of the system.

Another important component of the Agile methodology was the continuous involvement of stakeholders. Throughout the development process, the business owner and selected staff members of Kape Tubong regularly reviewed the developed system features and provided feedback regarding their functionality and usability. Their comments and suggestions enabled the researchers to refine the dashboard design, improve report presentation, simplify user workflows, and enhance the analytical capabilities of the system. Continuous stakeholder participation ensured that the developed platform remained aligned with the actual business processes and operational requirements of the organization.

After all system modules had been completed, the researchers conducted User Acceptance Testing (UAT) with selected staff members of Kape Tubong. During this stage, the participants evaluated the system based on its functionality, usability, efficiency, and effectiveness in supporting daily business operations. The participants were asked to perform actual system tasks and provide feedback regarding the overall performance of the platform. The researchers documented the comments, recommendations, and observations gathered during the evaluation process. Based on the evaluation results, the necessary revisions

and enhancements were implemented to improve the final version of the system before deployment.

The Agile methodology also promoted continuous improvement throughout the project. At the end of every sprint, the researchers reviewed the completed work, evaluated the progress of the project, and identified opportunities for improvement. Lessons learned from each iteration were applied during the succeeding development cycles to improve development efficiency and software quality. This iterative process allowed the researchers to respond effectively to changing business requirements and stakeholder expectations without disrupting the overall progress of the project.

The Agile System Development Methodology provided a structured yet flexible framework for developing the BREWINSIGHT BI: Revenue Pattern Analytics System. Its iterative approach enabled the researchers to continuously plan, design, develop, test, evaluate, and improve the system throughout the project. The methodology ensured that every system component was carefully validated before implementation and that stakeholder feedback was incorporated into every stage of development. As a result, the completed system was reliable, user-friendly, and capable of meeting the operational and analytical requirements of Kape Tubong while supporting more effective and data-driven business decision-making.

Planning Phase

The first phase of the Agile System Development Methodology was the **Planning Phase.** During this phase, the researchers established the project scope, objectives, system requirements, development schedule, and implementation strategies for the **BREWINSIGHT BI: Revenue Pattern Analytics System.** This phase served as the foundation of the system development process because it provided the necessary direction and structure for all succeeding activities. Through proper planning, the researchers ensured that the development process remained organized, focused, and aligned with the operational requirements of Kape Tubong.

During this phase, the researchers conducted interviews, consultations, and discussions with the business owner, **Bryan Tacayon**, and selected operational staff members of Kape Tubong. These activities were performed to gather relevant information regarding the existing business operations, sales recording procedures, reporting practices, and decision-making processes. The researchers also examined how sales information was collected, stored, processed, and presented to management across different kiosk locations. Through this assessment, the researchers gained a better understanding of the current workflow and identified areas where improvements were needed.

The information gathered from the stakeholders allowed the researchers to identify the existing operational challenges of Kape Tubong. These challenges included fragmented sales records, manual consolidation of reports, delays in preparing business summaries, limited visibility of branch performance, and the absence of a centralized platform for analyzing revenue patterns. These identified issues became the basis for determining the necessary functions and capabilities of the proposed Business Intelligence system.

After analyzing the current business processes, the researchers established the objectives and expected outcomes of the project. The primary objective was to develop a centralized analytics platform that could collect, organize, and analyze sales information from multiple Kape Tubong kiosks. The system was planned to provide interactive dashboards, automated reports,

revenue trend analysis, product performance monitoring, and branch comparison features. These capabilities were intended to support management in making faster and more informed business decisions using accurate and accessible information.

The researchers identified and documented the functional requirements needed for the development of the BREWINSIGHT BI system. These requirements included user authentication, account management, centralized dashboard visualization, sales data monitoring, revenue analysis, product performance evaluation, branch performance comparison, automated report generation, filtering functions, data export capabilities, and role-based access control. Each requirement was evaluated based on its importance in addressing the operational concerns identified during the initial assessment.

Aside from functional requirements, the researchers also determined the non-functional requirements necessary to ensure the quality and effectiveness of the system. These requirements included usability, reliability, security, accessibility, maintainability, scalability, and system responsiveness. The researchers considered the importance of developing a system that could be easily used by Kape Tubong employees while maintaining accurate data processing and protecting confidential business information. Security measures, such as user authentication and access restrictions, were included to prevent unauthorized access to sensitive records.

The researchers also identified the technical resources needed for the development of the system. The required programming languages, software tools, database technologies, and development environment were evaluated based on their compatibility, performance, reliability, and ability to support future system improvements. The selected technologies provided the necessary foundation for developing a web-based Business Intelligence platform capable of managing and analyzing business data efficiently.

A development schedule was prepared during the Planning Phase following the Agile approach. The researchers divided the project

into several sprints, where each iteration focused on specific system components and objectives. Each sprint included planned activities such as requirement review, design, development, testing, and evaluation. This approach allowed the researchers to monitor progress continuously and make improvements based on testing results and stakeholder feedback.

Feature prioritization was also performed to determine which system components should be developed first. The researchers identified critical features such as database integration, user authentication, sales monitoring, dashboard visualization, and report generation as high-priority components because these directly addressed the primary needs of Kape Tubong. Other enhancements and additional functionalities were scheduled for later iterations after the essential modules had been completed and evaluated.

Documentation was maintained throughout the Planning Phase to ensure proper organization and consistency during the development process. The researchers recorded the identified requirements, project objectives, feature priorities, development schedules, and planned system capabilities. These documents served as references during the succeeding Agile iterations and helped ensure that all development activities remained aligned with the original project goals.

The Planning Phase provided the necessary preparation and direction for the development of the BREWINSIGHT BI system. Through stakeholder consultation, requirement analysis, project scheduling, and feature prioritization, the researchers established a structured approach that guided the succeeding stages of development. This phase reduced possible development risks and ensured that the proposed system was designed according to the actual operational needs of Kape Tubong. The information and decisions gathered during this stage became the foundation for the design, development, testing, and implementation of the Business Intelligence platform.

Analysis Phase

The Analysis Phase served as the second stage of the Agile System Development Methodology for the development of the BREWINSIGHT BI: Revenue Pattern Analytics System. During this phase, the researchers focused on gathering, examining, and analyzing the operational, technical, and organizational information needed for system development. This phase aimed to develop a clear understanding of Kape Tubong's existing business processes, data management practices, reporting procedures, and decision-making activities before proceeding to the design and development stages.

The researchers conducted a detailed assessment of the current operations of Kape Tubong to determine the existing challenges and identify areas where the proposed system could provide improvements. The assessment focused on how sales transactions were recorded, stored, processed, and reported across different kiosk locations. The researchers examined the current methods used for monitoring sales performance, preparing business reports, evaluating branch operations, and analyzing product demand. Through this assessment, the researchers gained a better understanding of the organization's workflow and identified limitations affecting operational efficiency.

The analysis revealed several issues within the existing process, including manually prepared reports, scattered sales records, inconsistent data formats between branches, difficulties in comparing branch performance, and limited access to analytical information. These challenges affected the speed and accuracy of business reporting and made it difficult for management to obtain

timely insights for decision-making. The findings from this assessment became the basis for determining the system requirements and identifying the features needed for the BREWINSIGHT BI platform.

To obtain detailed information regarding the organization's requirements, the researchers conducted interviews, consultations, and discussions with the business owner, Bryan Tacayon, together with selected branch managers, administrative personnel, and operational staff. These activities allowed the researchers to understand the experiences, responsibilities, and challenges encountered by the users during daily operations. The stakeholders provided information regarding existing workflows, reporting procedures, sales monitoring practices, and the type of business information needed for effective decision-making.

The researchers also analyzed the information requirements of the organization by identifying the reports, indicators, and analytical tools needed by management. The stakeholders provided insights regarding important business measurements, such as revenue trends, product performance, branch comparisons, and sales patterns. Their feedback helped determine how information should be presented within the system, including the use of dashboards, charts, graphs, summary indicators, and automated reports. These requirements ensured that the developed platform would provide useful and understandable business insights.

During this phase, the researchers identified and documented the data requirements necessary for the operation of the system. The required data included sales transactions, product information, pricing details, branch records, revenue information, transaction

dates, product categories, and other performance indicators. The researchers analyzed how these data should be collected, organized, stored, and processed to ensure accuracy and consistency. The relationships between different data elements were also examined to support the development of an efficient database structure.

The quality of the existing business records was also evaluated during the Analysis Phase. The researchers assessed the completeness, accuracy, and consistency of available sales information and identified possible data issues, such as duplicate records, incomplete entries, and inconsistent formatting. Based on these findings, the researchers established data validation procedures and standardization methods to ensure that information collected from different branches could be properly integrated into a centralized database.

The researchers also analyzed the different user roles and responsibilities within Kape Tubong to determine the appropriate access privileges for the system. Users were classified according to their responsibilities, including administrators, managers, and operational staff. This analysis helped establish the requirements for user authentication and role-based access control. Through this approach, the researchers ensured that authorized users could access the information and features necessary for their tasks while protecting confidential business data.

The functional and non-functional requirements identified during the analysis were refined and documented for use in the succeeding phases of development. Functional requirements

included dashboard visualization, sales monitoring, revenue analysis, product performance evaluation, branch comparison, automated reporting, data filtering, and user management. Meanwhile, non-functional requirements included usability, reliability, security, maintainability, scalability, compatibility, and system performance. These requirements served as the basis for designing and developing the BREWINSIGHT BI system.

To further understand the existing business processes, the researchers created process models, workflow diagrams, and data flow diagrams. These visual representations illustrated how information moved throughout the organization and helped identify areas where automation could improve efficiency. The diagrams also provided a clear comparison between the current manual processes and the proposed system-supported workflow. Through these models, the researchers were able to identify opportunities for reducing processing time, improving data accuracy, and enhancing reporting procedures.

The feasibility of implementing the proposed system was also assessed during this phase. The researchers evaluated the available technological resources of Kape Tubong, including hardware availability, internet connectivity, software requirements, and user readiness. The technical capabilities and digital familiarity of the intended users were considered to ensure that the system design would be practical and suitable for the organization’s environment.

The researchers also examined the business rules, operational procedures, and organizational constraints that could affect

system implementation. Factors such as reporting processes, management structure, branch operations, and data handling practices were analyzed to ensure that the proposed system would support existing workflows rather than create unnecessary changes. This assessment helped improve the practicality and acceptance of the developed platform.

Requirement prioritization was performed to determine which system features should be developed first during the Agile iterations. The researchers identified core functionalities, such as centralized sales monitoring, dashboard analytics, database management, and automated reporting, as high-priority features because these directly addressed the main operational problems of Kape Tubong. Prioritizing these requirements allowed the researchers to organize sprint activities effectively and focus development efforts on the most valuable system components.

The researchers also considered future improvements and possible system expansion during the analysis process. The scalability requirements of the platform were examined to determine whether future integrations, additional branches, and expanded analytical capabilities could be supported. This consideration helped ensure that the system architecture would remain flexible and adaptable as the organization continued to grow.

Documentation was maintained throughout the Analysis Phase to record all gathered information, identified requirements, stakeholder feedback, workflow assessments, and technical considerations. Documents such as software requirement specifications, process diagrams, data requirements, and feasibility assessments served as references during the

succeeding design and development stages. These records helped maintain consistency, improve communication among stakeholders, and ensure that all development activities remained aligned with the objectives of the project.

The Analysis Phase provided the necessary foundation for the succeeding stages of the Agile development process. Through operational assessment, stakeholder consultation, requirement identification, and feasibility evaluation, the researchers ensured that the BREWINSIGHT BI system was developed according to the actual needs and conditions of Kape Tubong. The results obtained from this phase guided the design and implementation of a Business Intelligence platform that aimed to improve sales monitoring, enhance reporting efficiency, and support data-driven decision-making.

Design Phase

The **Design Phase** was the stage of the Agile System Development Methodology where the researchers transformed the gathered requirements, operational findings, and technical specifications from the previous phases into detailed designs for the **BREWINSIGHT BI: Revenue Pattern Analytics System**. This phase served as the bridge between the analysis and development stages by converting the identified business needs into structured technical solutions and visual system representations. During this stage, the researchers planned the system architecture, database structure, user interface design, analytical components, reporting modules, and security features that would guide the

actual implementation of the platform. The design process ensured that the proposed system was organized, functional, efficient, and aligned with the operational requirements of Kape Tubong.

The researchers began the Design Phase by establishing the overall system architecture of the BREWINSIGHT BI platform. The system architecture defined the relationship and interaction between the major components of the system, including the user interface, application layer, database, analytics modules, and reporting features. The researchers designed how data would enter the system, how information would be processed, how calculations would be performed, and how results would be displayed to users. This architectural planning provided a clear structure for the development process and helped ensure that all system components would function properly as an integrated platform.

The system architecture was designed based on the requirements identified during the Analysis Phase. The researchers considered the operational workflow of Kape Tubong when determining how the different modules should communicate with one another. The design focused on creating a centralized system where sales information from different kiosk locations could be collected, stored, processed, and presented through a single platform. This approach addressed the existing problem of scattered business records and allowed management to access consolidated information for monitoring and analysis. The planned architecture also considered future improvements by ensuring that additional features and system enhancements could be incorporated without requiring major changes to the entire platform.

A detailed database design was developed during this phase to support the storage and management of business information. The researchers identified the required data entities, attributes, relationships, and constraints needed for the proper operation of the system. Important entities such as sales transactions, products, branches, users, inventory information, transaction dates, revenue records, and performance indicators were defined and organized. The researchers established the relationships between these entities to ensure that information could be retrieved accurately and efficiently during system operations.

The database structure was further refined by defining primary keys, foreign keys, and data validation rules. These elements were included to maintain data consistency, reduce duplication, and improve the reliability of stored information. The researchers also considered database organization and optimization to ensure that the system could handle increasing amounts of sales data as the business continued to grow. Entity-Relationship Diagrams (ERDs) were created to illustrate the relationship between database entities, while Data Flow Diagrams (DFDs) were developed to represent the movement of information between users, processes, and data storage components. These diagrams served as technical references during the succeeding development activities.

The researchers also focused on designing the user interface and dashboard structure of the system. Interface prototypes, wireframes, and design models were prepared to visualize how users would interact with the BREWINSIGHT BI platform. The design emphasized simplicity, organization, and accessibility to ensure

that users could easily understand and navigate the system. Since the platform was intended to support business monitoring and decision-making, the researchers prioritized the clear presentation of important information through dashboards, charts, tables, summary cards, and interactive filters.

The dashboard design was planned based on the analytical requirements identified during the previous phases. The researchers designed visual components that displayed important business information, including revenue trends, product performance, branch comparisons, and key performance indicators. Appropriate visualization methods were selected to make complex business data easier to interpret. Line graphs were planned for monitoring revenue patterns over time, bar charts were considered for comparing product and branch performance, and summary indicators were included to provide quick views of important business metrics. These design decisions aimed to help management identify trends, evaluate performance, and make informed decisions more efficiently.

The navigation structure and user interaction flow were also designed during this phase. The researchers organized the system menus, page layouts, search functions, filtering options, and user actions to provide a smooth and efficient user experience. The design considered the different responsibilities of system users and ensured that frequently used features could be accessed easily. The researchers also applied responsive design principles to allow the system to function properly on different devices, including desktop computers, laptops, tablets, and mobile devices. This approach improved accessibility and allowed users

to view business information conveniently regardless of the device being used.

Security design was also incorporated into the system architecture during this phase. The researchers planned the authentication process, user roles, access permissions, and data protection mechanisms required to secure business information. Different levels of access were designed for administrators, managers, and staff members based on their responsibilities within the organization. The system was planned to restrict access to sensitive information and ensure that only authorized users could perform specific actions. Security considerations such as password protection, controlled access, and proper user management were included to protect sales records and other confidential business information.

The internal processing logic of the system was also designed during this phase. The researchers defined how the system would collect, validate, process, and transform raw business data into meaningful analytical information. Business rules related to revenue computation, sales analysis, product ranking, branch comparison, and automated report generation were identified and documented. These rules ensured that the analytical outputs generated by the system would be accurate and consistent. The researchers also planned the interaction between system modules to ensure that information would flow correctly throughout the platform.

Performance and scalability considerations were included in the system design to ensure that the BREWINSIGHT BI platform could support future operational growth. The researchers considered

efficient database structures, optimized data retrieval processes, and organized system modules to improve system responsiveness. The design also allowed additional branches, products, analytical features, and reporting capabilities to be incorporated in the future. This modular approach supported easier system maintenance and reduced the complexity of implementing future improvements.

Design validation was conducted throughout this phase to ensure that the proposed system design met the expectations of stakeholders. The researchers presented interface designs, dashboard layouts, workflow models, and planned system features to the business owner, **Bryan Tacayon**, and selected Kape Tubong personnel for evaluation. The feedback gathered from these reviews was used to improve the design of the platform. Adjustments were made to the interface arrangement, dashboard presentation, navigation flow, and system features based on the suggestions and observations provided by the stakeholders.

The researchers also reviewed the usability of the proposed design by examining whether users could easily understand the interface and perform required tasks. Elements such as color selection, typography, icons, notifications, and system messages were considered to create a consistent and user-friendly environment. The researchers ensured that the design supported the actual workflow of Kape Tubong and avoided unnecessary complexity that could affect user adoption. This validation process allowed potential design issues to be identified and corrected before the implementation stage.

Documentation was continuously maintained throughout the Design Phase to record the technical plans and design decisions made during the process. The researchers documented the system architecture, database structure, interface specifications, workflow diagrams, security plans, and module descriptions. These documents served as references during coding, testing, deployment, and future maintenance activities. Proper documentation also improved communication among the researchers and stakeholders by ensuring that all planned system features and technical requirements were clearly defined.

The researchers applied a modular design approach to ensure that the BREWINSIGHT BI platform remained flexible and maintainable. Each major system component was designed as an independent module that could be updated, improved, or expanded without significantly affecting other parts of the system. This approach supported future enhancements, such as additional analytical tools, integration with other business applications, and expansion to more branches. The modular structure allowed the platform to adapt to changing business requirements while maintaining system stability and performance.

The Design Phase established the technical and visual foundation required for the development of the BREWINSIGHT BI system. Through detailed planning of the architecture, database, interface, security features, analytical functions, and system processes, the researchers created a comprehensive blueprint for implementation. The outputs produced during this phase guided the succeeding development activities and ensured that the system would be developed according to the identified requirements of

Kape Tubong. The completed designs provided the necessary structure for creating a reliable, scalable, secure, and user-centered Business Intelligence platform capable of supporting improved sales monitoring and data-driven decision-making.

Implementation Phase

The Implementation Phase was the stage of the Agile System Development Methodology where the approved system designs were converted into a functional software application. During this phase, the researchers developed, integrated, configured, and prepared the BREWINSIGHT BI: Revenue Pattern Analytics System based on the requirements, system architecture, database structure, and design specifications established during the previous phases. While the Planning, Analysis, and Design phases focused on identifying requirements and preparing the technical foundation of the system, the Implementation Phase focused on transforming these plans into an operational web-based Business Intelligence platform that addressed the sales monitoring and analytical needs of Kape Tubong.

The researchers followed an incremental development approach through Agile sprint cycles during the implementation process. The system development activities were divided into smaller iterations, with each sprint focusing on the completion of specific modules and functionalities. This approach allowed the researchers to develop and evaluate system components gradually instead of waiting until the entire platform was completed. Each

iteration included development, testing, review, and improvement activities to ensure that every implemented feature met the required standards. The sprint-based approach also allowed changes and recommendations from stakeholders to be incorporated during development, ensuring that the system remained aligned with the operational requirements of Kape Tubong.

The implementation process began with the development of the front-end components of the system. The researchers created the user interface elements based on the approved designs from the Design Phase. These components included the main dashboard, navigation menus, sales monitoring pages, report generation interfaces, data filtering tools, charts, graphs, summary cards, and visualization panels. The researchers applied responsive web development techniques to ensure that the system could function properly on different devices, including desktop computers, laptops, tablets, and mobile devices.

Special attention was given to creating an interface that was simple, organized, and easy to understand. Since the system was intended for business users with different levels of technical experience, the researchers designed the interface to minimize complexity and improve usability. Interactive elements such as search functions, filtering options, date selection tools, and dynamic charts were implemented to allow users to customize the information displayed according to their specific monitoring and reporting needs. These features enabled users to access important business information more efficiently and interpret sales data more effectively.

Alongside the development of the front-end components, the researchers implemented the back-end functionality of the system. The back-end handled the main processing operations, including business logic, database communication, user authentication, report generation, data processing, and system security management. The researchers developed the necessary functions that allowed the system to receive, process, store, and retrieve business information accurately. User authentication and role-based access control were also implemented to ensure that users could only access features and information permitted according to their assigned responsibilities.

The researchers developed different user permissions based on the organizational roles identified during the Analysis Phase. Administrators were provided with access to system management functions, while managers and operational staff were provided access to business information relevant to their tasks. This access control structure helped protect sensitive sales and operational records while maintaining efficient information availability for authorized users. Security practices such as input validation, password protection, and controlled access management were incorporated throughout the implementation process to strengthen system reliability and data protection.

A major activity during the Implementation Phase was the development of the Business Intelligence and analytical features of the platform. The researchers implemented the necessary processes that transformed raw business data into meaningful information for decision-making. Sales transactions, product information, branch records, and operational data were processed

and converted into analytical outputs that could be displayed through dashboards and reports.

The analytical modules developed during this phase included revenue trend analysis, product performance evaluation, branch comparison, sales pattern monitoring, and peak business period identification. The researchers created processes that allowed the system to generate daily, weekly, monthly, and annual sales summaries based on available transaction records. These features enabled management to evaluate business performance, identify trends, and recognize areas requiring improvement. Automated report generation was also implemented to reduce manual reporting tasks, improve reporting consistency, and provide users with timely access to business information.

Database integration was another important activity completed during this phase. The researchers connected the developed application with the database structure designed during the Design Phase. This integration allowed the system to store, retrieve, and process business data efficiently. Database operations such as inserting new records, updating information, retrieving reports, and managing user data were implemented to support the functionality of the platform.

To maintain data accuracy and reliability, validation mechanisms were applied during the processing of information. The researchers implemented procedures to check incoming records, maintain consistent data formats, and preserve relationships between database tables. These measures helped prevent inaccurate information from affecting system outputs and ensured that the dashboards and reports displayed reliable business insights.

Database optimization techniques, including efficient query design and organized data retrieval processes, were also applied to improve system performance and reduce processing delays.

Each developed module underwent continuous testing and evaluation throughout the implementation process. The researchers tested individual components before integrating them into the complete system. Functional testing was performed to verify that each feature operated according to its intended purpose, while integration testing ensured that different system modules communicated properly with one another. Errors, inconsistencies, and unexpected behaviors identified during testing were documented, corrected, and retested before proceeding to the next sprint.

The Agile approach also encouraged continuous collaboration with stakeholders during the implementation stage. The business owner, Bryan Tacayon, and selected Kape Tubong personnel participated in reviewing partially completed system components and providing feedback regarding functionality and usability. Sprint review sessions were conducted to present completed features, discuss observations, and identify improvements needed for succeeding iterations. The feedback collected from these activities helped the researchers refine system workflows, improve interface designs, and ensure that the developed features reflected actual business requirements.

The researchers also performed technical improvements throughout the implementation process. Activities such as code optimization, debugging, refactoring, version control management, and system integration were conducted to improve the quality and

maintainability of the application. Development frameworks and supporting libraries were utilized when necessary to improve system performance, enhance visualization capabilities, and simplify implementation tasks. These technical improvements contributed to creating a more stable and efficient platform.

Security and system protection measures were continuously maintained throughout the development process. The researchers implemented mechanisms such as secure authentication, access restrictions, input validation, and protection against unauthorized system activities. These measures helped safeguard important business information and ensured that the platform maintained confidentiality and reliability during operation.

Technical documentation was also updated throughout the Implementation Phase. The researchers recorded implemented features, system configurations, database updates, coding practices, and development decisions. These records served as references during testing, deployment, maintenance, and future enhancement activities. Proper documentation also supported better communication among the researchers and stakeholders by providing a clear record of the system development process.

The Implementation Phase transformed the planned designs of the BREWINSIGHT BI system into a working Business Intelligence platform capable of supporting the operational requirements of Kape Tubong. Through incremental development, continuous testing, stakeholder collaboration, and technical refinement, the researchers ensured that the completed system met the required standards for functionality, security, reliability, and usability. The activities conducted during this phase prepared

the platform for the succeeding testing and evaluation stage, where its effectiveness and performance were further assessed.

Testing Phase

The **Testing Phase** served as the quality assurance stage of the Agile System Development Methodology for the **BREWINSIGHT BI: Revenue Pattern Analytics System**. During this phase, the researchers systematically evaluated the developed platform to determine whether it satisfied the established functional requirements, technical specifications, usability standards, and operational expectations of Kape Tubong. The primary objective of this phase was to verify that all implemented components performed correctly, produced accurate results, maintained system reliability, and provided an effective solution to the identified business challenges. Testing activities were conducted to ensure that the completed system was stable, secure, responsive, and ready for deployment within the organization's operational environment.

Unlike a single testing activity conducted only after development completion, testing was integrated throughout the Agile development process. Each completed module from every sprint underwent evaluation before being incorporated into the complete system. This iterative testing approach allowed the researchers to identify defects, inconsistencies, and functional issues at an early stage. Problems discovered during testing were documented, analyzed, corrected, and retested to verify that the implemented solutions resolved the identified concerns without affecting

existing system functionalities. This continuous evaluation process improved the quality, reliability, and stability of the BREWINSIGHT BI platform throughout its development lifecycle.

The researchers initially conducted functional testing to verify that each system feature operated according to the approved requirements and design specifications. Individual modules, including user authentication, dashboard visualization, sales monitoring, data filtering, branch comparison, revenue analysis, product performance evaluation, and automated report generation, were tested to determine whether they performed their intended functions correctly. Each feature was evaluated using different scenarios to confirm that the system produced accurate outputs, handled user requests properly, and responded appropriately to different operational conditions.

The researchers also examined the accuracy of the analytical outputs generated by the system. Dashboard components, including charts, graphs, tables, and summary indicators, were tested to ensure that the displayed information accurately represented the data stored in the centralized database. Revenue calculations, sales summaries, product rankings, branch performance comparisons, and trend analyses were verified by comparing system-generated results with expected values and manually calculated records. These activities ensured that the Business Intelligence features of the platform provided reliable information that could support business monitoring and managerial decision-making.

Integration testing was conducted to evaluate the interaction between the different system components. The researchers examined

the communication between the front-end interface, back-end processes, database management system, analytical modules, and reporting functions. This testing process verified that data could move correctly between different parts of the platform without errors or inconsistencies. Issues related to data synchronization, module communication, database connections, and system workflow were identified and resolved to ensure that the platform operated as a complete and unified system.

The researchers also performed performance testing to evaluate the speed, efficiency, and responsiveness of the BREWINSIGHT BI platform. System response times were measured during activities such as dashboard loading, report generation, data retrieval, and analytical processing. The researchers tested the system using different amounts of business data to determine whether it could maintain acceptable performance while handling increasing transaction records. These tests helped ensure that users could access business information efficiently without experiencing significant delays during daily operations.

Load and stress testing were also performed to evaluate how the system responded under increased workloads. Simulated conditions involving multiple users accessing the platform simultaneously, large data requests, continuous dashboard updates, and complex report generation tasks were conducted. The researchers observed the system's stability, resource usage, and response behavior under these conditions. The results of these tests helped determine whether the platform could support current operational demands and accommodate future business expansion.

Compatibility testing was conducted to confirm that the system functioned properly across different devices, browsers, and operating environments. The researchers tested the platform using desktop computers, laptops, tablets, and mobile devices to evaluate interface responsiveness, navigation behavior, and consistency of displayed information. Different web browsers were also assessed to ensure that dashboards, charts, menus, and other interface components operated correctly. This testing ensured that authorized users from different Kape Tubong locations could access the system effectively regardless of their preferred device or browser environment.

Security testing was performed to evaluate the effectiveness of the implemented protection mechanisms. The researchers assessed user authentication, password security, session management, role-based access control, and input validation procedures. Testing was conducted to verify that unauthorized users could not access restricted information or perform prohibited actions. The researchers also evaluated how the system handled incorrect login attempts, invalid inputs, and improper user actions to ensure that sensitive business information remained protected.

The researchers conducted additional validation using incorrect and unexpected data inputs to evaluate the system's error-handling capabilities. Examples included incomplete forms, invalid data formats, incorrect user credentials, and unsupported actions. The researchers verified whether the system provided appropriate notifications, prevented inaccurate information from being stored, and maintained normal operation despite user

errors. These activities improved the reliability of the platform by ensuring that unexpected situations could be handled properly.

Data accuracy validation was also performed to confirm the reliability of the analytical information generated by the system. The researchers compared system-generated reports, revenue calculations, sales trends, branch performance results, and product analyses with existing transaction records and manual computations. This process verified that the system accurately processed business information and produced dependable outputs. Ensuring data accuracy was essential because management decisions would rely on the insights provided by the platform.

User Acceptance Testing (UAT) was conducted with selected stakeholders of Kape Tubong, including management representatives, branch personnel, administrative staff, and operational users. During this activity, participants performed actual business-related tasks using the system, such as viewing dashboards, generating reports, monitoring sales performance, and analyzing business trends. The participants evaluated the platform based on usability, functionality, convenience, and effectiveness in supporting their daily responsibilities.

Feedback collected during the User Acceptance Testing was documented and analyzed by the researchers. Suggestions regarding interface improvements, navigation adjustments, report presentation, and additional functionality were reviewed and considered for system refinement. The involvement of actual users allowed the researchers to determine whether the platform successfully addressed the practical needs of Kape Tubong and

whether it could be effectively adopted within the organization's workflow.

Regression testing was conducted whenever modifications, corrections, or improvements were applied to the system. The researchers verified that newly implemented changes did not negatively affect previously completed functionalities. This process maintained system stability and ensured that improvements introduced during development continued to support the intended objectives of the platform.

Comprehensive testing documentation was maintained throughout the entire phase. The researchers recorded testing procedures, identified issues, corrective actions, validation results, and system improvements. These records provided evidence that the BREWINSIGHT BI platform underwent systematic evaluation before deployment. The documentation also served as a reference for future maintenance, troubleshooting, and enhancement activities.

The Testing Phase confirmed that the BREWINSIGHT BI system satisfied the required standards for functionality, reliability, security, usability, and performance. Through continuous testing, evaluation, and refinement, the researchers improved the quality of the platform and reduced potential risks before implementation. The results of this phase provided assurance that the system was capable of delivering accurate business analytics and supporting operational decision-making within Kape Tubong.

Deployment Phase

The **Deployment Phase** involved the transition of the completed BREWINSIGHT BI: Revenue Pattern Analytics System from the development environment into the actual operational setting of Kape Tubong. During this phase, the researchers installed, configured, and prepared the system for use by authorized personnel. The activities focused on ensuring that the platform operated correctly within the organization's environment, maintained secure access, and provided reliable business intelligence capabilities for daily operations.

Before deployment, the researchers conducted final preparation activities to verify that all system components were ready for implementation. Server configurations, database connections, security settings, user accounts, and application requirements were reviewed to ensure proper system operation. The researchers confirmed that the production environment met the technical requirements needed to support the platform and that all necessary configurations were completed before making the system available to users.

The researchers migrated the required business data into the deployed system to ensure that users could immediately access relevant information and generate meaningful reports. Historical sales records, product information, branch details, and other necessary operational data were transferred, checked, and validated to maintain accuracy and consistency. Data verification procedures were performed after migration to confirm that the transferred information matched the original records and that dashboards and reports displayed correct results.

User account configuration was also completed during this phase. The researchers created user profiles and assigned appropriate permissions based on the roles identified during the Analysis and Design phases. Administrators, managers, and operational users were provided with access privileges according to their responsibilities. This process ensured that users could efficiently perform their assigned tasks while maintaining proper protection of confidential business information.

User orientation and training activities were conducted to familiarize Kape Tubong personnel with the newly implemented platform. The researchers provided instructions regarding system navigation, dashboard interpretation, report generation, filtering functions, and other important features. Training materials and user documentation were prepared to assist employees in understanding the system's capabilities and applying the platform effectively in their daily operations.

A structured deployment procedure was followed to minimize interruptions to existing business activities. The researchers conducted final system checks involving database integrity, application connectivity, security configurations, and overall system functionality before full implementation. Backup procedures were also established to protect business information and provide recovery options in case unexpected issues occurred after deployment.

After deployment, the researchers monitored the performance of the system within the actual business environment. System behavior, response times, database performance, and user interactions were observed to identify possible issues that might

not have appeared during the testing stage. Minor adjustments and improvements were implemented when necessary to enhance usability, stability, and operational effectiveness.

The researchers also collected feedback from users during the initial implementation period. Observations regarding system navigation, dashboard interpretation, report usage, and workflow integration were documented and analyzed. This feedback supported continuous improvement by identifying areas where additional adjustments or enhancements could improve the user experience and system effectiveness.

Technical documentation related to deployment procedures, configuration settings, security policies, user management, and system maintenance was maintained throughout this phase. These records provided future references for system updates, troubleshooting, and possible expansion. Proper documentation also supported the long-term sustainability of the BREWINSIGHT BI platform by ensuring that future administrators and developers could understand the deployed environment.

The Deployment Phase successfully introduced the BREWINSIGHT BI platform into the operational environment of Kape Tubong. Through careful installation, data migration, user preparation, monitoring, and continuous support, the researchers ensured that the system transitioned smoothly from development into practical use. The deployed platform provided the organization with a centralized Business Intelligence solution capable of supporting sales analysis, revenue monitoring, performance evaluation, and data-driven decision-making.

Maintenance Phase

The Maintenance Phase represented the final stage of the Agile System Development Methodology and focused on sustaining the performance, reliability, security, and long-term effectiveness of the BREWINSIGHT BI: Revenue Pattern Analytics System after deployment. During this phase, the researchers continued monitoring the system, addressing operational concerns, correcting identified issues, and implementing improvements based on actual user experiences and changing business requirements. Unlike the previous phases that focused on development and implementation, this phase concentrated on ensuring that the deployed platform continued to function properly and remained valuable for the daily operations and decision-making processes of Kape Tubong.

Continuous technical support was provided throughout the Maintenance Phase to address issues encountered during actual system usage. The researchers monitored reported concerns related to system behavior, interface performance, data accuracy, calculation results, and overall functionality. Any software defects, logical errors, unexpected system responses, or inconsistencies identified by users or discovered through monitoring activities were documented, analyzed, and resolved. After corrections were implemented, affected features were tested again to verify that the improvements successfully addressed the identified issues without negatively affecting other system components.

The researchers conducted regular performance monitoring to ensure that the BREWINSIGHT BI platform remained stable and responsive as the volume of business data and number of users increased. System performance indicators, including page loading speed, database response time, report generation efficiency, and dashboard responsiveness, were observed to determine whether the platform continued to meet operational expectations. Monitoring activities allowed the researchers to identify possible performance limitations and apply necessary improvements before they affected user experience or business operations.

Software updates and technical improvements were also applied during this phase whenever necessary. The researchers reviewed the system environment and implemented required updates to maintain compatibility, security, and reliability. Framework updates, library improvements, security patches, and technical adjustments were applied carefully to prevent system instability. Before applying major changes, the researchers evaluated their possible effects on existing functionalities to ensure that system operations remained uninterrupted.

User feedback served as an important source of information for continuous system improvement during the Maintenance Phase. The researchers collected comments, recommendations, and observations from Kape Tubong personnel regarding system usability, dashboard presentation, reporting requirements, and workflow improvements. These suggestions were evaluated based on their relevance, technical feasibility, and contribution to operational efficiency. Approved enhancements were incorporated into future updates to ensure that the platform continued to reflect the

evolving needs of the organization.

Minor interface improvements and functional adjustments were implemented based on operational requirements and stakeholder recommendations. Improvements included modifications to dashboard arrangements, enhancement of visualization components, refinement of report formats, and adjustments to user interaction processes. These improvements aimed to make the platform easier to use while maintaining consistency with the organization's existing workflows. The researchers ensured that any modifications maintained the accuracy, reliability, and usability standards established during the previous development phases.

Compatibility maintenance was also performed to ensure that the system continued to operate correctly despite changes in the technological environment. The researchers regularly verified that the platform remained functional across updated web browsers, operating systems, devices, and hosting environments. These activities prevented compatibility-related issues and ensured that users from different locations and devices could continue accessing the platform effectively.

Backup and recovery procedures were reviewed and improved during this phase to protect important business information. The researchers evaluated the effectiveness of existing backup mechanisms and ensured that critical sales records, user information, and analytical data could be restored in case of unexpected failures, data corruption, or system interruptions. Regular backup validation helped maintain data availability and reduced the risk of operational disruptions.

The Maintenance Phase also supported the scalability and future expansion of the BREWINSIGHT BI platform. As Kape Tubong continued to improve its operations, introduce new products, or require additional analytical capabilities, the system was designed to accommodate these changes through controlled enhancements. Additional data fields, reporting categories, analytical indicators, and system features could be integrated without requiring major modifications to the existing platform structure. This flexibility allowed the system to remain useful as organizational requirements developed over time.

Database maintenance activities were also performed to preserve efficient data management. The researchers reviewed database performance by analyzing query execution, storage utilization, and data organization. Optimization procedures such as query refinement, indexing improvements, and database cleanup were conducted when necessary to maintain efficient information retrieval. These activities ensured that increasing volumes of historical sales records would not significantly affect system responsiveness or analytical performance.

Regular system audits and performance evaluations were conducted to determine whether the platform continued to satisfy operational objectives and technical standards. The researchers reviewed system logs, usage records, and error reports to identify recurring problems, possible security concerns, and opportunities for optimization. These evaluations provided valuable information regarding the actual performance of the platform and supported evidence-based decisions for future improvements.

Preventive maintenance activities were also conducted to minimize possible system failures and performance degradation. Security assessments, system health checks, and operational reviews were scheduled periodically to ensure that the platform remained secure and reliable. User manuals, technical documentation, and instructional materials were updated whenever new features, interface changes, or revised procedures were introduced. Maintaining updated documentation supported effective system usage and simplified future troubleshooting activities.

The researchers also ensured that the system remained aligned with changes in Kape Tubong's business strategies and operational practices. Adjustments were considered whenever the organization introduced new pricing methods, modified reporting requirements, expanded operational processes, or required additional performance indicators. These improvements ensured that the analytical outputs generated by the platform remained relevant and useful for management decision-making.

Future enhancement opportunities were also identified during the Maintenance Phase. Based on user feedback and operational observations, the researchers identified possible improvements such as predictive analytics, automated forecasting, advanced reporting capabilities, and additional visualization tools. Although these enhancements were beyond the initial scope of the project, identifying them provided direction for possible future development and expansion of the BREWINSIGHT BI platform.

All maintenance activities, system modifications, configuration changes, and improvement procedures were properly documented to maintain traceability throughout the system lifecycle. These

records included details regarding implemented updates, resolved issues, performance evaluations, and technical adjustments. Documentation supported future maintenance activities, troubleshooting procedures, and possible system expansion by providing a complete history of the platform's development and improvement.

Through continuous monitoring, technical support, optimization, and enhancement activities, the Maintenance Phase ensured that the BREWINSIGHT BI platform remained reliable, adaptable, and effective after deployment. The researchers maintained the system's operational quality by addressing issues promptly, improving existing features, and ensuring that the platform continued to support accurate business analysis and data-driven decision-making within Kape Tubong.

3.3 System Architecture

The BREWINSIGHT BI: Revenue Pattern Analytics System was designed using a client-server architecture as the primary framework supporting its web-based operation. This architecture separated the system into three major layers: the client layer, the server layer, and the database layer. The separation of these components improved system organization, processing efficiency, security management, and scalability. The architecture allowed users to access the platform through web browsers while maintaining centralized processing and data management on the server side.

The client layer represented the interface through which authorized users interacted with the system. Users accessed the

platform using web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge without requiring additional software installation on their devices. Through the browser interface, users were able to log into the system, view dashboards, generate reports, analyze sales performance, and monitor revenue patterns. The client layer functioned as the presentation component of the platform by displaying processed information through charts, graphs, tables, and summary indicators.

The client-side interface was designed with emphasis on usability, accessibility, and user convenience. The researchers structured the interface layout to present business information clearly and logically, allowing users to interpret analytical results without requiring advanced technical knowledge. Interactive dashboard components, including filters, date selection tools, search features, and comparison controls, were integrated to provide flexibility when analyzing business information. These features enabled management personnel to customize displayed information based on specific operational requirements and reporting needs.

The responsive design approach allowed the client interface to adjust according to different screen sizes and device types. Users accessing the system through desktop computers, laptops, tablets, or mobile devices were able to experience consistent navigation and functionality. This accessibility supported flexible business monitoring by allowing authorized personnel to review performance information regardless of their physical location.

The server layer served as the central processing component of

the BREWINSIGHT BI platform. It handled application logic, user authentication, business calculations, report generation, and communication between the user interface and database. The system utilized Apache Web Server through XAMPP as the hosting and development environment. Apache processed incoming requests from users and returned the required information or system responses through the browser interface.

When users requested information, such as revenue reports, dashboard updates, or filtered sales analysis, the server processed the request by executing the required application logic. Backend processes performed calculations, retrieved necessary records, and prepared the information before presenting it to the client interface. Centralizing these processes improved consistency because all users received results generated using the same processing rules and analytical methods.

The server architecture was designed using a modular approach where different system functions operated as separate but connected components. Modules responsible for authentication, reporting, analytics, and data processing were organized independently to improve maintainability and future expansion. This structure allowed developers to modify or enhance specific features without significantly affecting other parts of the system.

The database layer utilized MySQL as the relational database management system responsible for storing and managing business information. The database served as the centralized repository for sales transactions, product information, branch records, user accounts, timestamps, revenue summaries, and other operational

data required by the system. Organizing information within a centralized database eliminated problems commonly associated with fragmented records and manual reporting processes.

The researchers structured the database using relational tables connected through defined relationships. Primary keys, foreign keys, and database constraints were implemented to maintain data integrity and ensure accurate connections between related records. Database normalization practices were applied to reduce redundancy, improve consistency, and support efficient retrieval of information. These database design practices allowed the system to generate accurate analytical outputs and reliable reports.

The database supported various analytical operations required by the BREWINSIGHT BI platform, including filtering, sorting, aggregation, comparison, and trend analysis. Through SQL queries, the system retrieved relevant information and transformed stored records into meaningful business insights displayed through dashboards and reports. Backup procedures were also established to protect important business data and support recovery in case of unexpected failures.

The communication process within the architecture followed a structured request-response model between the client, server, and database. When users performed actions through the dashboard interface, requests were transmitted to the server for processing. The server then communicated with the database to retrieve or modify the required information. After processing the request, the server returned the results to the client interface

where they were displayed in an understandable format.

This layered communication structure improved system organization by separating user interaction, processing operations, and data storage responsibilities. It simplified troubleshooting, improved security, and ensured consistent application of business rules throughout the platform. The separation of responsibilities between layers also supported easier maintenance and future system enhancement.

The client-server architecture provided several operational advantages for Kape Tubong's multi-kiosk environment. Since all information was stored and processed centrally, management could monitor sales performance across multiple kiosk locations using a single platform. The architecture reduced the need for manual consolidation of reports and improved the consistency of business information used for decision-making.

Centralized processing also allowed system updates and improvements to be implemented more efficiently. Changes made on the server became immediately available to authorized users without requiring individual installation on every device. This reduced maintenance effort and ensured that all users accessed the latest version of the platform.

The selected architecture provided a practical foundation for the successful operation of the BREWINSIGHT BI platform. By separating the presentation, processing, and data management layers, the system achieved improved organization, security, performance, and scalability. The client-server architecture supported the operational requirements of Kape Tubong by enabling

centralized business intelligence, efficient data management, and accessible analytical monitoring across multiple kiosk locations.

The architecture also prepared the system for future expansion and integration. Additional branches, advanced analytical features, and possible connections with other business applications could be incorporated without requiring significant restructuring of the existing platform. This flexibility ensured that the BREWINSIGHT BI system remained adaptable and capable of supporting the changing needs of Kape Tubong.

3.4 Tools and Technologies Used

- Frontend: HTML, CSS and JavaScript
- Backend: Python Flask
- Database: MySQL
- Development Tools: Visual Studio Code and XAMPP

3.5 Data Gathering Techniques

- Structured interviews with owner (Bryan Tacayon)
- Observation of sales processes at kiosk and main shop
- Document analysis of sales records, receipts, and manual logs
- Surveys with staff on technology comfort and reporting preferences

3.6 Testing Procedures

To evaluate the quality, reliability, functionality, security, and usability of the BREWINSIGHT BI: Revenue Pattern Analytics System, the researchers conducted a comprehensive testing process

consisting of Unit Testing, System Testing, and User Acceptance Testing (UAT). These testing procedures were performed throughout the development lifecycle and before the final implementation of the system to ensure that all components operated according to their intended functions and design specifications. The testing process aimed to verify that the developed platform could effectively support the business intelligence and revenue analytics requirements of Kape Tubong by providing accurate, reliable, and meaningful information for operational monitoring and managerial decision-making.

The testing procedures followed a structured and progressive approach, beginning with the evaluation of individual system components and progressing toward the assessment of the complete integrated platform. This method allowed the researchers to identify and resolve errors at an early stage, preventing possible issues from affecting other system components. Each identified issue was documented, analyzed, corrected, and subjected to retesting to confirm that the implemented solutions successfully addressed the problems without affecting existing functionalities.

The testing process focused on validating the system’s ability to perform essential operations, including user authentication, dashboard visualization, revenue analysis, sales monitoring, report generation, data filtering, branch comparison, and product performance tracking. The researchers also assessed the accuracy of analytical outputs by comparing generated reports, dashboard summaries, and revenue computations against verified transaction records and manual calculations. Through repeated testing, the

researchers ensured that the system produced consistent and dependable results suitable for supporting business operations.

Performance and reliability testing were also conducted by evaluating the system under different operational scenarios. The researchers examined the platform's response when processing different data volumes, performing multiple analytical operations, generating reports, and handling simultaneous user activities. These evaluations measured system responsiveness, processing efficiency, database performance, and overall stability to ensure that the platform remained functional during normal and increased workloads.

Security and validation testing were performed to ensure that business information remained protected and that only authorized users could access restricted features and data. Authentication mechanisms, user permissions, input validation, error handling, and data protection procedures were evaluated to confirm that the system maintained confidentiality, integrity, and reliability. Invalid inputs, incomplete records, incorrect credentials, and unexpected user actions were also tested to determine whether the system could properly handle errors without affecting stored information or overall functionality.

The researchers also evaluated the usability and practicality of the platform through User Acceptance Testing (UAT). Selected representatives of Kape Tubong assessed the system by performing actual business-related tasks such as viewing dashboards, generating reports, analyzing sales trends, and monitoring branch performance. Their feedback regarding interface design, navigation, accessibility, and usefulness of analytical outputs

was considered in refining the final version of the system before deployment.

Through the implementation of Unit Testing, System Testing, and User Acceptance Testing, the researchers established that the BREWINSIGHT BI: Revenue Pattern Analytics System met the required technical, operational, and usability standards. The testing procedures provided evidence that the system was accurate, stable, secure, and capable of delivering valuable business insights that supported data-driven decision-making within Kape Tubong.

Unit Testing

Unit Testing was conducted to evaluate individual components and modules of the BREWINSIGHT BI: Revenue Pattern Analytics System before integrating them into the complete platform. This testing stage focused on verifying whether each system function operated correctly according to its intended purpose and technical specifications. By examining individual modules separately, the researchers were able to identify errors, inconsistencies, and functional issues at an early stage, allowing corrections to be applied before they affected other parts of the system.

The researchers tested various system components, including the user authentication module, database connection functions, dashboard visualization components, data filtering features, report generation processes, user account management functions, data input forms, and analytics processing modules. Each module

was evaluated using different test scenarios and input conditions to verify whether expected outputs, calculations, and system responses were produced accurately and consistently.

The primary objective of Unit Testing was to detect programming errors, logical inconsistencies, calculation errors, database communication problems, and unexpected system behavior within individual modules. Testing components separately simplified troubleshooting by allowing the researchers to identify the specific source of errors and implement appropriate solutions without affecting other system functionalities.

Input validation testing was performed to verify that the system correctly processed valid information while preventing incorrect or incomplete data from being stored. The researchers tested different conditions, including incorrect login credentials, missing required fields, invalid numerical values, duplicate records, and improperly formatted data entries. These tests ensured that the platform maintained data accuracy and prevented unreliable information from affecting analytical results.

The researchers also evaluated error-handling mechanisms to determine whether the system provided appropriate responses when users encountered operational issues. Warning messages, validation prompts, and system notifications were tested to ensure that users received clear guidance when performing incorrect actions. Exception-handling procedures were also evaluated to confirm that unexpected errors, database connection interruptions, or invalid operations did not cause system crashes or data loss.

Performance testing of individual modules was conducted by evaluating response times, processing efficiency, and stability during repeated operations. Dashboard components, report generation functions, filtering tools, and database queries were tested to determine whether they could process requests efficiently while maintaining accurate outputs. These assessments ensured that each component performed reliably before integration.

Compatibility checks were also performed to verify that individual modules properly interacted with supporting technologies, including the web interface, database server, and integrated software components. The researchers confirmed that data transferred between modules remained accurate, synchronized, and free from corruption during processing.

Edge-case testing was included to evaluate the robustness of individual components under unusual conditions. The researchers tested boundary values, empty fields, invalid credentials, unsupported inputs, incorrect filter combinations, and duplicate entries to determine whether the system maintained proper functionality even when exposed to unexpected situations.

Regression testing was performed whenever modifications or corrections were introduced. Each updated component was retested to ensure that implemented fixes resolved identified problems without introducing new errors or affecting previously functional features. This process maintained system stability throughout the development cycle.

All Unit Testing activities, including test conditions,

identified issues, corrective actions, and retesting results, were documented by the researchers. These records served as references for troubleshooting, maintenance, and future enhancement activities. The completion of Unit Testing confirmed that each individual component of the BREWINSIGHT BI platform functioned correctly, providing a reliable foundation for the succeeding integration and system-level testing phases.

System Testing

After completing Unit Testing, System Testing was conducted to evaluate the performance, functionality, reliability, and stability of the fully integrated BREWINSIGHT BI: Revenue Pattern Analytics System. Unlike Unit Testing, which focused on individual modules, System Testing assessed the complete platform as a unified system to determine whether all components worked together properly and satisfied the functional and operational requirements established during the development process.

The researchers evaluated the interaction between the user interface, backend application logic, database management system, analytics modules, dashboard visualization components, and reporting functions. The primary objective of this testing phase was to verify that data flowed correctly across all system components without errors, duplication, loss, or inconsistencies. This ensured that information entered into the system was accurately processed, stored, analyzed, and presented through the appropriate dashboard displays and generated reports.

During System Testing, the researchers performed various operational scenarios that represented the actual activities conducted within Kape Tubong. These activities included user login and authentication, accessing dashboards, retrieving sales records, applying data filters, generating reports, analyzing revenue patterns, comparing branch performance, monitoring product performance, and viewing summary analytics. Each process was evaluated to confirm that the system produced accurate results and responded correctly to user requests.

The researchers verified the consistency and synchronization of data throughout the entire platform. Updates made to transaction records were examined to ensure that corresponding changes were accurately reflected in dashboard visualizations, analytical summaries, and generated reports. The testing process confirmed that the system maintained reliable communication between the database and application modules, allowing users to access updated business information efficiently.

System Testing also evaluated the accuracy of analytical computations and generated outputs. Revenue calculations, sales summaries, trend analysis, branch comparisons, and product rankings were compared against verified transaction data and manual computations. This validation process ensured that the system transformed stored business information into accurate and meaningful analytical insights that could support managerial decision-making.

The researchers assessed the stability and reliability of the integrated platform by performing repeated operations under different conditions. The system was tested while executing

multiple activities, such as generating reports, retrieving large datasets, updating dashboard information, and performing filtering operations. These evaluations determined whether the platform could maintain consistent performance during continuous usage without experiencing errors, delays, or unexpected interruptions.

Performance testing was also included during this phase to measure the responsiveness and efficiency of the system under normal operational conditions. The researchers monitored dashboard loading speed, report generation time, database query performance, and overall system response. These measurements helped determine whether the platform provided sufficient performance for daily business monitoring and analytical activities.

To evaluate the system’s ability to support multiple users, simulated concurrent access scenarios were performed. Several user activities were tested simultaneously, including dashboard viewing, report generation, data retrieval, and account access operations. The researchers observed whether increased user activity affected system responsiveness, data accuracy, or overall stability. The results confirmed whether the platform could accommodate regular business operations with minimal performance impact.

Compatibility testing was also conducted to verify that the BREWINSIGHT BI platform functioned properly across different devices, browsers, operating systems, and screen sizes. The researchers evaluated the system using commonly used web browsers and devices to ensure that interface elements, dashboards,

charts, tables, navigation menus, and reports displayed correctly. This testing ensured that users could access and operate the system consistently regardless of their preferred device or browser environment.

The responsiveness and adaptability of the user interface were further evaluated through different screen resolutions and device configurations. The researchers examined whether dashboard components, buttons, tables, filters, and navigation elements adjusted properly while maintaining readability and usability. These assessments ensured that the web-based platform remained accessible and convenient for users across different environments.

Security testing was conducted as part of the System Testing process to verify the effectiveness of user authentication, access control, and data protection mechanisms. The researchers tested login procedures, user permissions, session management, and restricted access features to ensure that confidential business information was only available to authorized personnel. The system was also evaluated against unauthorized access attempts and incorrect user actions to confirm that security controls operated effectively.

Error handling and recovery mechanisms were assessed by intentionally introducing invalid operations, incomplete inputs, interrupted processes, and unexpected user actions. The researchers verified whether the system displayed appropriate notifications, prevented incorrect data processing, and recovered properly from operational issues. These tests ensured that the platform maintained stability and protected data integrity even

when unexpected situations occurred.

Data integrity testing was performed to confirm that information remained accurate throughout all system processes. The researchers compared stored database records with displayed dashboard results and generated reports to verify consistency. Calculations, data retrieval processes, and analytical outputs were examined to ensure that no information was altered, lost, or incorrectly processed during system operations.

All findings obtained during System Testing were documented, including detected issues, applied solutions, testing conditions, and retesting results. These records provided evidence of system behavior, performance, and reliability while serving as references for future maintenance and enhancement activities.

The successful completion of System Testing confirmed that the BREWINSIGHT BI: Revenue Pattern Analytics System operated as an integrated, stable, and functional platform. The testing process verified that all modules communicated effectively, analytical outputs were accurate, security mechanisms were functional, and system performance met the operational requirements of Kape Tubong. The results provided confidence that the platform was ready for evaluation by actual users through the User Acceptance Testing phase.

User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted to determine whether the BREWINSIGHT BI: Revenue Pattern Analytics System satisfied

the operational needs, expectations, and requirements of its intended users. This testing phase focused on evaluating the system from the perspective of actual users within Kape Tubong and determining whether the platform was practical, understandable, efficient, and suitable for daily business operations.

Unlike Unit Testing and System Testing, which primarily focused on technical functionality and system performance, User Acceptance Testing emphasized the overall user experience, usability, and effectiveness of the platform in supporting business activities. The purpose of this phase was to confirm that the completed system could be effectively utilized by management and staff for monitoring sales performance, analyzing revenue patterns, generating reports, and supporting data-driven decision-making.

During this phase, the business owner, Bryan Tacayon, together with selected staff members, branch personnel, and administrative users of Kape Tubong participated in the evaluation of the system. The participants performed actual business-related tasks using the BREWINSIGHT BI platform, including logging into the system, viewing dashboards, filtering sales information, generating reports, reviewing revenue trends, comparing branch performance, and analyzing product-related data.

The researchers observed how users interacted with the system and evaluated whether the platform’s features, navigation structure, and analytical presentations were aligned with the users’ operational requirements. Particular attention was given to determining whether users could easily locate necessary

information, understand displayed analytics, and complete required tasks without experiencing significant difficulties.

The usability evaluation focused on several important factors, including ease of navigation, clarity of dashboard displays, accessibility of reports, system responsiveness, accuracy of information presented, and overall user satisfaction. The participants assessed whether charts, tables, summary indicators, and analytical visualizations were presented in a clear and understandable manner. Since the primary users were business personnel rather than technical specialists, the researchers ensured that the system provided information in a simple, organized, and user-friendly format.

The researchers also evaluated the learning ability of users when interacting with the platform. Users were observed while performing activities such as navigating through system modules, applying filters, generating reports, and interpreting analytical outputs. This process helped determine whether the interface design, terminology, menus, and system controls were intuitive and appropriate for users with different levels of technical experience.

Feedback was collected through direct observation, interviews, discussions, and evaluation forms. Participants provided comments and recommendations regarding the system’s interface design, dashboard arrangement, report format, feature usefulness, accessibility, and overall performance. The researchers documented user feedback to identify areas that required improvement before final deployment.

The researchers also evaluated whether the analytical outputs generated by the system were meaningful and useful for business monitoring and decision-making. Users assessed the clarity and relevance of revenue summaries, sales trends, branch comparisons, and product performance reports. This evaluation confirmed whether the platform successfully transformed business data into understandable information that could assist management in evaluating performance and planning future strategies.

Based on the feedback gathered during UAT, necessary improvements and refinements were implemented. Adjustments were made to improve dashboard organization, report presentation, navigation flow, and overall system usability. Each modification was reviewed and tested again to ensure that improvements addressed user concerns without affecting existing system functionality.

The researchers also determined the level of training and support required for effective system adoption. Based on user interactions during testing, recommendations were developed regarding user manuals, instructional materials, and orientation sessions to help employees become familiar with the platform. These preparations supported smoother implementation and encouraged effective long-term system utilization.

The completion of User Acceptance Testing confirmed that the BREWINSIGHT BI: Revenue Pattern Analytics System was acceptable, practical, and suitable for deployment within Kape Tubong. The involvement of actual users ensured that the final system reflected the organization’s operational practices, user expectations, and business requirements.

Testing Conclusion

The implementation of Unit Testing, System Testing, and User Acceptance Testing provided a comprehensive evaluation of the BREWINSIGHT BI: Revenue Pattern Analytics System. Each testing phase contributed to verifying different aspects of the platform, including module functionality, system integration, performance reliability, security, accuracy of analytical outputs, and usability.

Unit Testing confirmed that individual system components operated correctly and consistently before integration. System Testing verified that all modules functioned together as a complete platform and maintained accurate communication between the user interface, application logic, database, analytics engine, and reporting features. User Acceptance Testing validated that the system was practical, understandable, and capable of supporting the actual operational needs of Kape Tubong.

The testing procedures also confirmed that the platform effectively addressed the identified challenges related to sales monitoring, revenue analysis, report generation, branch performance comparison, and business information management. Through continuous evaluation, correction, and refinement, the researchers ensured that the system achieved the required standards for functionality, reliability, security, and usability.

The results of the testing process demonstrated that the BREWINSIGHT BI: Revenue Pattern Analytics System was capable of producing accurate business intelligence outputs, maintaining

stable performance, protecting organizational data, and providing a user-friendly environment for management and staff. The documentation of testing activities, identified issues, corrective actions, and evaluation results also established a foundation for future maintenance, improvements, and system expansion.

Overall, the successful completion of the testing procedures confirmed that the BREWINSIGHT BI platform was technically reliable, operationally effective, and ready for implementation within Kape Tubong. The comprehensive evaluation ensured that the developed system could support revenue analysis, performance monitoring, and informed decision-making while providing long-term value to the organization.

3.7 Evaluation Criteria

The developed BREWINSIGHT BI: Revenue Pattern Analytics System was evaluated using a set of criteria that determined its effectiveness, quality, and suitability for implementation within Kape Tubong’s operational environment. The evaluation process aimed to verify whether the system achieved its intended objectives and satisfied the requirements established during the planning, analysis, and development phases.

The selected evaluation criteria were based on software quality characteristics relevant to business intelligence systems. These criteria focused on the system’s ability to support business operations, improve decision-making processes, manage organizational data, and provide accurate analytical information

for Kape Tubong.

The evaluation considered the specific operational requirements of the organization, including sales monitoring, transaction tracking, revenue analysis, inventory-related information, and reporting activities. It also considered the characteristics of the coffee shop industry, such as changing customer preferences, product performance variations, seasonal sales trends, and revenue patterns that influence business decisions.

Through the evaluation process, the researchers were able to determine the strengths, limitations, and areas for improvement of the developed platform. The assessment verified whether BREWINSIGHT BI was functional, reliable, efficient, secure, and convenient for its intended users. It also examined the system's ability to process business information, generate meaningful insights, and support management decisions.

The evaluation framework included the following criteria: functionality, usability, reliability, efficiency, and security. These criteria provided a structured basis for measuring the quality and readiness of BREWINSIGHT BI before its implementation in the organization.

Functionality refers to the ability of a system to perform its intended tasks accurately, completely, and according to specified requirements. This criterion measures whether the system provides the necessary features and services required to achieve its intended purpose.

The functionality of BREWINSIGHT BI was evaluated based on its

ability to support Kape Tubong's sales monitoring, revenue analysis, reporting, and decision-making processes. The researchers assessed whether the system's major components, including interactive dashboards, sales tracking, branch performance comparison, revenue trend analysis, product performance monitoring, and report generation, operated properly according to the defined requirements.

The researchers examined the accuracy of information generated by the system, including revenue calculations, sales summaries, product analysis, and comparative reports. The evaluation verified whether the platform produced correct and consistent results based on the transaction records and business data stored within the database.

The system's ability to perform analytical operations was also assessed. These operations included applying filters, selecting date ranges, comparing branch performance, grouping product categories, and generating customized reports. The researchers evaluated whether these functions produced complete and accurate outputs under different conditions.

Functional completeness was evaluated by verifying whether all required features identified during the planning and design stages were successfully implemented. Each module was examined to determine whether it performed its intended function and contributed to the overall purpose of the system.

The researchers also assessed the system's data integration capability. Information gathered from different branches and operational sources was evaluated to determine whether it could

be consolidated into a centralized platform while maintaining accuracy, consistency, and proper organization.

Automated functions such as sales consolidation, dashboard updating, report generation, and performance calculations were included in the evaluation. These features were assessed based on their ability to reduce manual processes, minimize errors, and improve the efficiency of business analysis activities.

The user access functions of the system were also evaluated to verify whether authentication and role-based permissions operated correctly. The researchers examined whether administrators, managers, and staff members could access only the information and features appropriate to their assigned responsibilities.

The reporting capabilities of BREWINSIGHT BI were assessed based on the accuracy, clarity, and usefulness of generated dashboards, charts, tables, and analytical summaries. The evaluation determined whether these outputs provided valuable information that supported business monitoring and decision-making.

The researchers also considered the system’s ability to support future growth. The platform was evaluated to determine whether it could accommodate increasing transaction records, additional users, and future analytical requirements without reducing system performance.

The functionality evaluation determined whether BREWINSIGHT BI successfully provided the required business intelligence features and accurately supported Kape Tubong’s operational and managerial needs.

Usability refers to the ease, efficiency, and convenience experienced by users when learning, navigating, and operating a system. This criterion measures whether the system provides an interface that is understandable, accessible, and user-friendly for individuals with different levels of technical knowledge and experience.

The usability of BREWINSIGHT BI was evaluated based on how easily users could access system features, locate required information, interpret analytical results, and complete necessary tasks without excessive difficulty. The researchers assessed different interface elements, including menu organization, dashboard layout, labels, navigation buttons, charts, graphs, and filtering options, to determine whether they supported effective user interaction.

The clarity and organization of the system interface were examined during the evaluation. The researchers assessed whether information displayed through dashboards, tables, graphs, and summaries was presented in a clear and understandable manner. A properly organized interface helped users interpret business information efficiently and reduced confusion during system operation.

The learning process of first-time users was also evaluated. The researchers determined how quickly users became familiar with the system features and completed basic tasks without requiring extensive assistance. A simple and understandable interface reduced the time needed for training and supported easier adoption of the platform.

Navigation efficiency was assessed by examining the number of steps required to perform common activities, such as generating reports, viewing sales information, applying filters, and comparing business performance. The researchers evaluated whether users could complete these tasks smoothly without unnecessary procedures.

The consistency of the system interface was included in the usability assessment. The researchers examined whether similar functions, buttons, menus, and responses followed consistent patterns throughout the platform. Consistent design helped users develop familiarity with the system and improved interaction efficiency.

The readability and accessibility of displayed information were also evaluated. Factors such as text arrangement, visual presentation, color usage, and organization of analytical data were examined to determine whether users could understand information comfortably and accurately.

The researchers assessed the effectiveness of system feedback and error-handling features. Notifications, validation messages, and system responses were evaluated based on whether they helped users understand successful actions and correct mistakes during operation.

The ability of BREWINSIGHT BI to improve user productivity was also considered. Features such as automated reports, data filtering, organized dashboards, and simplified workflows were assessed to determine whether they reduced manual effort and

allowed users to complete tasks more efficiently.

The adaptability of the system for different user roles was evaluated. The researchers examined whether managers, administrators, and staff members could access appropriate features based on their responsibilities without being presented with unnecessary information or functions.

The visual design of the platform was also considered during the evaluation. The use of colors, spacing, icons, typography, and graphical elements was assessed to determine whether the system provided a professional and comfortable user experience while maintaining functionality.

The researchers evaluated the system's ability to support users in recovering from mistakes. Features such as confirmation prompts, validation controls, and editable inputs were examined to determine whether they helped prevent errors and maintain smooth workflow continuity.

The usability evaluation determined whether BREWINSIGHT BI provided a convenient, understandable, and efficient platform that users could confidently operate in supporting Kape Tubong's daily activities and decision-making processes.

Reliability refers to the ability of a system to perform consistently, accurately, and dependably under expected operating conditions. This criterion measures whether the system can maintain stable performance, produce correct results, and continue functioning properly during regular usage.

The reliability of BREWINSIGHT BI was evaluated based on its

ability to operate without frequent failures, interruptions, or processing errors. The researchers tested important system functions, including dashboard loading, report generation, data retrieval, and filtering operations, to determine whether the platform maintained consistent performance during repeated usage.

The researchers assessed the system's capability to handle different levels of workload. The evaluation examined whether BREWINSIGHT BI remained responsive and stable while processing increasing amounts of transaction data, generating analytical reports, and supporting multiple user activities.

The accuracy and consistency of generated outputs were also evaluated. The researchers verified whether repeated execution of similar processes produced the same results when using identical data inputs and conditions. Revenue summaries, sales reports, product performance analysis, and branch comparisons were examined to ensure that the information provided by the system remained reliable.

The system's ability to respond properly to unexpected situations was assessed as part of the reliability evaluation. The researchers examined how the platform handled invalid inputs, incomplete records, and minor operational issues without causing system failure, data corruption, or interruption of important processes.

Database integrity was also evaluated to determine whether stored information remained complete, accurate, and consistent. The researchers assessed whether sales records, transaction histories, and analytical data were properly maintained without

duplication, loss, or unauthorized modification.

The availability of the system was included in the assessment. The researchers evaluated whether users could access dashboards, reports, and analytical features whenever needed with minimal downtime or service interruption.

The capability of BREWINSIGHT BI to support multiple users simultaneously was also examined. The researchers tested whether concurrent user activities affected system speed, stability, accuracy, or data consistency during operation.

Data synchronization processes were evaluated to determine whether information from different sources was updated and reflected accurately within the centralized system. The researchers examined whether updated records appeared correctly and within an acceptable period to support timely decision-making.

Backup and recovery procedures were also assessed to determine whether important business information could be protected and restored after unexpected failures, technical issues, or accidental data loss. The effectiveness of these procedures contributed to maintaining continuous system operation and protecting valuable organizational records.

The reliability evaluation determined whether BREWINSIGHT BI functioned as a stable and dependable business intelligence platform capable of supporting accurate reporting, continuous operations, and informed decision-making within Kape Tubong.

Efficiency refers to the ability of a system to perform its

required functions within an acceptable amount of time while effectively utilizing available resources. This criterion measures how quickly and effectively BREWINSIGHT BI processes information, retrieves data, generates reports, and presents analytical results to users.

The efficiency of BREWINSIGHT BI was evaluated based on the system's response time during common operations, including dashboard loading, report generation, data filtering, and retrieval of sales information. The researchers assessed whether the platform provided fast and smooth performance that allowed users to access important business insights without unnecessary delays.

The researchers evaluated the optimization of database operations and data processing procedures. The assessment determined whether the system could manage increasing amounts of transaction records while maintaining acceptable performance. Efficient data processing was necessary to ensure that the platform remained useful as Kape Tubong's business information continued to expand.

The system's data synchronization and updating processes were also examined. The researchers assessed whether sales information from different sources could be processed and reflected accurately within the dashboard within a reasonable period. Efficient data updating supported timely monitoring and improved business decision-making.

The performance of analytical operations was evaluated through functions such as revenue computation, trend analysis, product comparison, and report generation. The researchers determined

whether these processes were completed efficiently without excessive delays, system slowdown, or interruptions.

The responsiveness of the user interface was also assessed. Navigation menus, dashboards, charts, filters, and reporting tools were evaluated based on how quickly they responded to user commands. A responsive interface improved user productivity and provided a smoother interaction experience.

The researchers examined the system's capability to support multiple users performing different activities at the same time. The evaluation determined whether simultaneous access affected system speed, stability, or the accuracy of generated results.

System initialization and loading performance were included in the evaluation. The researchers assessed the time required for user authentication, dashboard preparation, and retrieval of relevant information to determine whether users could access the system efficiently.

The efficiency of data storage and retrieval mechanisms was also evaluated. The researchers examined whether proper database organization and processing methods allowed users to search, retrieve, and analyze information quickly even as the volume of stored data increased.

Automated processes, including report generation, sales consolidation, and dashboard updates, were assessed based on their ability to reduce manual workload and improve operational efficiency. These automated functions supported faster analysis and allowed users to focus on important business activities.

The researchers also evaluated the system's resource utilization. The assessment considered whether BREWINSIGHT BI operated effectively while maintaining reasonable use of computing resources required for processing, storage, and system maintenance.

The efficiency evaluation determined whether BREWINSIGHT BI provided fast, responsive, and resource-effective performance while supporting Kape Tubong's analytical requirements and daily operational activities.

Security was considered one of the most important evaluation criteria because it determined the capability of BREWINSIGHT BI to protect Kape Tubong's confidential business information and maintain safe system operations. The evaluation assessed whether the platform properly implemented security measures, including user authentication, authorization, access control, data protection, and prevention of unauthorized activities. The researchers verified that sensitive information such as sales records, financial data, transaction histories, and operational reports remained protected from unauthorized access, modification, or loss.

The assessment results showed that the system maintained a secure environment through controlled user permissions, reliable authentication procedures, and proper management of user access levels. The platform ensured that administrators, managers, and staff members could only access features and information relevant to their assigned responsibilities. These security controls reduced the possibility of unauthorized data viewing, accidental

changes, and misuse of system resources.

The evaluation also confirmed that BREWINSIGHT BI maintained data integrity and confidentiality throughout its operations. Security mechanisms such as input validation, secure data handling procedures, activity monitoring, and backup processes helped preserve the accuracy and reliability of stored information. The system was able to prevent common operational risks that could affect business records and ensured that important data remained available for analysis and decision-making.

Security testing further demonstrated that the platform remained stable and protected even when multiple users accessed the system at the same time. The implementation of role-based permissions, controlled transactions, and proper data management practices helped prevent conflicts and unauthorized modifications during concurrent usage. These measures strengthened the reliability of the platform as a centralized business intelligence solution for Kape Tubong.

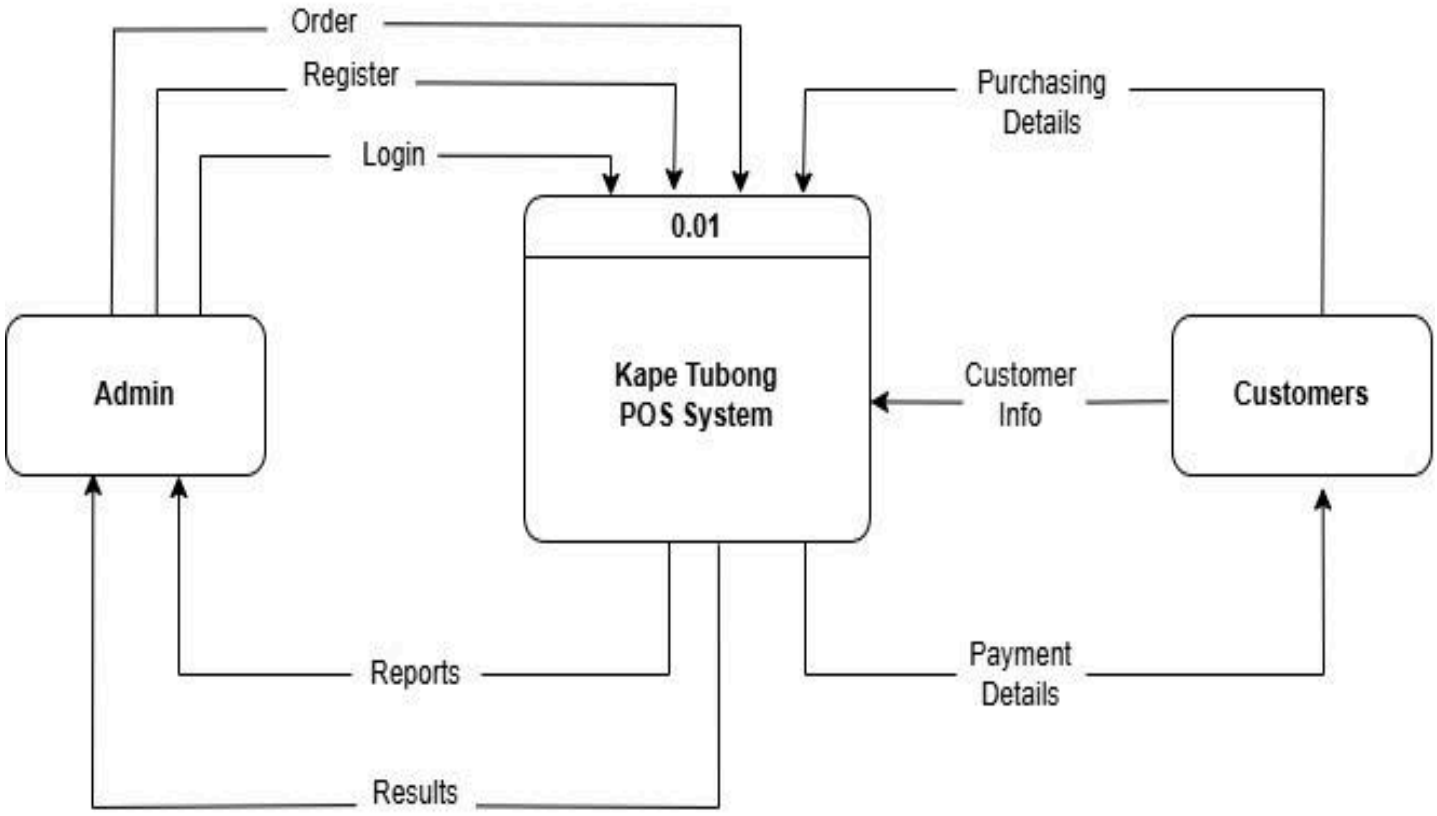
The evaluation results provided a clear basis for determining the effectiveness, quality, and readiness of the developed BREWINSIGHT BI: Revenue Pattern Analytics System. Through the assessment of functionality, usability, reliability, efficiency, and security, the researchers verified that the platform satisfied the identified requirements and supported the operational needs of Kape Tubong. The findings showed that the system was capable of providing accurate business insights, improving data management processes, and assisting management in

making informed decisions. The evaluation also identified opportunities for future improvements, ensuring that the system remained adaptable, useful, and sustainable for the organization’s changing needs.

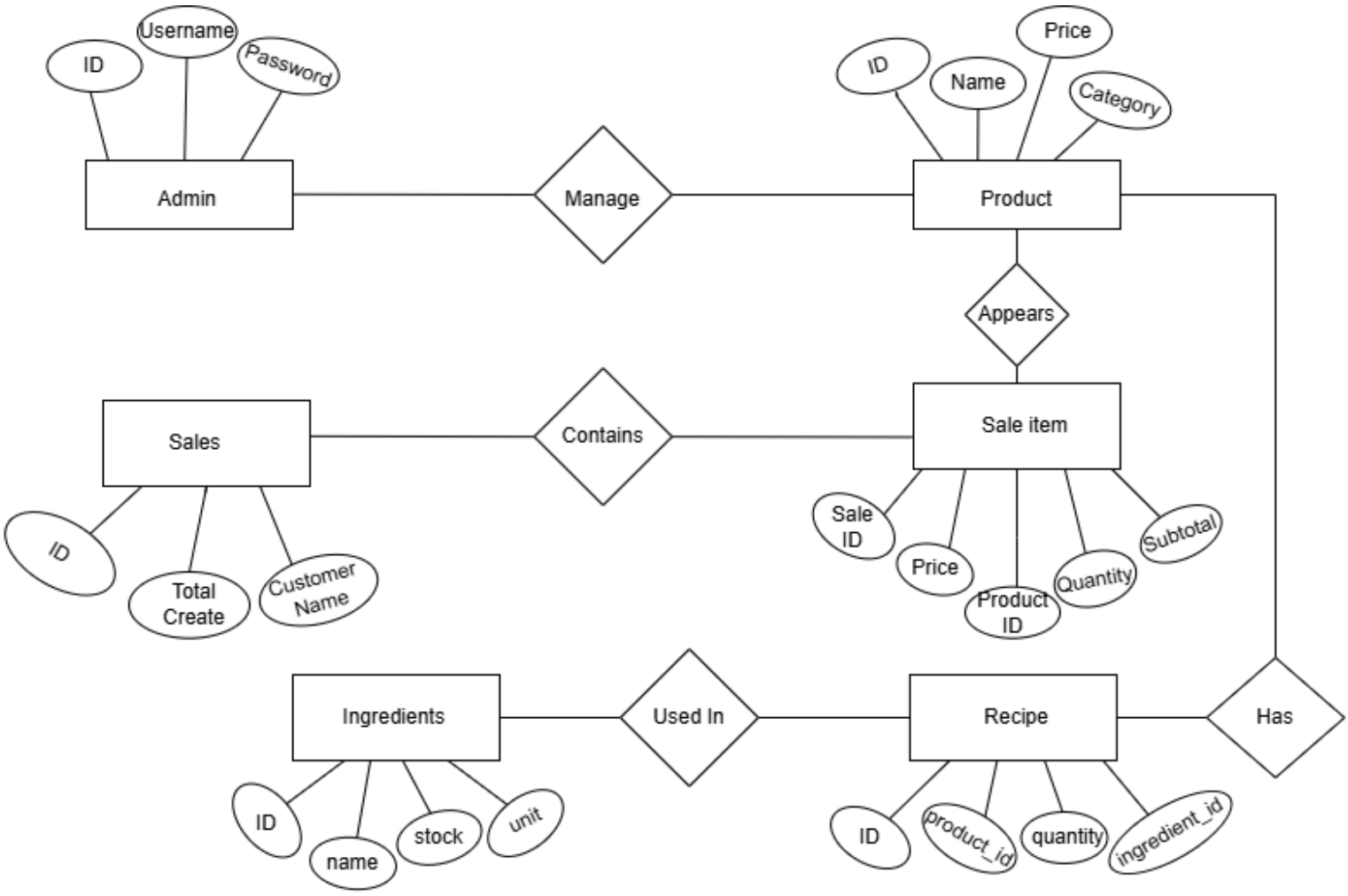
SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 0)

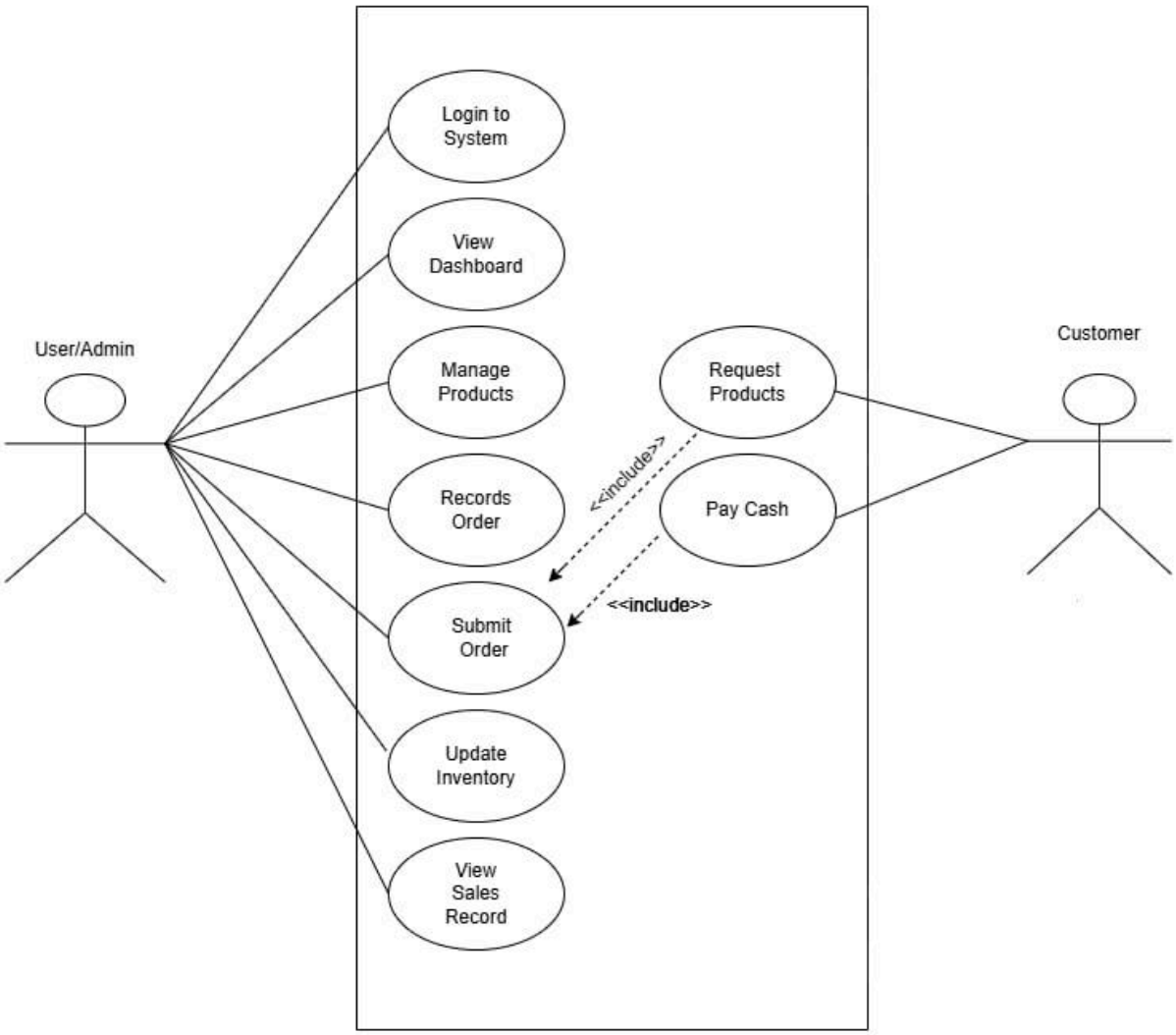
Admin → Pos System→ Customers



Entity Relationship Diagram (ERD)



Use Case Diagram



CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION,TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The Kape Tubong POS System was developed as a web-based Point-of-Sale (POS) and inventory management system to improve the daily operations of Kape Tubong. It was designed to replace the manual process of recording sales, managing products, monitoring ingredient inventory, and preparing reports. The system provided a faster and more organized way of handling business transactions while reducing errors in recording and inventory monitoring.

Before the system was developed, sales transactions and inventory updates were recorded manually. This process took more time and sometimes resulted in inaccurate records, delayed inventory updates, and difficulties in monitoring product sales and available ingredients. Preparing reports also required checking transaction records one by one, making the process less efficient.

The developed system solved these problems by providing a secure login for administrators and users. After entering a valid

username and password, users were directed to the dashboard, where they could access the different features of the system. If incorrect login credentials were entered, the system displayed a login failed message and asked the user to try again.

The dashboard served as the main page of the system and allowed users to perform the different business operations.

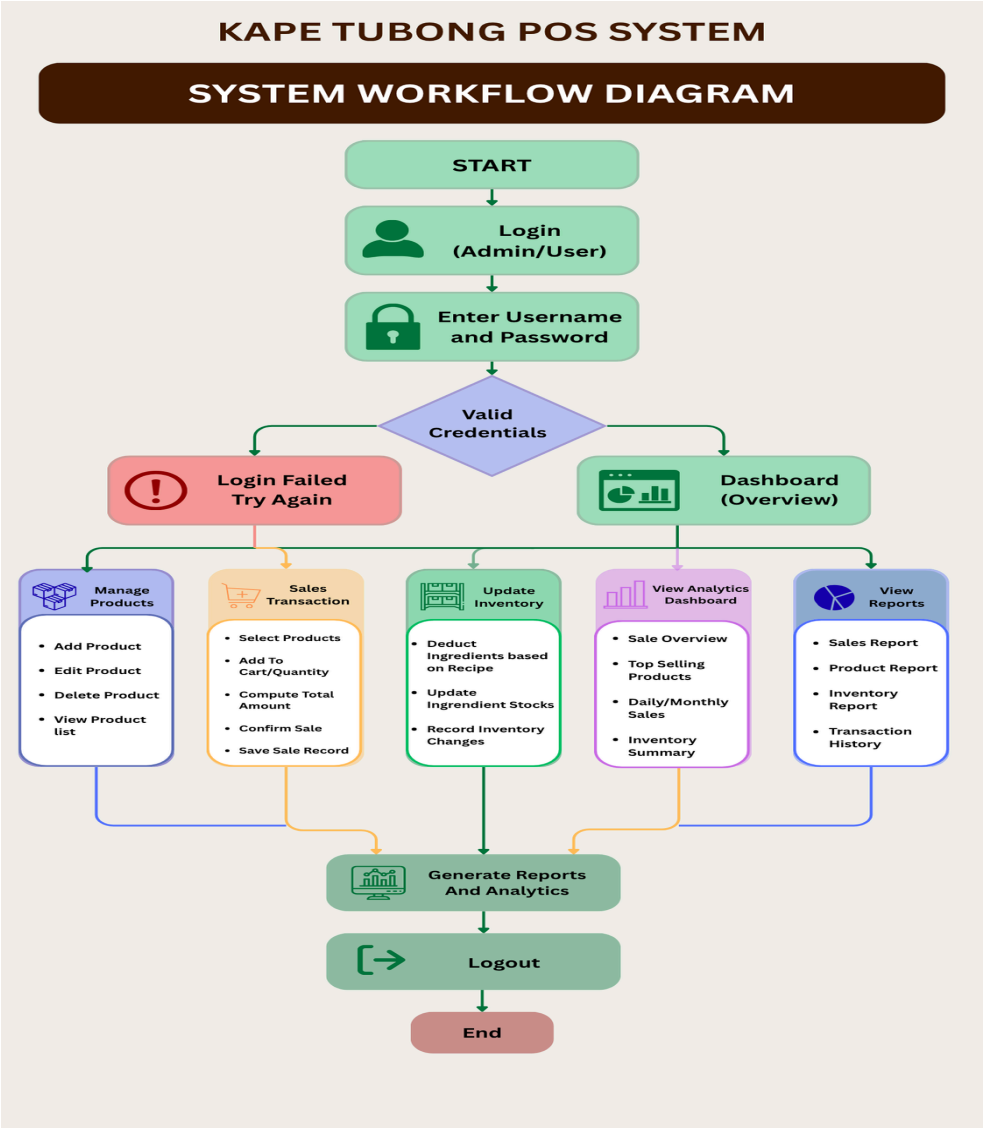
The system included the following modules:

- User Authentication, for verifying the username and password of administrators and users before granting access to the system.
- Dashboard, for displaying an overview of sales, inventory, and other important business information.
- Product Management, for adding, editing, deleting, and viewing available products.
- Sales Transaction, for selecting products, adding items to the cart, calculating the total amount, confirming the sale, and saving transaction records.
- Inventory Management, for automatically deducting ingredient stocks based on the product recipe, updating inventory levels, and recording inventory changes after every sale.
- Analytics Dashboard, for showing the sales overview, top-selling products, daily and monthly sales, and inventory summary.
- Reports Module, for generating sales reports, product reports, inventory reports, and transaction history.

All of these modules worked together to make the system function properly. Every time a sale was completed, the transaction was

saved automatically, the ingredient inventory was updated, and the necessary reports were generated. Because of this, the business operations became more organized, record-keeping became more accurate, and less time was spent on manual tasks. The system also made it easier for the business owner to monitor sales, inventory, and overall business performance.

Figure 4.1 illustrates the overall workflow of the developed system.



The workflow connected the administrator and users through a centralized system, allowing product management, sales transactions, inventory monitoring, report generation, and sales analytics to be completed in one platform. This process made the daily operations of Kape Tubong more organized, efficient, and accurate by reducing manual work and improving record management.

4.3 System Development Methodology

The development of the Kape Tubong Point of Sale (POS) System adopted the Agile Software Development Methodology. This methodology provided an iterative and flexible approach that allowed the researchers to develop the system in small and manageable phases while continuously evaluating its progress. It also enabled the researchers to incorporate feedback from the business owner throughout the development process, ensuring that the system effectively addressed the operational needs of Kape Tubong.

The Agile methodology divided the project into several development phases. Each phase focused on a specific activity that contributed to the successful completion of the system.

Product Backlog

The researchers prepared the Product Backlog by listing all the features, system requirements, and priorities needed for the Kape Tubong POS System. The backlog included order taking, inventory management, sales reporting, customer transactions, and revenue

analytics. The features were prioritized according to their importance to the business.

Planning

During the Planning phase, the researchers identified the features that would be developed during the current sprint. The development tasks were organized into manageable activities, while the project schedule and objectives were established to guide the implementation of the system.

Requirements Analysis

The researchers analyzed the selected features to determine their functional requirements, business rules, workflows, and system specifications. This phase helped ensure that every feature was clearly defined before development began.

Design

During the Design phase, the researchers designed the user interface, database structure, and overall system workflow. The layout of the application was planned to provide a simple and user-friendly interface while ensuring that transaction records, inventory information, and sales data were properly organized within the database.

Development

The researchers developed the approved system modules based on the completed design and requirements. The coding process

included the implementation of order processing, inventory management, sales recording, user account management, and revenue analytics. Each module was completed incrementally following the Agile sprint plan.

Testing

After the development of each module, the researchers conducted testing to verify that every feature functioned correctly. The system was tested to ensure the accuracy of sales calculations, inventory updates, report generation, transaction processing, and user authentication. Errors that were identified during testing were corrected before the next phase.

Review and Feedback

After completing each sprint, the researchers presented the developed modules to the business owner for evaluation. Comments, suggestions, and recommendations were collected and were used to improve the functionality and usability of the system before deployment.

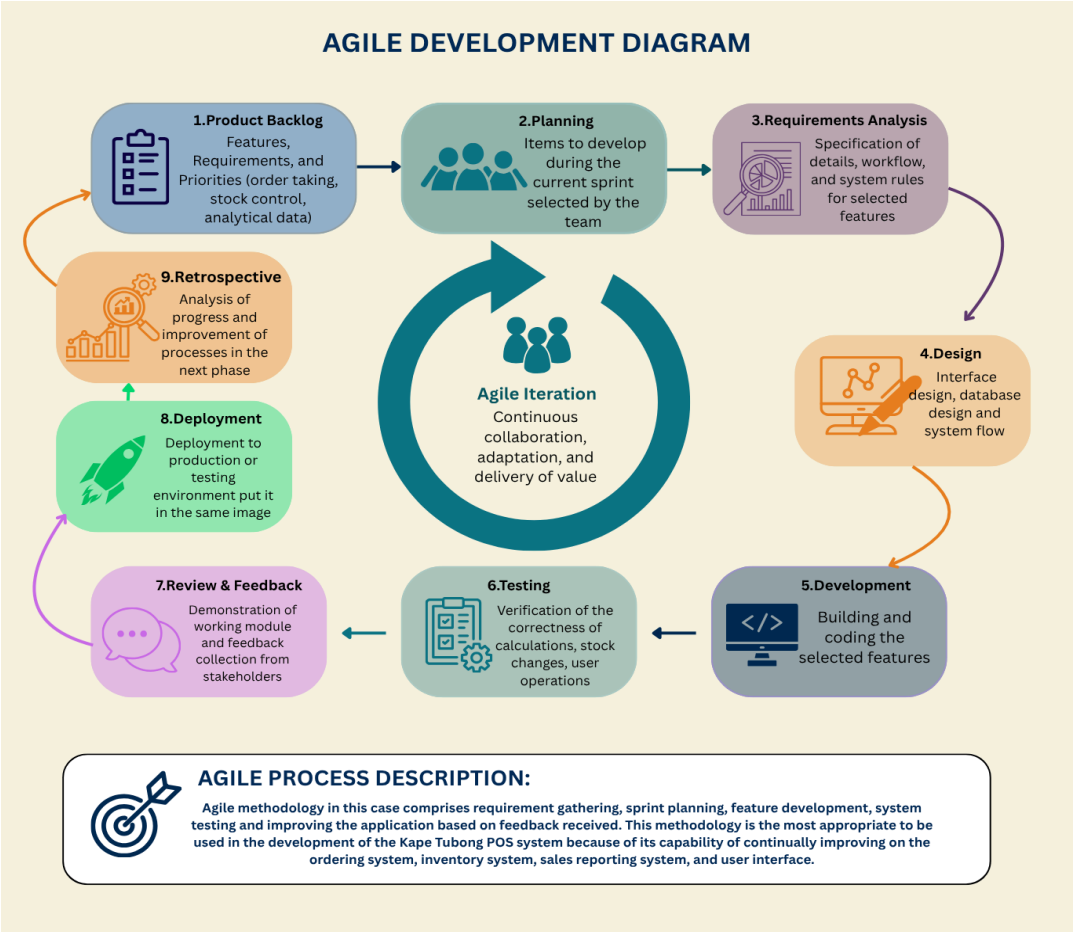
Deployment

Once the developed modules passed the required testing and evaluation, the researchers deployed the system to the intended environment at Kape Tubong. The application was installed, the database was configured, and the system was verified to ensure that all features operated properly before implementation.

Retrospective

After deployment, the researchers evaluated the completed sprint by reviewing the overall development process. The challenges encountered were discussed, successful practices were identified, and areas for improvement were determined. The lessons learned were applied to improve future development activities and maintain the quality of the Kape Tubong POS System.

Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.



Agile Process Description

The Agile process was a software development approach that focused on developing the system through small and manageable iterations called sprints. Rather than completing the entire application in a single development cycle, the researchers implemented the system features incrementally while continuously gathering feedback from stakeholders. This approach enabled the researchers to identify issues at an early stage, implement necessary improvements, and verify that the developed system met the operational requirements of Kape Tubong.

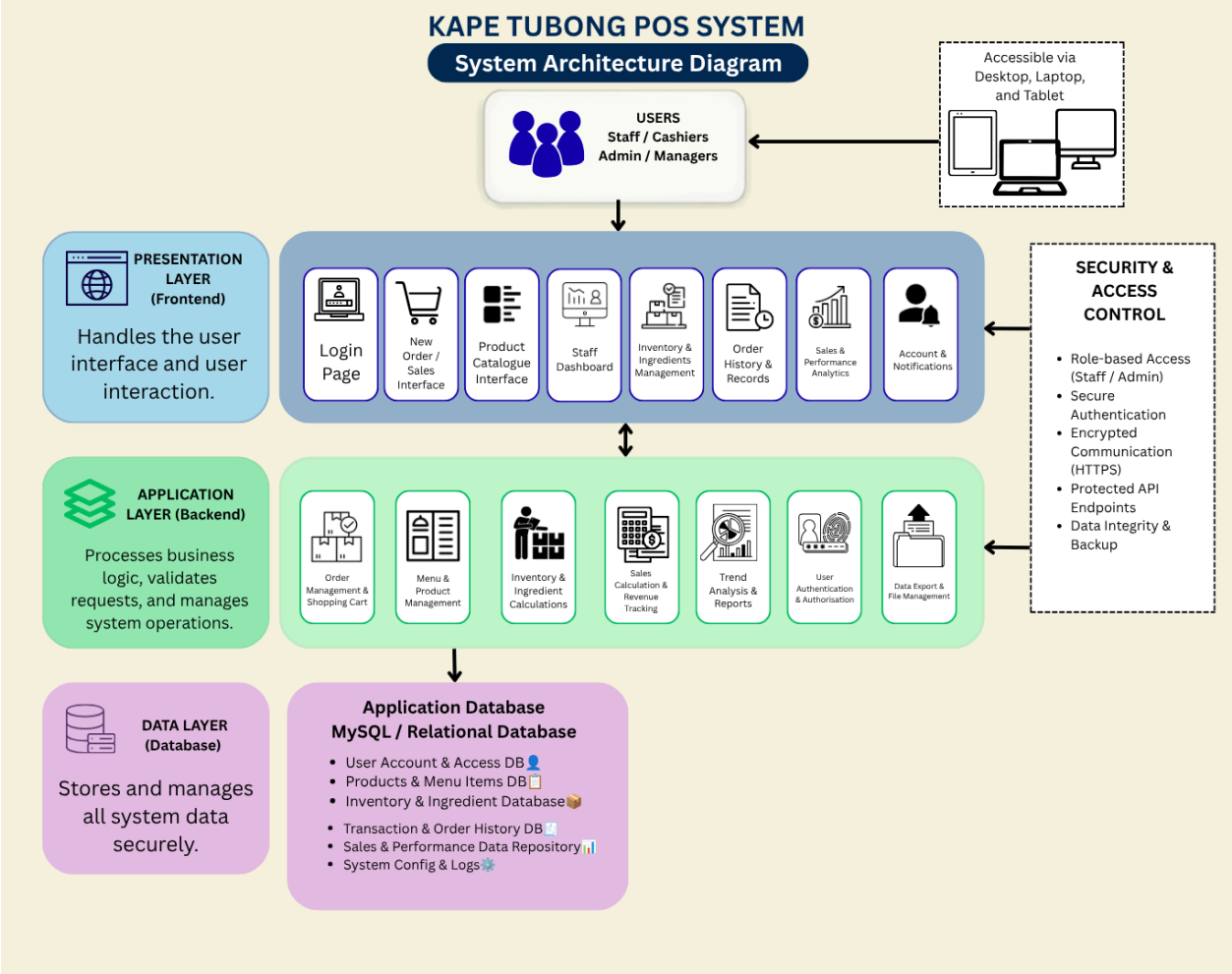
For the development of the Kape Tubong POS System, the Agile methodology was selected because it provided flexibility in accommodating changes in business requirements while promoting continuous collaboration and system improvement. Throughout the development process, the researchers prepared the product backlog, planned each sprint, analyzed the system requirements, designed the application, developed the system modules, conducted testing, gathered stakeholder feedback, deployed the completed features, and evaluated each sprint through retrospectives. By following this methodology, the researchers developed a reliable, efficient, and user-friendly Point of Sale System that supported order processing, inventory management, sales reporting, and revenue analytics for Kape Tubong.

4.4 System Architecture

The developed system followed a three-layer architecture that separated the presentation, application, and data layers into

distinct components. This architectural design allowed each layer to perform its assigned responsibilities independently while communicating with the other layers when necessary. The separation of functions improved system organization, simplified maintenance, enhanced security, and allowed future modifications to be implemented without significantly affecting the other layers. It also provided a structured framework that supported the efficient processing, storage, and presentation of sales information.

Figure 4.3 illustrates the architecture of the developed system.



The architecture consists of the following components:

Presentation Layer

The Presentation Layer served as the part of the system that users directly interacted with. It consisted of the visible interfaces that allowed staff, cashiers, and administrators to access the system and perform their assigned tasks. This layer displayed information retrieved from the lower layers while collecting user inputs such as sales transactions, inventory updates, and report requests. The collected information was then forwarded to the Application Layer for processing. The interface was developed using a responsive design to ensure that the system functioned properly on tablets and smartphones used during daily business operations. It could also be accessed through desktop and laptop computers when needed.

The major interface components included:

- Login Page
- New Order / Sales Interface
- Product Catalogue Interface
- Staff Dashboard
- Inventory and Ingredients Management Interface
- Order History and Records Interface
- Sales and Performance Analytics Interface
- Account and Notifications Interface

Application Layer

The Application Layer served as the core processing component of

the system. It received requests from the Presentation Layer, validated user inputs, applied the necessary business rules, and communicated with the Data Layer to retrieve or store information. This layer processed customer orders, calculated transaction totals, updated inventory records, generated sales reports, analyzed revenue patterns, and managed user authentication and authorization. It also ensured that the information exchanged between the user interface and the database remained accurate, consistent, and reliable throughout the system's operation.

The major functional components included:

- Order Management and Shopping Cart
- Menu and Product Management
- Inventory and Ingredient Usage Calculations
- Sales Calculation and Revenue Tracking
- Trend Analysis and Report Generation
- User Authentication and Authorization
- Data Export and File Management

Data Layer

The Data Layer managed the storage, organization, and retrieval of all system information using a MySQL relational database. It securely stored user accounts, product information, inventory records, sales transactions, and other business data required by the system. This layer maintained data integrity by processing database requests efficiently and ensuring that only authorized users could access stored information. It also maintained system

configuration records and activity logs that supported system monitoring and troubleshooting.

The key data storage components included:

- User Accounts and Access Database
- Products and Menu Items Database
- Inventory and Ingredients Database
- Transaction and Order History Database
- Sales and Performance Data Repository
- System Configuration and Logs

4.5 Hardware and Software Requirements

This section presented the hardware and software requirements used in the development and implementation of the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong. These requirements ensured that the system operated properly during development, testing, and actual implementation. The hardware specifications provided the necessary resources for smooth system performance, while the software requirements identified the tools and technologies used to develop, test, and maintain the system.

4.5.1 Hardware Requirements

Table 4.1 presented the minimum and recommended hardware specifications used in the development and implementation of the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong. These specifications ensured that the system performed efficiently during data processing, report generation, and

dashboard visualization.

Table 4.1 Hardware Requirements

Components	Minimum Specification	Recommended Specification
Processor	Quad-core 1.8 GHz	Octa-core 2.0 GHz or Higher
Memory (RAM)	3 GB	4 GB or higher
Storage	32 GB	64 GB or higher
Display	8-inch HD (1280x800)	10-inch Full HD (1920x1080)
Internet Connection	10 Mbps	25 Mbps or higher
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provided sufficient processing capability to execute analytical operations, generate reports, and display graphical dashboards without significant delays. It also improved the overall user experience by reducing loading times and enhancing system responsiveness.

4.5.2 Software Requirements

Table 4.2 presented the software requirements used in the development and implementation of the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong. These software tools were utilized for system development, database management, user interface design, testing, data visualization, and version control.

Table 4.2 Software Requirements

Software	Purpose
Window 11	Operating System
Visual Studio Code	Sources Code Editor
Python 3.11	Backend Programming Language
Flash	Web Application Framework
MySQL	Database Management System
HTML,CSS,and JavaScript	Frontend Development
Bootstrap 5	Responsive User Interface Development

Chart.js	Data Visualization and Analytics Charts
Google Chrome	Web Browser and System Testing
Git	Version Control
XAMPP	Local Development Environment (Apache and MySQL)

The software tools were selected because of their reliability, compatibility, and suitability for web application development. They were used for developing the system, managing the database, creating the user interface, testing the application, and supporting version control throughout the development process.

4.6 Database Design

The database for the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong was designed using a relational database structure composed of six interconnected tables. Each table represented an important entity within the system and worked together to manage sales transactions, products, inventory, and user information. The database design followed the principles of relational database management to minimize data redundancy, maintain data integrity, and improve the efficiency of storing and retrieving business records.

At the center of the database design was the users table, which stored the account information of system users, including staff and administrators. This table was connected to the sales table through the user_id foreign key, allowing every recorded transaction to be associated with the user who processed the sale. This relationship made it possible to monitor sales activities and maintain accountability for every transaction recorded in the system.

The sales table stored the details of each completed transaction, including the total amount, payment method, sale date, and the user responsible for the transaction. It was linked to the sale_items table through the sale_id foreign key, establishing a one-to-many relationship where a single sales transaction could contain multiple purchased products. This structure allowed the system to record complete transaction details while maintaining an organized database.

The sale_items table served as the bridge between sales transactions and products. It stored the quantity, price, and subtotal of every product included in a transaction. Each record was connected to the products table through the product_id foreign key, ensuring that every item sold corresponded to an existing product in the database.

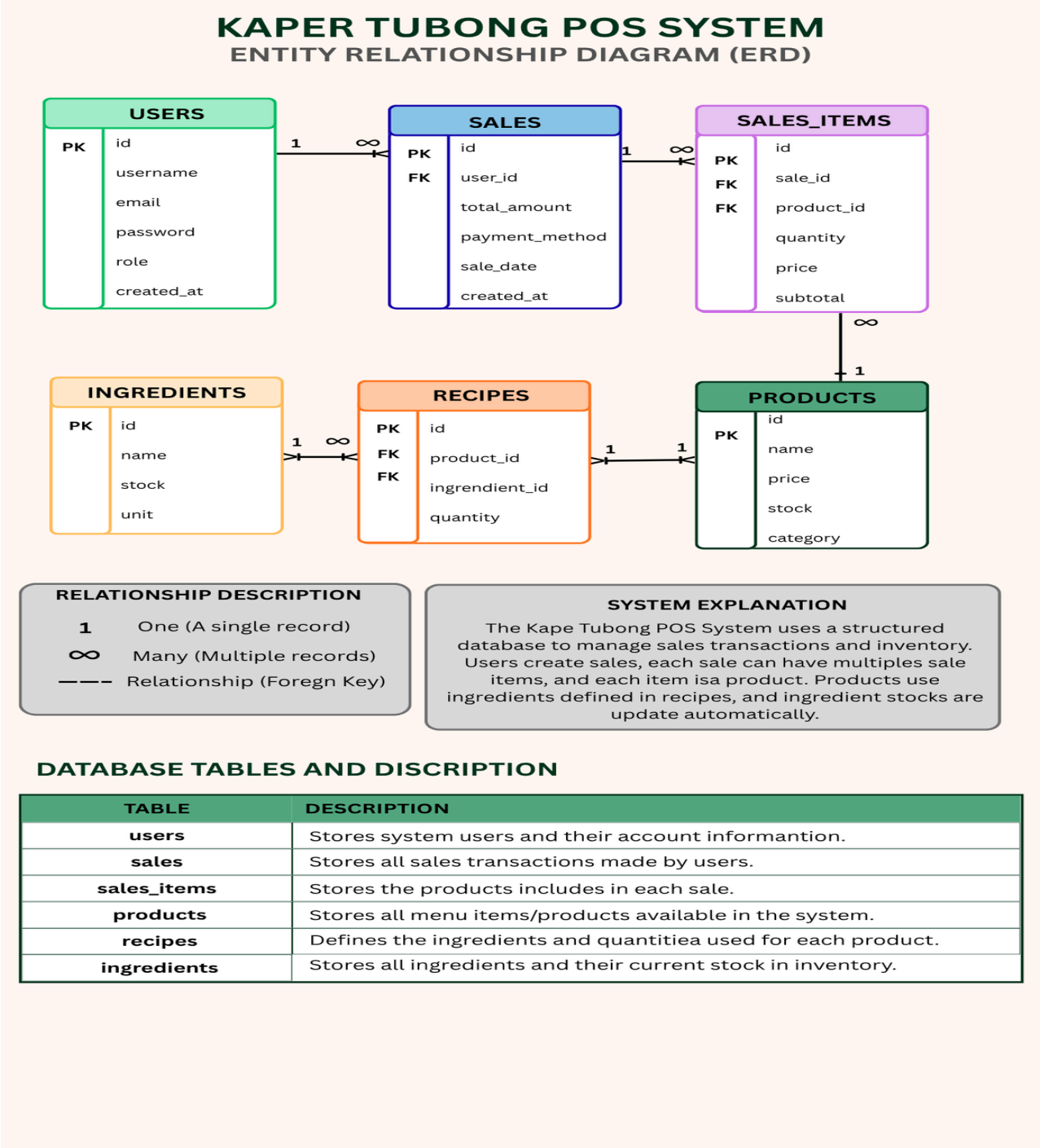
The products table stored all menu items offered by Kape Tubong, including their names, prices, stock levels, and product categories. This table was connected to the recipes table, which

defined the ingredients and corresponding quantities required for each product. The recipes table was also linked to the ingredients table through the ingredient_id foreign key. These relationships allowed the system to automatically deduct ingredient quantities from inventory whenever a product was sold, ensuring that stock records remained accurate and up to date.

The ingredients table maintained the inventory records of all raw materials used in preparing products. It stored information such as ingredient names, available stock, and units of measurement. By connecting the ingredients table to the recipes table, the system was able to monitor ingredient consumption and support inventory management during daily operations.

The database design established clear relationships among all six tables by using primary keys and foreign keys to maintain data consistency and integrity. This structure supported accurate transaction processing, inventory monitoring, sales reporting, and revenue analytics while providing a reliable foundation for the core functions of BrewInsight BI.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



DATABASE TABLES AND DISCRIPTION

TABLE	DESCRIPTION
users	Stores system users and their account informantion.
sales	Stores all sales transactions made by users.
sales_items	Stores the products includes in each sale.
products	Stores all menu items/products available in the system.
recipes	Defines the ingredients and quantitiea used for each product.
ingredients	Stores all ingredients and their current stock in inventory.

The primary entities of the database include:

- Users
- Sales
- Sale Items

- Products
- Recipes
- Ingredients

Users Table

This table stores account credentials and user information of all authorized users of the system.

Field	Description
id	Primary Key
Username	Username of the system user
Email	User email address
password	User password
Role	User role (Admin or Staff)
Create_at	Date the account was create

Sales Table

This table stored all sales transactions processed by the system.

Field	Description
id	Primary Key
user_id	Foreign Key to User
total_amount	Total amount of the transaction
payment_method	Payment method used
sale_date	Date of the sale
create_at	Date the transaction was recorded

Sales Items Table

This table stored the individual products included in each sales transaction.

Field	Description
id	Primary Key
sales_id	Foreign Key to Sales
product_id	Foreign Key to Products
quantity	Quantity of the product sold
price	Selling price of the product
subtotal	Total amount of the selected product

Products Table

This table stored all menu items available in the system.

Field	Description
Id	Primary Key
name	Product name
price	Selling price of the product
stock	Available stock quantity
Category	Product category

Recipes Table

This table stored the ingredients required for preparing each product.

Field	Description
id	Primary Key

product_id	Foreign key to Products
ingredients_id	Foreign Key to ingredients
quantity	Quantity of the ingredient required for the product

Ingredients Table

This table stored all ingredients used in preparing menu items.

Field	**Description**
id	Primary Key
name	Ingredient name
stock	Available stock quantity
unit	Unit of measurement (e.g., Grams, milliliters, pieces)

Database Summary

The system database was designed around six interconnected entities: Users, Sales, Sale Items, Products, Recipes, and Ingredients. These entities were connected through primary keys and foreign keys to maintain data integrity and establish clear relationships among system records. The database structure supported user management, sales transaction processing, product management, recipe management, and inventory monitoring. It also enabled the system to generate accurate reports, analyze sales performance, and automatically update ingredient inventory whenever a sales transaction was completed. This relational database design provided a reliable and organized foundation for the core functions of the BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the conclusion of the study entitled BrewInsight BI: Revenue Pattern Analytics System for Kape Tubong. It summarized the major findings obtained from the development, implementation, testing, and evaluation of the system. The findings were based on the results of system testing, user acceptance evaluation, and feedback provided by the business owner during the implementation of the developed system.

The study focused on developing a Business Intelligence-based system that improved the management of sales transactions, inventory records, and revenue analysis. Before the implementation of the developed system, Kape Tubong relied on traditional methods of monitoring sales and preparing reports, which required more time and effort. These manual processes also increased the possibility of inaccurate records, duplicate entries, and delays in generating business reports.

The developed system provided a centralized database that stored business information securely and organized sales, inventory, and product records in one platform. It also included interactive dashboards and analytical reports that presented revenue patterns, sales performance, product performance, and inventory

summaries. These features supported faster and more informed decision-making by allowing the business owner to monitor business performance using real-time information.

This chapter also presented the recommendations that may improve the system in the future. Possible enhancements, future research opportunities, and implementation strategies were discussed to support the continuous improvement of the developed system. The chapter also explained the contributions of the study to Kape Tubong, future researchers, developers, and other small businesses that intended to implement Business Intelligence solutions.

5.2 Summary of Findings

The findings of the study showed that BrewInsight BI successfully achieved the objectives of developing a Business Intelligence-based Revenue Pattern Analytics System for Kape Tubong. The developed system provided an organized platform for recording sales transactions, monitoring inventory, managing products, and generating analytical reports.

The implementation of the login module improved the security of the system by allowing only authorized users to access business information. Different user roles were managed through secure authentication, which helped protect confidential business records from unauthorized access.

The sales transaction module simplified the process of recording customer purchases. Users were able to select products, calculate the total amount automatically, confirm completed sales, and save transaction records into the database. This process reduced manual computations and minimized errors during daily business operations.

The product management module allowed administrators to add new products, update product information, delete unavailable products, and manage menu items efficiently. This feature helped maintain accurate product records and ensured that available products were properly displayed during sales transactions.

The inventory management module automatically updated ingredient quantities based on recorded sales transactions. The use of recipes connected products with their corresponding ingredients, allowing inventory levels to decrease automatically whenever a sale was completed. This feature improved inventory monitoring and reduced the possibility of stock inconsistencies.

The reporting and analytics module generated different business reports, including sales reports, inventory reports, transaction history, and product performance reports. These reports reduced the time required to prepare business documents manually and improved the accessibility of business information.

The Business Intelligence dashboard successfully transformed collected sales data into meaningful visual information. Revenue

trends, best-selling products, slow-moving products, sales summaries, and inventory information were presented through charts and graphical reports. These visualizations enabled the business owner to understand business performance more easily and identify areas that required improvement.

The developed database design provided a structured relationship among users, products, ingredients, recipes, sales, and sale items. The Entity Relationship Diagram ensured that business information was stored consistently while reducing data redundancy. The relationships between database tables improved the accuracy and reliability of stored information.

The system workflow also supported efficient business operations. From user authentication to transaction processing, inventory updates, report generation, and logout procedures, each function followed an organized process that minimized unnecessary steps during system operation.

The Agile software development methodology contributed to the successful completion of the project. The development process included product backlog preparation, sprint planning, requirement analysis, system design, coding, testing, deployment, user evaluation, and continuous improvement. This approach allowed system features to be improved based on user feedback received during development.

The evaluation results indicated that the developed system

functioned correctly according to its intended purpose. The business owner found the system easy to use, reliable, and appropriate for daily business operations. The generated reports and dashboards supported better monitoring of business performance and reduced the effort required in preparing sales reports manually.

The findings demonstrated that BrewInsight BI improved sales management, inventory monitoring, business reporting, and revenue analysis. The system also supported better decision-making by transforming sales records into useful business information through Business Intelligence techniques.

5.3 Conclusion

Based on the findings of the study, BrewInsight BI: Revenue Pattern Analytics System successfully addressed the problems encountered by Kape Tubong in managing sales transactions, inventory records, and revenue analysis.

The developed system provided a centralized platform that integrated user authentication, product management, inventory management, sales processing, report generation, and Business Intelligence analytics into one application. This integration improved the efficiency of daily business operations and reduced the dependence on manual record-keeping.

The implementation of the system improved the accuracy of sales

recording by storing transaction information directly in the database. Automatic inventory updates ensured that ingredient quantities reflected completed sales, reducing inventory monitoring errors and improving stock management.

The Business Intelligence dashboard provided valuable insights by presenting sales performance, revenue trends, product performance, and inventory summaries through graphical reports. These visual reports enabled the business owner to understand business performance quickly and make informed decisions regarding inventory planning, pricing strategies, promotional activities, and future business improvements.

The developed database structure improved data organization by maintaining proper relationships among business records. The system architecture also supported efficient communication between the presentation layer, application layer, and database layer, resulting in stable and reliable system performance.

The Agile development methodology contributed significantly to the successful development of the system. Continuous planning, testing, and user feedback allowed the developed application to satisfy the operational requirements of Kape Tubong while maintaining flexibility during development.

The evaluation results confirmed that the developed system was functional, user-friendly, reliable, and suitable for daily business operations. The business owner successfully used the

developed features to manage sales transactions, monitor inventory, review business reports, and analyze revenue patterns.

The objectives of the study were achieved because the system successfully automated important business processes while providing useful analytical information through Business Intelligence techniques. The implementation of BrewInsight BI demonstrated that Business Intelligence systems could improve operational efficiency, reduce manual work, and support better business decisions for small businesses.

5.4 Recommendations

Based on the findings and conclusions of the study, several recommendations were proposed to improve the developed system and support future development.

System Improvements

- The developed system may include an automated notification feature that alerts users whenever ingredient stocks reach minimum levels, unusual sales activities occur, or products become slow-moving. These notifications may assist business owners in responding immediately to operational concerns.
- Additional security measures may also be implemented by strengthening password policies, providing account recovery options, limiting failed login attempts, and enabling automatic logout after periods of inactivity. These improvements may increase the protection of confidential

business information.

- The reporting module may be enhanced by allowing reports to be exported in PDF, Excel, and CSV formats. These additional export options may improve documentation, report sharing, and business record keeping.
- Additional dashboard visualizations may also be developed to compare monthly revenue, seasonal sales performance, product categories, and inventory movement. These analytical features may provide deeper business insights and improve strategic planning.
- The system may also include employee activity logs that record user actions performed within the application. This feature may improve accountability and system monitoring.
- A customer management module may also be integrated to record customer information, purchasing history, and customer preferences. This information may support personalized marketing activities and customer relationship management.

Future Research

- Future researchers may incorporate predictive analytics techniques capable of estimating future sales based on historical transaction records, seasonal demand, and previous revenue patterns.
- Machine learning algorithms may also be explored to improve sales forecasting accuracy and identify hidden purchasing trends that are difficult to observe through descriptive

analytics alone.

- Future studies may evaluate the developed system over a longer implementation period to determine its long-term effects on business growth, operational efficiency, customer satisfaction, and revenue improvement.
- Researchers may also optimize the application for smartphones by improving mobile responsiveness, navigation, and user interface design. Since mobile devices are commonly used by small businesses, improved accessibility may increase system usability.
- Additional business modules such as supplier management, expense monitoring, payroll management, customer loyalty programs, online ordering, and customer feedback analysis may be integrated into future versions of the system.
- Future studies may also implement the developed system in other coffee shops or small businesses to evaluate its adaptability under different business environments.

Implementation Strategies

- Future users should receive proper orientation and training before operating the system to ensure accurate use of all available features.
- Regular database backups should be scheduled to protect business information from accidental data loss or system failures.
- System maintenance should be conducted periodically to correct software errors, improve performance, and update security features.
- Sales records and inventory information should be reviewed

regularly to maintain data accuracy and ensure that business reports remain reliable.

- A user manual containing screenshots and step-by-step instructions should be prepared to guide future users during system operation.
- Business owners should also encourage employees to record transactions immediately after every sale to maintain accurate and updated business records.

5.5 Impact of the Study

The study provided significant benefits to Kape Tubong by introducing BrewInsight BI, a Business Intelligence-based Revenue Pattern Analytics System that improved sales management, inventory monitoring, and revenue analysis. The centralized database organized business records while reducing errors commonly associated with manual record keeping.

The developed system enabled faster sales recording, automatic inventory updates, organized product management, and efficient report generation. These improvements reduced the workload of the business owner and provided more time to focus on business operations and customer service.

The Business Intelligence dashboard transformed collected sales data into visual reports that simplified the interpretation of business performance. Revenue trends, product performance, and inventory summaries supported informed business planning and

operational decision-making.

The study also demonstrated that affordable Business Intelligence solutions could be implemented successfully in small businesses without requiring expensive software or complex technologies. The system architecture supported accessibility through desktop computers, laptops, and tablets, making the application suitable for different business environments.

The developed study also served as a useful reference for students, researchers, and software developers who intended to develop similar Business Intelligence applications. The project demonstrated the practical application of Agile software development, database management, system analysis, and Business Intelligence concepts within a real business environment.

The implementation of BrewInsight BI also encouraged small business owners to adopt data-driven decision-making by using organized business information instead of relying only on personal observations and manual computations.

5.6 Summary

This chapter presented the conclusions, recommendations, and overall contributions of the study. The findings confirmed that BrewInsight BI successfully achieved the objectives of developing a Business Intelligence-based Revenue Pattern Analytics System for Kape Tubong.

The developed application improved sales management, inventory monitoring, report generation, and revenue analysis by integrating essential business functions into one centralized information system. The implementation of Business Intelligence dashboards provided meaningful visual reports that supported informed decision-making and improved business monitoring.

The developed system functioned according to its intended purpose and satisfied the operational requirements of Kape Tubong. The recommendations presented in this chapter may guide future improvements that will further enhance the functionality, security, usability, and analytical capabilities of the system.

The study demonstrated that Business Intelligence technology can be applied successfully in small businesses to improve operational efficiency, organize business information, and support data-driven management. The developed system also established a strong foundation for future researchers who intend to expand similar systems with advanced analytical features and additional business modules.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.